General Purpose Gear Units for Petroleum, Chemical, and Gas Industry Services

API STANDARD 677
SECOND EDITION, JULY 1997



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Manufacturing, Distribution and Marketing Department

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General Purpose Gear Units for Petroleum, Chemical, and Gas Industry Services

SECTION 1—GENERAL

1.1 Scope

1.1.1 This standard covers the minimum requirements for general-purpose, enclosed single and multistage gear units incorporating parallel shaft helical and right angle spiral bevel gears for the petroleum, chemical, and gas industries. Gears manufactured according to this standard shall be limited to the following pitchline velocities. Helical gears shall not exceed 60 meters per second (12,000 feet per minute), and spiral bevels shall not exceed 40 meters per second (8,000 feet per minute). Spiral bevel gearsets shall be considered matched sets. Other types of gears may be suitable for certain applications, but they are not covered by this standard.

Note: Gears manufactured to this standard are generally limited to 1,500 kilowatts (2,000 horsepower).

This standard includes related lubricating systems, instrumentation, and other auxiliary equipment. This standard is not intended to apply to gears in special-purpose service, which are covered in API 613; to gears integral with other equipment; to epicyclic gear assemblies; or gears with non-involute tooth forms.

Note: A bullet (•) at the beginning of a paragraph indicates that either a decision is required or further information is to be provided by the purchaser. This information should be indicated on the data sheets (see Appendix A); otherwise it should be stated in the quotation request or in the order.

1.1.2 General-purpose gears are applied in equipment trains that are usually spared, or are in noncritical service.

Typical applications for which this standard is intended are cooling tower water pump systems, forced and induced draft fan systems, and other general-purpose equipment trains.

1.2 Alternative Designs

The vendor may offer alternative designs. Equivalent metric dimensions, fasteners, and flanges may be substituted as mutually agreed upon by the purchaser and the vendor.

1.3 Conflicting Requirements

In case of conflict between this standard and the inquiry or order, the information included in the order shall govern.

1.4 Definition of Terms

Terms used in this standard are defined in 1.4.1 through 1.4.19.

1.4.1 axially (horizontally) split: Refers to casing joints that are parallel to the shaft centerline.

1.4.2 bending stress number (S): Defined in 2.4.4.2.

1.4.3 gear: The lower speed element of a gear set.

1.4.4 gear rated power: The maximum power specified by the purchaser on the data sheets and stamped on the nameplate (see 2.4.1).

1.4.5 gear service factor (SF): The factor that is applied to the tooth pitting index and bending stress number, depending on the characteristics of the driver and the driven equipment, to account for differences in potential overload, shock load, and/or continuous oscillatory torque characteristics.

1.4.6 hunting tooth combination: Exists for mating gears when a tooth on the pinion does not repeat contact with a tooth on the gear until it has contacted all the other gear teeth.

1.4.7 maximum allowable speed (in revolutions per minute): The highest speed at which the manufacturer's design will permit continuous operation.

1.4.8 maximum continuous speed (in revolutions per minute): The speed at least equal to 105 percent of the rated pinion speed for variable-speed units and is the rated pinion speed for constant-speed units.

1.4.9 minimum allowable speed (in revolutions per minute): The lowest speed at which the manufacturer's design will permit continuous operation.

1.4.10 normal transmitted power: The power at which usual operation is expected and optimum efficiency is desired. The normal transmitted power may be equal to or less than the gear rated power.

1.4.11 pinion: The higher speed element of a gear set.

1.4.12 rated input speed of the gear unit (in revolutions per minute): The specified (or nominal) rated speed of its driver, as designated by the purchaser on the data sheets.

1.4.13 rated output speed of the gear unit (in revolutions per minute): The specified (or nominal) rated speed of its driven equipment, as designated by the purchaser on the data sheets.

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Note: In selecting the number of teeth for the pinion and gear, it is often impractical for the vendor to match exactly both the rated input and rated output speeds designated on the data sheets. The purchaser will therefore indicate which of the two is specified (that is, must be exactly adhered to by the vendor) and which is nominal (that is, permits some variation). An S will be used to indicate the specified speed, and an N will be used to indicate the nominal speed. The purchaser will also indicate on the data sheets the allowable percentage of variation in the designated gear ratio.

1.4.14 standby service: A normally idle or idling piece of equipment that is capable of immediate automatic or manual start-up and continuous operation.

1.4.15 thermal capacity: The horsepower a unit will transmit continuously for 3 hours or more without exceeding an operating sump temperature rise of 45°C (80°F) above ambient.

1.4.16 trip speed (in revolutions per minute): The speed at which the independent emergency overspeed device operates to shut down a prime mover. (For steam turbines and reciprocating engines, the trip speed will be at least 110 percent of the maximum continuous speed. For gas turbines, the trip speed will be at least 105 percent of the maximum continuous speed.

1.4.17 tooth pitting index (K): Defined in 2.4.3.

1.4.18 unit responsibility: The responsibility for coordinating the technical aspects of the equipment and all auxiliary systems included in the scope of the order. It includes responsibility for reviewing such factors as the power requirements, speed, rotation, general arrangement, couplings, dynamics, noise, lubrication, sealing system, material test reports, instrumentation, piping, and testing of components.

1.4.19 The use of the word *design* in any term (such as design power, design pressure, design temperature, or design speed) should be avoided in the purchaser's specifications. This terminology should be used only by the equipment designer and the manufacturer.

1.5 Referenced Publications

1.5.1 This standard makes reference to American Standards. Other international or national standards may be used as mutually agreed between purchaser and vendor provided it can be shown that these other standards meet or exceed the referenced American Standards.

1.5.2 The editions of the following standards, codes, and specifications that are in effect at the time of publication of this standard shall, to the extent specified herein, form a part of this standard. The applicability of changes in standards, codes, and specifications that occur after the inquiry shall be mutually agreed upon by the purchaser and the vendor.

API

RP 520	*Sizing, Selection and Installation of Pressure-Relieving Devices in Refineries, Part I—Sizing and Selection and Part II—Installation*
RP 526	*Flanged Steel Safety Relief Valves*
Std 611	*General-Purpose Steam Turbines for Petroleum, Chemical, and Gas Industry Services*
Std 613	*Special-Purpose Gear Units for Refinery Service*
Std 614	*Lubrication, Shaft-Sealing, and Control Oil Systems for Special-Purpose Applications*
Std 670	*Vibration, Axial-Position, and Bearing-Temperature Monitoring Systems*
Std 671	*Special-Purpose Couplings for Refinery Service*

AFBMA[1]

Std 7	*Shaft and Housing Fits for Metric Radial Ball and Roller Bearings (Except Tapered Roller Bearings) Conforming to Basic Boundary Plans*
Std 9	*Load Ratings and Fatigue Life for Ball Bearings*
Std 11	*Load Ratings and Fatigue Life for Roller Bearings*
Std 20	*Radial Bearings of Ball, Cylindrical Roller and Spherical Roller Types, Metric Design; Basic Plan for Boundary Dimensions, Tolerances and Identification Code*

AGMA[2]

2001	*Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*
908	*Geometry Factors for Determining the Pitting Resistance and Bending Strength of Spur, Helical, Herringbone Gear Teeth*
6010	*Standard for Spur, Helical, Herringbone, and Bevel Enclosed Drives*
6011	*Specification for High Speed Helical Gear Units*
9002	*Bores and Keyways for Flexible Couplings*
1012	*Gear Nomenclature, Definitions of Terms with Symbols*

ASME[3]

B1.1	*Unified Inch Screw Threads (UN and UNR Thread Form*)
B16.1	*Cast Iron Pipe Flanges and Flanged Fittings*
B16.5	*Pipe Flanges and Flanged Fittings*
B16.11	*Forged Fittings Socket-Welding and Threaded*

[1]Anti-Friction Bearing Manufacturers Association, 1235 Jefferson Davis Highway, Arlington, Virginia 22202.
[2]American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, Virginia 22314.
[3]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

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B16.42 *Ductile Iron Pipe Flanges and Fittings*
B1.20.1 *General Purpose (Inch) Pipe Threads*
B31.3 *Chemical Plant and Petroleum Refinery Piping Boiler and Pressure Vessel Code*, Section VIII, "Rules for Construction of Pressure Vessels," and Section IX, "Welding and Brazing Qualifications"
Y14.2M *Line Conventions and Lettering*

SSPC[4]
SP6 *Commercial Blasting Cleaning*

ASTM[5]
A 6 *Standard Specification for General Requirements for Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use*
A 27 *Standard Specification for Carbon Steel Castings for General Application*
A 36 *Standard Specification for Structural Steel*
A 48 *Standard Specification for Gray Iron Casting*
A 106 *Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service*
A 131 *Standard Specification for Structural Steel for Ships*
A 192 *Standard Specification for Seamless Carbon Steel Boiler Tubes for High-Pressure Service*
A 269 *Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service*
A 275 *Standard Test Method for Magnetic Particle Examination of Steel Forgings*
A 283 *Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates*
A 284 *Standard Specification for Low and Intermediate Tensile Strength Carbon-Silicon Steel Plates for Machine Parts and General Construction*
A 285 *Standard Specification for Carbon Steel Pressure Vessel Plates, Low- and Intermediate-Tensile Strength*
A 312 *Standard Specification for Seamless and Welded Austenitic Stainless Steel Pipe*
A 388 *Standard Practices for Ultrasonic Examination of Heavy Steel Forgings*
A 395 *Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures*
A 515 *Standard Specification for Carbon Steel Pressure Vessel Plates for Intermediate- and Higher-Temperature Service*
A 516 *Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate and Lower-Temperature Service*
A 575 *Standard Specification for Carbon Steel Bars, Merchant Quality, M-Grades*
A 576 *Standard Specification for Carbon Steel Bars, Hot-Wrought, Special Quality*
E 125 *Standard Reference Photographs for Magnetic Particle Indications on Ferrous Castings*
E 709 *Standard Guide for Magnetic Particle Examination*

ISO[6]
3740 *Determination of Sound Power Levels of Noise Sources—Guidelines for the Use of Basic Standards and for the Preparation of Noise Test Codes*
3746 *Determination of Sound Power Levels of Noise Sources—Survey Method*
3744 *Determination of Sound Power Levels of Noise Sources Using Sound Pressure—Engineering Method in an Essentially Free Field Over a Reflecting Plane*

NFPA[7]
70 *National Electrical Code*

1.5.3 The purchaser and the vendor shall mutually determine the measures that must be taken to comply with any governmental codes, regulations, ordinances, or rules that are applicable to the equipment.

1.5.4 It is the vendors responsibility to invoke all applicable specifications to each subvendor.

1.5.5 For unit conversion, the factors in Chapter 15 of API 2564, Manual of Petroleum Measurement Standards were used to convert from Customary to SI units. The resulting exact SI units were then rounded off.

[4]Structural Steel Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213-2683.
[5]American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, Pennsylvania 19428.
[6]International Organization for Standardization. ISO publications are available from the American National Standards Institute, 11 West 42nd Street, New York, New York 10036.
[7]National Fire Protection Association, 1 Batterymarch Park, P.O.Box 9101, Quincy, Massachusetts 02269-9101.

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SECTION 2—BASIC DESIGN

2.1 General

2.1.1 Gear units purchased according to this standard shall conform to ANSI/AGMA 6010 or 6011 and to related standards referenced therein, except as modified or supplemented by this standard.

2.1.2 The equipment (including auxiliaries) covered by this standard shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation. It is recognized that this is a design criterion.

2.1.3 The vendor shall assume unit responsibility for all equipment and all auxiliary systems included in the scope of the order.

• **2.1.4** The purchaser will specify the equipment's normal transmitted power (see 2.4.1 for gear-rated power). This is the power associated with the normal operating point of the driven equipment.

• **2.1.5** Control of the sound pressure level (SPL) of all equipment furnished shall be a joint effort of the purchaser and the vendor. The equipment furnished by the vendor shall conform to the maximum allowable sound pressure level specified by the purchaser.

2.1.6 Unless otherwise specified, cooling water systems shall be designed for the following conditions:

Velocity over heat exchange surfaces	1.5–2.5 m/s	5–8 ft/s
Maximum allowable working pressure (MAWP)	≥5.2 bar (Note 2)	≥75 psig
Test pressure	≥1.5 MAWP	≥115 psig
Maximum pressure drop	1 bar	15 psi
Maximum inlet temp.	30°C	90°F
Maximum outlet temp.	50°C	120°F
Maximum temp. rise	20K	30°F
Minimum temp. rise	10K	20°F
Fouling factor on water side	0.35m^{2}K/kW	0.002 hr-ft^2-°F/Btu
Shell corrosion allowance	3.0 mm	0.125 in

Provisions shall be made for complete venting and draining of the system.

Note 1: The vendor shall notify the purchaser if the criteria for minimum temperature rise and velocity over heat exchange surfaces result in a conflict. The criterion for velocity over heat exchange surfaces is intended to minimize water-side fouling; the criterion for minimum temperature rise is intended to minimize the use of cooling water. The purchaser will approve the final selection.

Note 2: Gauge pressure.

2.1.7 Equipment shall be designed to run safely to the trip speed setting. Rotors for turbine-driven gear units shall be designed to operate safely at momentary speeds up to 130 percent of the rated speed.

2.1.8 The arrangement of the equipment, including piping and auxiliaries, shall be developed jointly by the purchaser and the vendor. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance.

• **2.1.9** Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division or zone) specified by the purchaser on the data sheets, and shall meet the requirements of NFPA 70, Articles 500, 501, 502, and 504 as well as local codes specified and furnished by the purchaser.

2.1.10 Oil reservoirs and housings that enclose moving lubricated parts (such as bearings, shaft seals, highly polished parts, instruments, and control elements) shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation and idleness.

2.1.11 All equipment shall be designed to permit rapid and economical maintenance. Major parts such as casing components and bearing housings shall be designed (shouldered or cylindrically doweled) and manufactured to ensure accurate alignment on reassembly.

2.1.12 The combined performance of gear unit (with driver and driven machines) after installation shall be the joint responsibility of the purchaser and the vendor.

2.1.13 In designing the gear unit, the vendor shall take into account that many factors (such as piping loads, alignment at operating conditions, supporting structure, and handling during shipment and installation) may adversely affect site performance.

• **2.1.14** The purchaser will specify whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), as well as the weather and environmental conditions in which the equipment must operate (including maximum and minimum temperatures, unusual humidity, and dusty or corrosive conditions).

2.1.15 Gear units shall not require a break-in period.

2.1.16 The gearing shall be designed to withstand all internal and external loads inherent to geared, rotating machinery systems. The gearing shall be capable of withstanding the specified additional external loads (thrust, lube-oil piping and so forth) transmitted across the gear mesh while the unit is operating at the gear rated power specified by the purchaser.

2.1.17 Spare parts for the machine and all furnished auxiliaries shall meet all the criteria of this standard.

2.1.18 A guide to gear nomenclature will be found in AGMA 1012.

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• 2.2 Shaft Assembly Designation

The purchaser will specify on the data sheets the appropriate shaft assembly designation selected from the combinations listed in Table 1 and illustrated in Figure 1. The purchaser may alternatively circle one or more of the assembly designations on a copy of Figure 1 and submit the copy with the quotation request. If the shaft arrangement has not been finalized at the time of the quotation request, the purchaser will designate all of the combinations under consideration.

Note: The material for Table 1 and Figure 1 was reprinted from ANSI/AGMA 6010 with permission of the publisher. Refer to Figure 1 for explanatory notes.

Table 1—Shaft Assembly Combinations
Parallel Shaft Assembly Combination (see Figure 1-A)

High Speed Shaft	Low Speed Shaft
L	R
R	L
L	L
R	R
R	LR
L	LR
LR	L
LR	R
LR	LR

Bevel Gear Assembly Combinations Horizontal Shafts (see Figure 1-B)

High Speed Shaft	Low Speed Shaft
1	L
1	R
1	LR
2	L
2	R
2	LR

Bevel Gear Assembly Combinations Vertical Low Speed Shafts (see Figure 1-C)

High Speed Shaft	Low Speed Shaft
1	U
2	U
1	D
2	D
1	UD
2	UD

2.3 Shaft Rotation

2.3.1 The rotational direction of high-speed and low-speed shafts is either clockwise (CW) or counterclockwise (CCW), as viewed from the coupling ends of the respective shafts.

2.3.2 On the data sheets and in drawings and tables, the shaft rotational direction shall be designated by the abbreviations CW or CCW, as indicated by the circular arrows in Figure 2.

• **2.3.3** The purchaser will specify on the data sheets the rotational direction of both the high-speed and the low-speed shafts. When either or both shafts have an extension at each end, the purchaser may alternatively indicate the rotational directions on the appropriate assembly designation (see Figure 1) and submit a copy of the figure with the quotation request.

2.3.4 In finalizing the data for purchase, the purchaser shall prepare a sketch that shows the direction of rotation for each item in the train.

2.4 Rating

2.4.1 GEAR-RATED POWER

• **2.4.1.1** The gear-rated power of the unit will normally be specified by the purchaser. For gear units located next to the driver, the gear-rated power will be the maximum installed power of the driver. For electric motor drivers, the gear-rated power will be the motor nameplate rating multiplied by the motor service factor. All modes of normal and abnormal operation shall be examined. Modes of operation may include the number of starts per unit of time, reduced load, removal, reversed load, reduced speed, and overload and overspeed conditions. For gear units between two items of driven equipment, the power rating of such gears should normally be not less than Items a or b below, whichever is greater.

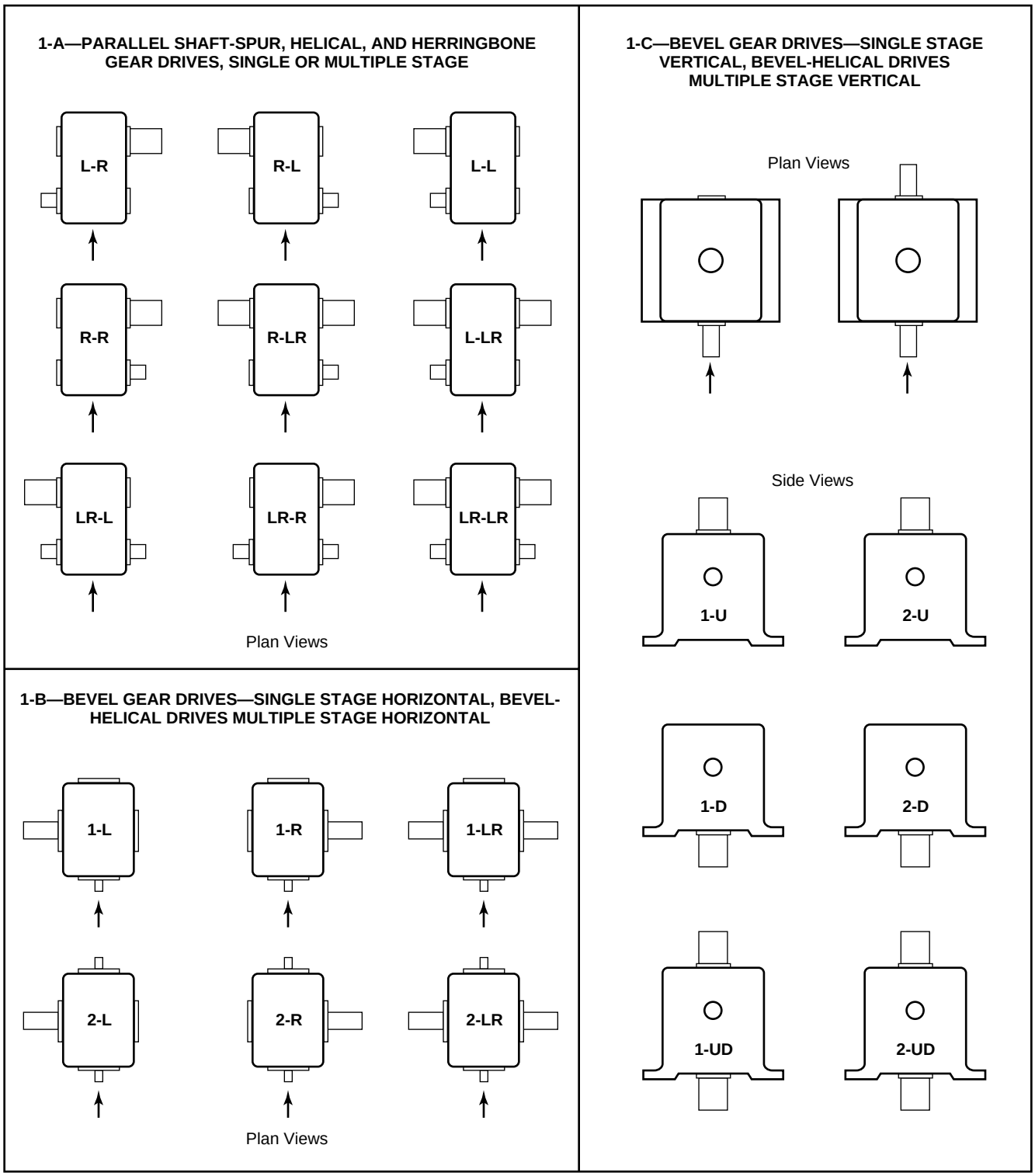
a. 110 percent of the maximum power required by the equipment driven by the gear.
b. The maximum power of the driver prorated between the driven equipment, based on normal power demands. If the maximum transmitted torque occurs at an operating speed other than the maximum continuous speed, this torque and its corresponding speed will be specified by the purchaser and shall be the basis for sizing the gear.

2.4.1.2 Gear unit rating shall be based on the lowest of the following capacities: tooth pitting resistance, tooth bending strength, or thermal capacity.

• 2.4.2 GEAR SERVICE FACTOR

The minimum gear service factor (SF) will be specified by the purchaser on the data sheets. In no case should the service factor be less than that required for the service by Table 2.

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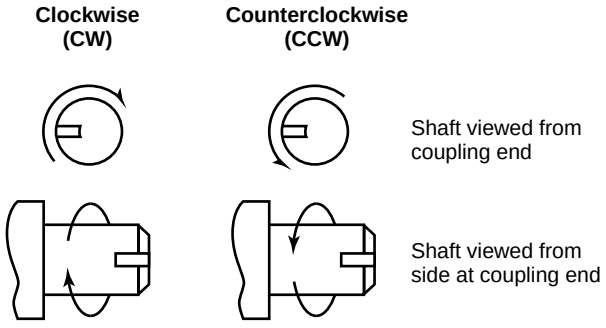
Notes:
1. Code: L = left; R = right; U = up position-low speed shaft; D = down position-low speed shaft.
2. Arrows indicate line-of-sight to determine direction of shaft extensions.
3. Numerals preceding the hyphen refer to number of high speed shaft extensions.
4. Letters preceding the hyphen refer to number and direction of high speed shaft extensions.
5. Letters following the hyphen refer to number and direction of low speed shaft extensions.

Figure 1—Shaft Assembly Designations

--`,,``,`,,,,`,``,,`,,,,`,,,`,`-`-`,,`,,`,`,,`---

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Figure 2—Shaft Rotation Designations

2.4.3 TOOTH PITTING INDEX

2.4.3.1 Gear elements shall be sized on the basis of a tooth pitting index called a *K* factor. This method includes factors to account for such considerations as the radii of curvature of the contacting tooth surfaces, extended life, high reliability, dynamic load effects, distribution of tooth loading across the face, and the strength of the materials in terms of pitting resistance. This simplified system for sizing the gear unit is consistent with AGMA 2001, with conservative assumptions for each variable in the more complex basic equations contained in that document.

2.4.3.2 The *K* factor is defined as follows:

$$K = [W_t / dF] \ [(R + 1)/R]$$

In SI units, W_t can be expressed as follows:

$$W_t = [(1.91 \times 10^7)\, P_g] / (N_p\, d)$$

In Customary units, this translates to:

$$W_t = (126{,}000\, P_g) / N_p$$

Where:

K = tooth pitting index, in pounds per square inch (megapascals).
W_t = transmitted tangential load at the operating pitch diameter, in newtons (pounds).
F = net face width, in millimeters (inches).
d = pinion pitch diameter, in millimeters (inches).
R = number of teeth in the gear divided by number of teeth in the pinion.
P_g = gear-rated power, in kilowatts (horsepower).
N_p = pinion speed, in revolutions per minute.

2.4.3.3 The allowable *K* factor at the gear-rated power will vary with the materials selected for the gear teeth, the tooth hardening processes used and the service factor.

Table 2—Minimum Gear Service Factors

Driven Equipment	Prime Mover		
	Motor	Turbine	Internal Combustion Engine
Agitators and mixers	1.7	1.7	2.0
Blowers			
Centrifugal	1.4	1.6	1.7
Rotary lobe	1.7	1.7	2.0
Vane	1.4	1.6	1.7
Compressors			
Rotary lobe (radial, axial, screw, and so forth)	1.7	1.7	2.0
Reciprocating	2.0	2.0	2.3
Conveyors			
Uniformly loaded and fed	1.4	1.6	1.7
Not uniformly fed	1.7	1.7	2.0
Shaker or reciprocating	2.0	2.0	2.3
Crusher (ore and stone)	2.0	2.0	2.3
Extruders			
General service	1.4	1.6	–
Blow molders and preplasticizers	1.7	1.7	–
Fans			
Centrifugal	1.4	1.6	1.7
Forced draft	1.4	1.6	1.7
Induced draft	1.7	2.0	2.2
Cooling tower	2.0	2.0	–
Feeders			
General service	1.7	1.7	2.0
Reciprocating	2.0	2.0	2.3
Generators	1.3	1.3	1.7
Pumps			
Centrifugal (all services except as listed below)	1.3	1.5	1.7
Centrifugal, boiler feed	1.7	2.0	–
Centrifugal, hot oil	1.7	2.0	–
High speed centrifugal (over 3,600 rpm)	1.7	2.0	–
Centrifugal, water supply	1.5	1.7	2.0
Rotary, axial flow (all types)	1.5	1.5	1.8
Rotary, gear	1.5	1.5	1.8
Reciprocating	2.0	2.0	2.3

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The allowable *K* factor is calculated as follows:

$$K_a = I_m / (SF)$$

Where:

K_a = allowable *K* factor.
I_m = material index number (from Table 3 and Figure 3).
SF = minimum gear services factor.

2.4.3.4 When rating bevel gearing, the bevel gearing must be converted to equivalent helical gearing per 2.4.3.2, calculated at the mean pitch diameter of the bevel set.

The mean pitch diameter of the pinion and gear is defined as follows:

$$MPD_p = PD_p–[(F) (\sin \Gamma)]$$
$$MPD_g = PD_g–[(F) (\sin \Gamma)]$$

Where:

MPD_p = mean pitch diameter of bevel pinion.
MPD_g = mean pitch diameter of bevel gear.
PD_p = bevel pinion pitch diameter.
PD_g = bevel gear pitch diameter.
N_p = number of teeth pinion.
N_g = number of teeth gear.
F = face width.
Γ = pitch cone angle.
Ψ = spiral angle = helix angle.
DP_1 = bevel linear pitch.
DP_n = bevel normal pitch at mean pitch diameter.
$DP_n = N_p / [(MPD_p) (\cos \Psi)]$.

2.4.3.5 Table 3 presents material index numbers and maximum length to diameter (*L*/*d*) ratios for several combinations of materials in current use. Index numbers for other materials and hardnesses can be determined by referring to Figure 3. It should be noted that the minimum hardness is specified. The normal heat treating practice requires a tolerance range of about 40 to 50 Brinell numbers. A gear on the high side of its range and a pinion on the low side of its range may therefore have the same or overlapping hardness. Such a combination is perfectly satisfactory.

Table 3—Material Index Numbers and Maximum *L*/*d* Ratios

Minimum Gear Hardness	Minimum Pinion Hardness	Material Index Number		Maximum Pinion *L*/*d* Ratio	
		Mega-pascals	Pounds per Square Inch	Double Helical	Single Helical
223 BHN	262 BHN	0.896	130	2.4	1.7
262 BHN	302 BHN	1.103	160	2.3	1.6
302 BHN	341 BHN	1.379	200	2.2	1.5
352 BHN	50 HRC[a]	1.793	260	2.0	1.45
50 HRC[a]	50 HRC[a]	2.068	300	1.9	1.4
55 HRC[b]	55 HRC[b]	2.827	410	1.7	1.35
58 HRC[b]	55 HRC[b]	3.034	440	1.6	1.3

Note: *L* = net face width plus gap, in millimeters (inches); *d* = pinion pitch diameter in millimeters (inches); BHN = Brinell hardness number; HRC = Rockwell hardness (C scale).
[a]Nitrided
[b]Carburized

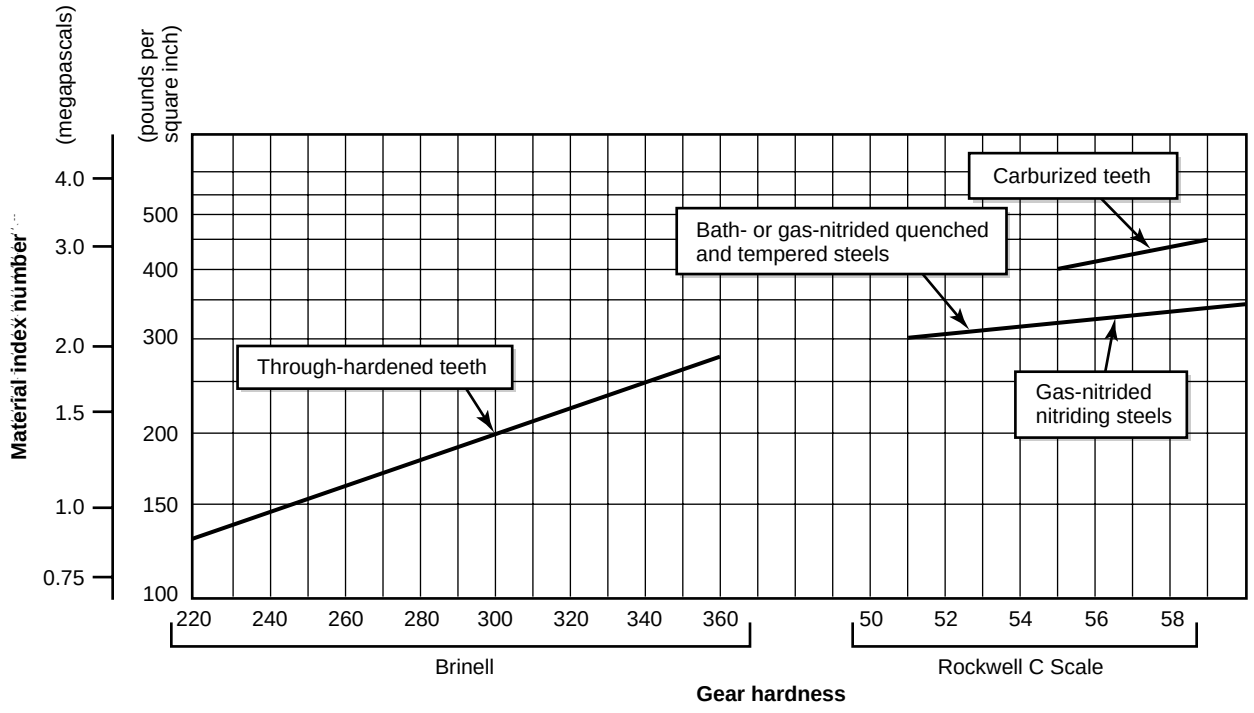


Figure 3—Material Index Numbers

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2.4.3.6 The *L*/*d* values shown in Table 3 apply to helical gears with unmodified leads when designed to transmit the rated power. Generally, higher *L*/*d* ratios can be allowed when the total lead mismatch caused by deflection (combined bending and torsional) across a helix is less than 0.038 millimeter (0.0015 inch) through hardened gears and 0.025 millimeter (0.0010 inch) on case hardened gears. The *L*/*d* values shown in Table 3 apply to helical gears only. The maximum face width for bevel gears is equal to 0.33 times the cone distance. This will result in *L*/*d* ratios lower than Table 3.

2.4.3.7 When a higher *L*/*d* ratio than tabulated in Table 3 is proposed, the gear vendor shall submit justification in the proposal for using the higher *L*/*d* ratio. Purchaser's approval is required when *L*/*d* ratios exceed those in Table 3. When operating conditions other than the gear rated power are specified by the purchaser, such as the normal transmitted power, the gear vendor shall consider in the analysis the length of time and load range at which the gear unit will operate at each condition so that the correct leads are to be furnished. Purchaser and vendor, shall agree on the tooth contact patterns obtained in the checking stand, housing, and test stand. The analysis of a gear tooth-load distribution versus lead modification is not within the scope of this standard.

2.4.4 TOOTH SIZE AND GEOMETRY

2.4.4.1 The size and geometry of the gear teeth shall be selected so that the bending stress number, as calculated using Equation 1, does not exceed the values in Figure 4. This method includes factors similar to those used to determine the allowable *K* factor. This simplified system for sizing gear teeth is consistent with AGMA 2001.

2.4.4.2 The vendor shall calculate the bending stress number for both the pinion and the gear. Where idlers are used, the calculated stress shall be limited to 70 percent of the value given in Figure 4. The bending stress number for helical gears is calculated as follows:

In SI units,

$$S = [(W_t / m_n F)] (SF) [(1.8 \cos \Psi) / J] \quad (1)$$

In Customary units,

$$[(W_t P_{nd}) / F] (SF) [(1.8 \cos \Psi) / J]$$

Where:

S = bending stress number.
P_{nd} = normal diametrical pitch, in 1/inches.
Ψ = helix angle.
J = geometry factor (from AGMA 908).
m_n = module number, in millimeters.

2.4.4.3 The bending stress number for bevel pinions and gears is calculated as follows:

In SI units,

$$S = [(W_t M_n) / F] (SF) [(3.9 \cos \Psi) / J]$$

In Customary units,

$$S = [(W_t P_{nd}) / F] (SF) [(3.9 \cos \Psi) / J]$$

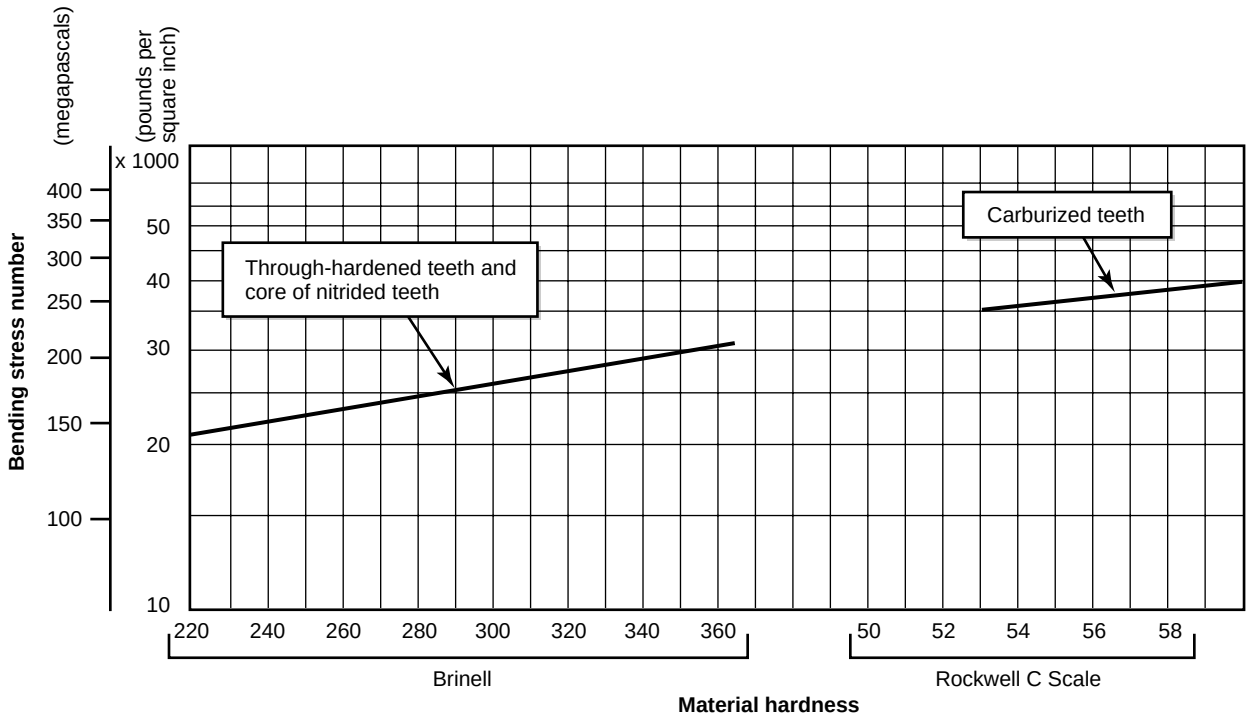


Figure 4—Bending Stress Numbers

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Where:

S = bending stress number.
P_{nd} = normal diametrical pitch, in 1/inches.
Ψ = spiral angle.
J = geometry factor (from AGMA 2003).
m_n = module number, in millimeters.

2.4.5 OVERHUNG LOADS

When a chain, gear, or belt drive is mounted on the input or output shaft, the overhung load shall be calculated by multiplying the transmitted force that is tangent to the pitch circle by the applicable factor from Table 4. Shafts, bearings, and bolting shall be sized according to the overhung load. All forces must be considered to be acting in the most unfavorable direction.

Table 4—Overhung Load Factors Applied to Parallel Shaft and Right Angle Gears

Drive Type	Factor
Single or multiple chain	1.00
Cut pinion run with cut gear	1.25
Timing belts	1.25
Single or multiple V-belt	1.50
Flat belt	2.50

2.4.6 DEVIATIONS

It is recognized that special cases will exist in which it may be desirable to deviate from the rating rules specified in 2.4.1 through 2.4.5. The vendor shall describe and justify such deviations in the proposal.

2.5 Casings

2.5.1 DESIGN PARAMETERS

2.5.1.1 Gear casings shall be either cast or fabricated and shall be designed and constructed to maintain rotor alignment under all load conditions when provided with supports and installed in accordance with the vendor's written instructions.

2.5.1.2 The equipment feet shall be provided with vertical jackscrews and shall be drilled with pilot holes at two locations that are accessible for use in final doweling. Dowel hole locations shall be selected to minimize the effects of casing distortion and misalignment with connected equipment.

Note: For parallel shaft units, these locations are typically as close as possible to the vertical plane of the highest speed pinion centerline.

2.5.1.3 Casing bores shall be machined to a sufficient degree of accuracy so that spare gear sets purchased with the gearbox or at a later date will have the gear-tooth contact and power rating of the original gear set.

2.5.1.4 All internal piping should preferably be welded and should preferably use flanges for all connections. Any threaded piping shall be a minimum of Schedule 80 and shall be seal welded at flanges (see 2.6.2.1d).

2.5.1.5 Where internal space does not permit the use of $^1/_2$-, $^3/_4$-, or 1-inch pipe, stainless steel tubing in accordance with 3.5.11 may be furnished.

2.5.1.6 The design of internal piping and oil pans shall achieve proper support and protection to prevent damage from vibration or from shipment, operation, and maintenance. Cantilevered piping shall include reinforcing gussets in two planes at all pipe-to-flange connections.

2.5.1.7 Casings shall be designed to permit rapid drainage of lube-oil and to minimize oil foaming that could lead to excessive heat rise of the oil.

2.5.1.8 A filter-breather shall be provided. The filter-breather shall be constructed of stainless steel with stainless steel or copper-nickel alloy internals, designed and located to prevent entrainment or discharge of oil to the atmosphere, pressure buildup in the casing, entrance of water during violent rainstorms, and entrance of dirt entrained in the air. The filter-breather shall be at least $^3/_4$-inch National Pipe Thread (NPT), and its construction shall permit easy disassembly for inspection and cleaning.

Note: Small gearboxes may not accommodate this size filter-breather. Alternate configurations shall be as mutually agreed between the purchaser and vendor.

2.5.1.9 A removable, gasketed inspection cover or covers shall be provided in the gear casing to permit direct visual inspection of the full-face width of the gear elements. The inspection opening or openings shall preferably be at least one-half the width of the gear face.

2.5.1.10 Permanent coatings or paint shall not be applied to the interior of the casing unless the purchaser approves in advance the material and method of application.

2.5.1.11 On units that have pressurized oil systems with pitch line velocities above 15 meters per second (3,000 feet per minute), the gearbox casing shall be designed so that the gears do not dip into the oil during operation or upon shutdown. Gear units at or below 15 meters per second (3,000 feet per minute) pitch line velocity may dip into the oil; however, above 10 meters per second (2,000 feet per minute) pitch line velocity, an oil pan shall be used to ensure rapid drainage and to minimize foaming.

2.5.2 JOINTS

Casing splitlines shall use a metal-to-metal joint (with a suitable joint compound) that is tightly maintained by suitable bolting. Gaskets (including string-type) shall not be used on splitlines for parallel shaft gears.

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Note: Due to the interaction of the rotating elements of spiral bevel gear units, it is normally required that the element's mounting distances be adjustable to achieve proper tooth mesh contact and backlash. When the adjustment is made by shifting axially, in line with the element's shaft center line of rotation by use of a combination housing cover-bearing carrier, shim type gaskets are permitted.

2.5.3 BOLTING

2.5.3.1 Case bolting may be of the through-bolt, studded, or cap-screw type. Threaded bolt holes shall not penetrate through the wall into the interior of the casing. Disassembly shall not require removal of studs.

2.5.3.2 Studded connections shall be furnished with studs installed. Blind stud holes should be drilled only deep enough to allow a preferred tap depth of $1^1/_2$ times the major diameter of the stud; the first $1^1/_2$ threads at the bottom end of each Class 1, 2, and 3 studs shall be removed. Class 4 and 5 studs shall be installed with a depth gauge and shall not bottom in holes.

2.5.3.3 Bolting shall be furnished as follows:

a. Details of threading shall conform to ASME B1.1.

b. Adequate clearance shall be provided at bolting locations to permit the use of socket or box wrenches.

c. Internal socket-type, slotted-nut, or spanner-type bolting shall not be used unless specifically approved by the purchaser.

2.5.4 ASSEMBLY AND DISASSEMBLY

It shall be possible to lift the upper half of horizontally split casings without disturbing the piping of the main oil supply to the lower half of the casing.

2.6 Casing Connections

2.6.1 SERVICE AND SIZE CRITERIA

2.6.1.1 General-purpose gear casings usually have a self-contained lube-oil system. When the lube-oil system is external and provided by the purchaser, a single supply and a single drain connection from the gear casing shall be provided by the vendor. The minimum drain pipe size for external systems shall be based on the total inlet flow to the gear casing, according to Table 5.

Table 5—Drain Pipe Sizes

Inlet Flow Rate		Minimum Drain Size	
Liters per Minute	Gallons per Minute	Millimeters	NPS[a]
25	7	75	3
55	15	100	4
170	45	150	6
380	100	200	8

[a]Nominal pipe size.

2.6.1.2 Openings for NPS of $2^1/_2$, $3^1/_2$, 5, 7, and 9 inches shall not be used. Openings for pipe sizes of $1^1/_4$ inches shall not be used for customer connections.

2.6.2 LUBE-OIL CONNECTIONS

• **2.6.2.1** Lube-oil inlet and drain connections, oriented as specified, shall be at least $^3/_4$-inch NPS and shall be flanged or machined and studded. Where flanged or machined and studded openings are impractical, threaded openings in sizes $^3/_4$- through $1^1/_2$-inch NPS are permissible. These threaded openings shall be installed as specified below:

a. A pipe nipple, preferably not more than 150 millimeters (6 inches) long, shall be screwed into the threaded opening.
b. Pipe nipples shall be a minimum of Schedule 160 seamless for sizes 1 inch and smaller and a minimum of Schedule 80 for a size of $1^1/_2$ inches.
c. The pipe nipple shall be provided with a welding-neck or socket-weld flange.
d. The threaded connection shall be seal welded; however, seal welding is not permitted on cast iron equipment, for instrument connections, or where disassembly is required for maintenance. Seal-welded joints shall be in accordance with ASME B31.3.
e. Tapped openings and bosses for pipe threads shall conform to ASME B16.5.
f. Pipe threads shall be taper threads conforming to ASME B16.5.

2.6.2.2 Tapped openings not connected to piping shall be plugged with solid, round-head steel plugs furnished in accordance with ASME B16.11. As a minimum, these plugs shall meet the material requirements of the casing. Plugs that may later require removal shall be of corrosion-resistant material. A lubricant that meets the proper temperature specification shall be used on all threaded connections. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted.

2.6.2.3 Flanges shall conform to ASME B16.1, B16.5, and B16.42 as applicable, except as specified in 2.6.2.3.1 and 2.6.2.3.2.

2.6.2.3.1 Cast iron flanges shall be flat faced and shall have a minimum thickness of Class 250 per ASME B16.1 for sizes 8 inches and smaller.

2.6.2.3.2 Flanges that are thicker or have a larger outside diameter than required by ASME B16.5 are acceptable.

2.6.2.3.3 Connections other than those covered by ASME B16.5 require purchaser's approval. Unless otherwise specified, mating parts for these nonstandard flanges shall be furnished by the vendor.

2.6.2.4 Machined and studded connections shall conform to the facing and drilling requirements of ASME B16.1 or

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B16.5. Studs and nuts shall be furnished installed. The first $1^1/_2$ threads at both ends of each stud shall be removed. Connections larger than those required by ASME shall meet the requirements of 2.6.2.3.3.

2.6.2.5 Alternative flange connections (for example, those of SAE[8]) are permitted for nonpurchaser connections unless otherwise specified.

2.7 Gear Elements

2.7.1 GENERAL

2.7.1.1 Gear-teeth profiles shall be manufactured by the generating or shaping process. Gears and pinions with pitch line velocities below 15 meters per second (3,000 feet per minute) may be finished unassembled with their shafts. Gears and pinions with pitch line velocities at or above 15 meters per second shall be finish cut or ground assembled with their respective shafts. The methods of generating and finishing provided in Table 6 are acceptable to the specified pitch line velocity limits. Shaving cutters and rotary hones shall have a hunting tooth combination with the work piece.

Table 6—Generating and Finishing Methods by Pitch Line Velocity Limits

Method	Pitch Line Velocity Limit	
	Meters per Second	Feet per Minute
Parallel shaft gearing		
Shape only	7.0	1,400
Precision hob only	20	4,000
Shape and shave	20	4,000
Shape and lap	20	4,000
Hob and grind[a]	Above 20	Above 4,000
Shape and grind	Above 20	Above 4,000
Precision hob and shave	Above 20	Above 4,000
Precision hob and lap	Above 20	Above 4,000
Precision hob and hone	Above 20	Above 4,000
Right angle gearing		
Shape and lap	10	2,000
Shape and hard cut	Above 10	Above 2,000
Shape and grind	Above 10	Above 2,000

[a]Grinding includes precision form finishing.

2.7.1.2 The tooth surface on loaded faces of completed gears shall have a minimum finish, as measured along the pitch line, of 0.8 micrometer (32 microinches) Ra above a 20 meters per second (4,000 feet per minute) pitch line velocity limit and 1.6 micrometers (64 microinches) Ra at or below a 20 meters per second (4,000 feet per minute) minimum pitch line velocity limit.

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2.7.1.3 The design of single-helical and right angle gears shall be such that the effects of the moments on the gear elements, resulting from axial tooth reaction at the gear mesh, will not impair the expected performance of the gear unit.

2.7.1.4 Hunting tooth combinations are required. To achieve this, it may be necessary for the purchaser to adjust the exact gear ratio. If this is impractical, the purchaser and the vendor shall negotiate a solution.

2.7.1.5 Each gear and each pinion of parallel shaft gear sets shall be supported between two bearings. Overhung designs are acceptable only below 15 meters per second (3,000 feet per minute) pitch line velocity and 225 kilowatts (300 horsepower). The purchaser's approval is required if these values are exceeded.

Note: Elements of right angle gear sets may require overhung designs at pitch line velocity limits above this criteria.

2.7.1.6 Mounting of Bevel Gears/Shafts Intersecting

There are two types of mountings for spiral bevel gear units, straddle and overhung. The mounting configuration is dictated by the size and ratio of the unit. When the gear diameter is large enough, one of the bearings shall be mounted inboard of the pitch cone (small end) or straddled. If the gear diameter will not permit straddle mounting, the element may be overhung. The vendor shall indicate on the data sheets the pinion mounting configuration, straddle or overhung.

2.7.2 QUALITY ASSURANCE

2.7.2.1 Each pair of mating gears for parallel shaft units shall have a contact check in the gear casing at the vendor's shop. Those gear sets operating above 20 meters per second (4,000 feet per minute) pitch line velocity shall be checked for contact on a contact checking stand and in the job casing at the vendor's shop. A thin coating of color transfer material (such as Prussian blue) shall be applied to four or more teeth of the dry degreased gear at three locations 120 degrees apart. The coated teeth shall be rotated through the mesh with a moderate drag torque applied in a direction that will cause the teeth to contact on the normally loaded faces. The color transfer shall show evidence of contact distributed across the face as prescribed by the vendor. Before the contact tests, the vendor shall make available to the purchaser a contact drawing or vendor engineering specification that defines the expected and acceptable contact. Unmodified tooth profile leads shall show a minimum contact of 70 percent along the axis and 30 percent radially. Contact drawings or specifications and the results of the contact checks shall be delivered with the unit documentation. The preferred method of preserving the results of the contact check is to lift the colors from a tooth by applying and peeling off a strip of clear adhesive tape and then sticking the tape to a notated sheet of paper.

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2.7.2.2 Spiral bevel gear tooth contact shall be checked per Appendix G.

2.7.2.3 The vendor shall have a manufacturing process control system to assure that the designed quality of gearing is produced. The process control system should include: manufacturing planning, machine tool maintenance and inspection programs, gear generating, hardening and finishing procedures, cutting tool selection and maintenance procedures, material control procedures, heat treatment control, inspection procedures, and quality assurance programs required to achieve and maintain the required gear quality.

2.7.2.4 The gear vendor shall demonstrate the axial stability of each meshing pair of double-helical gears operating with hydrodynamic radial bearings and pinion speeds above 1,800 revolutions per minute by either (a) measuring the unfiltered peak-to-peak shaft axial vibration, which shall not exceed 60 micrometers (.0025 inches) during full-speed testing, or (b) using indicators to make a slow rotation check. The preferred method for this slow rotation check is to hold one member (usually the gear) firmly in a fixed axial position and indicate the axial movement of the other member (usually the pinion) as the parts are rotated through at least one full revolution of the gear with a drag torque applied in a direction that will force the normally loaded tooth faces into contact. The total axial motion of the other member (pinion) relative to the fixed member (gear) shall not exceed 40 micrometers (0.0015 inches) for gear units having a 20 meters per second (4,000 feet per minute) pitch line velocity or greater and 60 micrometers (0.0025 inches) for gear units having less than a 20 meters per second pitch line velocity.

2.7.3 PINION FABRICATION

Pinions operating below 15 meters per second (3,000 feet per minute) pitch line velocity may be manufactured separate from their shafts. At 15 meters per second pitch line velocity and above, pinions shall be integrally forged with their shafts unless otherwise approved by the purchaser.

2.7.4 SHAFTS

2.7.4.1 Shafts shall be sized to transmit the gear rated power within the stress limits of AGMA 6011. Shafts shall be made of one-piece, heat-treated forged, or hot-rolled steel; shall be accurately machined throughout their entire length; and shall be suitably finished at their bearing surfaces.

• **2.7.4.2** When specified or when vibration and/or axial-position probes are furnished, the rotor shaft sensing areas to be observed by radial-vibration probes shall be concentric with the bearing journals. All shaft sensing areas (both radial vibration and axial position) shall be free from stencil and scribe marks or any other surface discontinuity, such as an oil hole or a keyway, for a minimum of one probe-tip diameter on each side of the probe. These areas shall not be metallized, sleeved, or plated. The final surface finish shall be a maximum of 1.0 micrometers (32 microinches) Ra, preferably obtained by honing or burnishing. These areas shall be properly demagnetized to the levels specified in API 670 or otherwise treated so that the combined total electrical and mechanical runout shall not exceed 25 percent of the maximum allowed peak-to-peak vibration amplitude or the following value, whichever is greater:

a. For areas to be observed by radial-vibration probes, 5 micrometers (0.25 mil).
b. For areas to be observed by axial-position probes, 10 micrometers (0.50 mil).

2.7.4.3 When coupling hubs are not integral with the shafts, the hubs shall be mounted with interference fits. The shaft end configuration shall conform to the requirements specified under 3.2.

2.8 Dynamics

2.8.1 CRITICAL SPEEDS

2.8.1.1 When the frequency of a periodic forcing phenomenon (exciting frequency) applied to a rotor-bearing support system coincides with a natural frequency of that system, the system may be in a state of resonance. A shaft rotational speed at which the rotor-bearing-support system is in a state of resonance with any exciting frequency associated with that speed, is called a *critical speed*.

2.8.1.2 A forcing phenomenon or exciting frequency may be less than, equal to, or greater than the rotational speed of the rotor. Potential exciting frequencies may include but are not limited to the following:

a. Unbalance in the rotor system.
b. Oil film instabilities (whirl).
c. Gear-tooth meshing and side-bands.
d. Coupling misalignment.
e. Loose rotor-system components.
f. Ball and race frequencies of antifriction bearings.
g. Start-up condition frequencies (for example, speed detents under inertial impedances or torsional deflections contributing to torsional resonances).

• **2.8.1.3** Critical speeds are typically not a problem for gear elements operating at 3,700 revolutions per minute (1,800 revolutions per minute for overhung bevel gears) or below. In addition, a lateral critical speed analysis is generally not required if the first bouncing mode is greater than 1.25 times the maximum continuous speed. The first bouncing mode is generally greater than:

$$\frac{100{,}000}{f} = \sqrt{Wr} \quad \text{rpm}$$

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Where:

W_r = weight of rotor, Kg.

In any case, the gear vendor shall provide a lateral critical speed analysis for any element operating at or above 7,200 revolutions per minute (3,000 revolutions per minute for overhung bevel gears) or when specified by the purchaser.

2.8.1.4 Gear and bearing housing resonances shall not occur within the specified operating speed range or specified separation margins.

2.8.1.5 Three lateral critical speed modes are generally of concern with gear units; the cylindrical (translational or bouncing) mode, the conical (pivoted or rocking) mode, and the first bending mode. The frequency at which these modes occur will vary as a function of the transmitted load, primarily due to the resulting change of stiffness of the bearing-oil film (see Appendix B). The gear rotors shall meet the requirements given in 2.8.1.5.1 through 2.8.1.5.2.

2.8.1.5.1 When operating at the maximum torque, the three defined critical speeds of each rotor shall not be less than 20 percent above the maximum continuous speed of that rotor.

• **2.8.1.5.2** When the operating torque is in the range of 50 to 100 percent of the maximum torque the separation margin above the maximum continuous speed of each rotor shall be 10 to 20 percent in proportion to the transmitted torque. Operating conditions at less than 50 percent of the maximum torque or less than 70 percent of maximum continuous speed will be specified by the purchaser.

2.8.1.6 Slow-roll, start-up, and shutdown of rotating equipment shall not cause any damage as critical speeds are passed.

2.8.1.7 If the lateral critical speed as calculated or revealed during mechanical testing falls within the specified range of operating speeds or fails to meet the separation margin requirements after practical design efforts have been exhausted, the unit vendor shall demonstrate an insensitive rotor design. This insensitivity must be proven by operation on the test stand at the critical speed in question with the rotor unbalanced. The unfiltered vibration shall not exceed the limits specified by Equation 2 in 2.8.3.2. Trip speed values may apply. During the sensitivity tests, the increment in vibration amplitude shall be based on the difference between the major axes of the orbits formed by synchronous x-y signals recorded during the balanced and unbalanced runs. Deflections of the rotor shall not exceed design rotor clearances or the allowable vibration limit specified in 2.8.3.2. The amount of rotor unbalance shall be calculated as follows:

In SI units,

$$U_A = 2.5 \text{ (residual unbalance limit)}$$
$$U_A = 2.5\ (6{,}350W/N)$$

In Customary units,

$$U_A = 2.5\ (4W/N)$$

Where:

U_A = residual unbalance, in gram-millimeters (ounce-inches).
W = journal static weight load, in kilograms (pounds).
N = maximum continuous speed, in revolutions per minute.

Modal analysis shall be used in the placement of the unbalance weights, as mutually agreed upon by the purchaser and the vendor.

• **2.8.1.8** When specified, the gear vendor shall supply to the driven-equipment vendor all necessary information for lateral and torsional vibration analyses. The driven-equipment vendor shall ensure that torsional modes of the complete unit shall be at least 10 percent below any operating speed or at least 10 percent above the trip speed or motor speed.

2.8.2 BALANCE

2.8.2.1 All gear elements shall be multiplane dynamically balanced after final assembly of the rotor. Rotors with single keys for couplings shall be balanced with half-keys in place. The maximum allowable residual unbalance per plane (journal) shall be calculated as follows:

In SI units,

$$U_{max} = 6{,}350W/N$$

In Customary units,

$$U_{max} = 4W/N$$

Where:

U_{max} = residual unbalance, in gram-millimeters (ounce-inches).
W = journal static weight load, in kilograms (pounds).
N = maximum continuous speed, in revolutions per minute.

2.8.2.2 Balancing shall be done on a recently calibrated machine and accuracy shall be within prescribed limits of the balancing machine manufacturer. The balancing machine shall have suitable sensitivity for use with the gear elements.

• **2.8.2.3** When specified, after the final balancing of each assembled rotating element has been completed, a residual unbalance check shall be performed in accordance with the residual unbalance work sheet (Appendix H).

2.8.3 VIBRATION

2.8.3.1 During the unloaded shop test of the assembled gear operating at its maximum continuous speed or at any other speed within the specified range of operating speeds,

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the vibration measured on the bearing caps shall not exceed the following values:

	Velocity	Acceleration
Frequency range	10Hz–2.5kHz	2.5kHz–10kHz
Unfiltered	5mm/sec (.2ips)	8 G's
Filtered	4mm/sec (.15ips)	

Note: Filtered readings shall be 1× operating speed for each rotor.

2.8.3.2 Where noncontacting vibration probes are specified, they shall be used during the shop test. In such cases, during the shop test of the assembled gear operating at maximum continuous speed or at any other speed within the specified operating speed range, the peak-to-peak amplitude of unfiltered vibration in any plane measured on the shaft adjacent to each radial bearing shall not exceed the following value or 50 micrometers (0.002 inches), whichever is less:

In SI units,

$$A = 25.4 \sqrt{\frac{12{,}000}{N}} \qquad (2)$$

In Customary units,

$$A = \sqrt{\frac{12{,}000}{N}} \qquad (2)$$

Where:

A = amplitude of unfiltered vibration, in micro-meters (mils) true peak to peak.

N = maximum continuous speed, in revolutions per minute.

For variable speed drivers, the vibration shall not exceed the above values plus 12 micrometers (0.0005 inches) at the trip speed. Mechanical and electrical runout for each probe area of each shaft shall (a) be documented in accordance with, and (b) be in compliance with the limits specified in API 670.

2.8.3.3 If the vendor can demonstrate that electrical or mechanical runout is present, a maximum of 25 percent of the test level calculated from Equation 2 or 6.5 micrometers (0.00025 inches), whichever is greater, may be vectorially subtracted from the vibration signal measured during the factory test.

2.9 Bearings

2.9.1 GENERAL

Radial and thrust bearings shall be of the hydrodynamic fluid film type or the oil-lubricated antifriction type. Antifriction-type bearings are subject to the limitations in Table 7.

Table 7—Maximum $d_m N$ Numbers for Antifriction Bearings

	Method of Lubrication	
Type of Bearing	Splash or Circulating Oil	Pressurized Oil
Ball or cylindrical roller	450,000	500,000
Tapered or spherical roller	300,000	350,000

With d_m = mean bearing diameter = $(d+D)/2$ (mm) and N = shaft speed (rpm).

2.9.2 RADIAL BEARINGS

2.9.2.1 Hydrodynamic Radial Bearings

2.9.2.1.1 Hydrodynamic radial bearings shall be split for ease of assembly, precision bored, and of the sleeve or pad type, with steel-backed, babbitted replaceable liners, pads, or shells. These bearings shall be equipped with antirotation pins and shall be positively secured in the axial direction.

2.9.2.1.2 The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping over the entire range of allowable bearing clearances to limit rotor vibration to the maximum specified amplitudes while the equipment is operating loaded or unloaded at specified operating speeds, including operation at any critical frequency.

2.9.2.1.3 The liners, pads, or shells shall be split and shall be replaceable without the removal of the coupling hub.

2.9.2.1.4 Bearings shall be designed to prevent installing backwards and/or upside down.

2.9.2.2 Antifriction Radial Bearings

2.9.2.2.1 Antifriction bearings shall be of a standard type and shall be selected to give 6 years (50,000 hours) minimum L_{10} rating life (see AFBMA Standards 9 and 11) with continuous operation at rated gear conditions but not less than 32,000 hours at maximum axial and radial loads and rated speed.

Note: The rating life is the number of hours at constant speed that 90 percent of a group of identical bearings will complete or exceed before the first evidence of failure.

2.9.2.2.2 Antifriction bearings shall be retained on the shaft and fitted into housings in accordance with the requirements of AFBMA Standard 7; however, the device used to lock ball thrust bearings to the shaft shall be restricted to a tongue-type lock washer, for example, Series W.

2.9.2.2.3 Except for the angular contact type, antifriction bearings shall have a loose internal clearance fit equivalent to AFBMA Symbol 3 as defined in AFBMA Standard 20. Single or double-row bearings shall be of the Conrad type (no filling slots).

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2.9.3 THRUST BEARINGS

2.9.3.1 General

Unless otherwise approved by the purchaser, thrust bearings shall be provided for all gear units. Thrust bearings shall be sized for continuous operation under the most adverse specified operating conditions including all external forces transmitted by the couplings.

2.9.3.1.1 For gear-type couplings, the external thrust force shall be calculated from the following formula:

In SI units,

$$F = \frac{(0.25)(9{,}550)P_r}{(N_r D)}$$

In Customary units,

$$F = \frac{(0.25)(63{,}000)P_r}{(N_r D)}$$

Where:

F = external force, in kilonewtons (pounds).
P_r = rated power, in kilowatts (horsepower).
N_r = rated speed, in revolutions per minute.
D = shaft diameter at the coupling, in millimeters (inches).

2.9.3.1.2 Thrust forces for flexible-element couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

2.9.3.1.3 If two or more rotor thrust forces are to be carried by one thrust bearing (such as with single helical gearing), the resultant of the forces shall be used provided the directions of the forces make them numerically additive; otherwise, the largest of the forces shall be used.

2.9.3.1.4 When gears are supplied without thrust bearings, limited end-float or diaphragm-type couplings shall be used to maintain positive axial positioning of the connected rotors.

Note: See Figure F-1, Panels A through F, Appendix F, for typical system arrangements in which thrust bearings may be eliminated from double-helical gears.

All gears without thrust bearings shall be supplied with locating collars on the low-speed shaft to prevent contact of the rotating elements with the gear casing. Axial float shall not be less than 12 millimeters ($^1/_2$ inch).

2.9.3.2 Hydrodynamic Thrust Bearings

2.9.3.2.1 Hydrodynamic thrust bearings shall be of the steel-backed, babbitt-faced type, designed for equal thrust capacity in both directions and arranged for continuous pressurized lubrication to each side. The maximum design criteria for babbit-faced hydrodynamic thrust bearings shall be 517 kilopascals (75 pounds per square inch) for flat-faced thrust bearings and 1.034 kilopascals (150 pounds per square inch) for tapered land thrust bearings.

2.9.3.2.2 When integral thrust collars are furnished, they shall be provided with a minimum of 3 millimeters ($^1/_8$ inch) of additional stock to enable refinishing if the collar is damaged. When replaceable collars are furnished (for assembly and maintenance purposes), they shall be positively locked to the shaft to prevent fretting.

2.9.3.3 Antifriction Thrust Bearings

2.9.3.3.1 Antifriction thrust bearings shall be rated for 6 years (50,000 hours) minimum L_{10} rating life with continuous operation at rated gear condition but not less than 32,000 hours at maximum operating axial and radial loads and rated speed.

2.9.3.3.2 If ball-type thrust bearings are used, they shall be of the duplex, single-row, 0.7 radian (40-degree), angular-contact type (7,000 series), installed back to back. The need for preload shall be determined by the vendor to suit the application and meet the bearing life requirements of 2.9.3.3.1.

2.9.4 BEARING HOUSING AND SHAFT SEALS

2.9.4.1 In this standard, the term bearing housing refers to all bearing enclosures, including the gear casing. Bearing housing shall be designed to minimize foaming. Housings shall have adequate drainage to prevent oil and foam from leaking past the shaft end seals. Gaskets shall not be used on housing end covers where the gasket thickness would affect the end play or clearance of the thrust bearing.

2.9.4.2 Bearing housings shall preferably be equipped with replaceable labyrinth-type end seals and deflectors where the shaft passes through the housing. With purchaser's approval, lip-type seals may be used for shaft speeds at the seal area not exceeding 4 meters per second (800 feet per minute). The seals and deflectors shall be made of nonsparking materials. The design of the seals and deflectors shall effectively retain oil in the housing and prevent entry of foreign material into the housing.

• **2.9.4.3** When specified, bearing housings shall be designed to accommodate noncontacting vibration and axial position probes as required by 3.4.5.

• **2.9.4.4** When specified, bearing housings shall be designed to mount acceleration or velocity transducers.

2.10 Lubrication

2.10.1 This section covers the following types of lubrication systems for enclosed gear drives:

a. Self-contained splash.

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b. Circulating (for increased thermal capacity).
c. Pressurized (to accommodate higher speeds).
d. Combined (with the driver and/or other driven equipment).

2.10.2 Unless otherwise specified, bearings and bearing housings shall be arranged for hydrocarbon oil lubrication.

• **2.10.3** Where a wide variable speed range is anticipated (such as is encountered with continuous slow-roll operation), these speeds shall be specified, and lubrication of the gear shall be given special consideration.

2.10.4 The gear unit shall be designed to limit the drain-oil or sump temperature to 80°C (180°F). The oil inlet temperature for pressurized or circulating systems shall not exceed 50°C (120°F).

2.10.5 Splash lubricated units are limited to those with pinion speeds not exceeding 1,800 revolutions per minute and pitch line velocities not exceeding 10 meters per second (2,000 feet per minute).

2.10.6 Oil flinger disks or oil rings shall have an operating submergence of 3.0 to 6.5 millimeters ($^1/_8$ to $^1/_4$ inch) above the lower edge of a flinger or above the lower edge of the bore of an oil ring. Oil flingers shall have mounting hubs to maintain concentricity and shall be positively secured to the shaft.

2.10.7 Where a pressurized or circulating lubrication system is required by the driver and/or driven equipment, the gear oil may be supplied from that system.

• **2.10.8** Where oil is supplied from a common system to two or more machines, the oil's characteristics will be specified by the purchaser on the basis of mutual agreement with all vendors supplying equipment served by the common oil system.

• **2.10.9** When specified by the purchaser or required by the vendor, a pressurized oil system shall be provided.

2.10.10 Pressure lubrication systems shall consist of the following items (see Appendix C for a typical schematic):

a. A main oil pump driven by a gear shaft, unless another source of pressurized oil is provided. Oil requirements shall be specified on the data sheets.

• b. A separately driven, automatically controlled full-capacity standby pump (same capacity as shaft-driven main pump), when specified by the purchaser.

• c. Unless otherwise specified, cast iron pump casings are acceptable. The purchaser will specify if steel pump casings are required for pumps not located above sump oil level or enclosed in a reservoir.

• d. An oil cooler shall be provided to maintain the oil supply temperature at or below 50°C (120°F). The cooler shall be of a water-cooled, shell-and-tube type with water on the tube side or a suitable air cooled type, as specified. The vendor shall include in the proposal details of the cooler. Cooling coils located directly in the oil sump are not acceptable. Fan cooling is not acceptable on gearbox cooling for thermal purposes unless approved by the purchaser.

e. Unless otherwise specified, a single full-flow filter with replaceable elements and filtration of 25 microns nominal or finer shall be supplied. When specified, duplex filters shall be provided. The filters shall be located down-stream of the cooler. For positive displacement pumps, filter cases and heads shall be suitable for operation at a pressure not less than the relief valve setting. (Filters shall not be equipped with a relief valve or an automatic bypass). Filter cartridge materials shall be corrosion resistant. Flow shall be from the outside toward the center of the filter cartridge. Stackable filters shall not be used. The pressure drop for clean filter elements shall not exceed 15 percent of the total allowable dirty pressure drop, or 0.35 bar (5 pounds per square inch) at an operating temperature of 38°C (100°F) and normal flow. A filter pressure differential indicator shall be provided. Cartridges shall have a minimum collapsing differential pressure of 5 bar (70 pounds per square inch).

• f. When specified, a removable steam-heating element external to the reservoir or a thermostatically controlled electric immersion heater with a sheath of AISI Standard Type 300 stainless steel shall be provided for heating the charge capacity of the oil before start-up in cold weather. The heating device shall have sufficient capacity to heat the oil in the reservoir from the specified minimum site ambient temperature to the manufacturer's required start-up temperature within 12 hours. If an electric immersion heater is used, the watt density shall not exceed 20 watts per square centimeter (15 watts per square inch).

g. An oil reservoir that, unless otherwise specified, may be located in the gear casing and supplied with the following characteristics:

1. The capacity to adequately settle moisture and foreign matter, to avoid frequent refilling, and to provide adequate allowance for rundown.
2. Provisions to eliminate air and to minimize flotation of foreign matter to the pump suction.
3. Connections for fill and for complete drainage.

h. Thermometers (with thermowells) before and after the oil cooler, and a temperature indicator at each bearing.

i. A pressure gauge (valved for removal) for oil pressure at the oil inlet to the unit and upstream of the filters

• j. When specified, an alarm switch signaling low oil pressure.

k. Low-oil-pressure shutdown switch.

• l. When specified, an alarm switch signalling high oil temperature.

m. An oil pressure regulating valve.

2.10.11 Unless otherwise specified, circulation systems used to enhance the thermal rating of gearboxes equipped with antifriction bearings may utilize a 40 micron filter with a

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built-in thermal rating bypass and a high-temperature shutdown switch in lieu of a low-pressure shutdown switch.

2.11 Materials

2.11.1 GENERAL

2.11.1.1 Materials of construction shall be the manufacturer's standard for the specified operating conditions, except as required or prohibited by the data sheets or this standard (see 3.5.4 for requirements for auxiliary piping materials). The metallurgy of all major components shall be clearly stated in the vendor's proposal.

2.11.1.2 Materials shall be identified in the proposal and on parts lists with their applicable ASTM, AISI[9], ASME, or SAE numbers, including the material grade (see Appendix D). When no such designation is available, the vendor's material specification, giving physical properties, chemical composition, and test requirements, shall be included in the proposal.

• **2.11.1.3** The vendor shall specify the ASTM optional tests and inspection procedures that may be necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. The purchaser may consider specifying additional tests and inspections, especially for materials used in critical components.

2.11.1.4 Gear and pinion materials shall be selected to meet the tooth pitting index and strength criteria outlined in 2.4. In selecting the material, the vendor shall consider whether the gear and pinion are to be through hardened, case hardened, forged, or welded.

• **2.11.1.5** The purchaser will specify any corrosive agents present in the environment, including constituents that may cause stress corrosion cracking.

2.11.2 WELDING

2.11.2.1 All welds shall be made by operators qualified on the materials being welded. The qualifying procedure shall be mutually agreed upon by the purchaser and the vendor. The procedures given in Section IX of the ASME Code are suggested as a guide.

2.11.2.2 All welds shall be continuous full-penetration welds. All welds shall be double welded, except when only one side is accessible; in such instances, a backup ring, a consumable insert, or an inert gas shield weld with an internal gas purge backup shall be used.

2.11.2.3 The vendor shall be responsible for the review of all repairs and repair welds to ensure that they are properly heat treated and nondestructively examined for soundness and compliance with the applicable qualified procedures.

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2.11.2.4 Repair welding in the area of the gear teeth is prohibited.

2.11.3 HEAT TREATMENT

2.11.3.1 After the gear materials have been rough machined and heat-treated, the tooth area shall be checked for proper hardness.

2.11.3.2 For surface-hardened gearing a suitably sized coupon heat-treated with the part shall be utilized to verify proper surface hardening.

2.11.3.3 Casings, whether of cast or fabricated construction, shall be stress-relieved before final machining and after any welding, including repairs.

2.11.4 CASTINGS

2.11.4.1 The vendor shall specify on the data sheets the material grade of the castings.

2.11.4.2 Castings shall be sound and free from porosity, hot tears, shrink holes, blow holes, cracks, scale, blisters, and other similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, shot blasting, chemical cleaning, or any other standard method. Mold-parting fins and remains of gates and risers shall be chipped, filed, or ground flush.

2.11.4.3 Fully enclosed cored voids, including voids closed by plugging, are prohibited.

• 2.11.5 LOW TEMPERATURE

For operating temperatures below –30°C (–20°F) or when specified for other low ambient temperatures, steels shall have, at the lowest specified temperature, an impact strength sufficient to qualify under the minimum Charpy V-notch impact energy requirements of Section VIII, Division 1, UG-84, of the ASME Code. For materials and thicknesses not covered by the code, the purchaser will specify the requirements on the data sheets.

2.12 Nameplates and Rotation Arrows

2.12.1 A nameplate shall be securely attached at a readily visible location on the equipment and on any other major piece of auxiliary equipment.

2.12.2 Rotation arrows shall be cast in or attached at a readily visible location. Nameplates and rotation arrows (if attached) shall be of AISI Standard Type 300 stainless steel or of nickel-copper alloy (Monel or its equivalent). Attachment pins shall be of the same material. Welding is not permitted.

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2.12.3 The following data shall be clearly stamped on the nameplate:

a. The vendor's name.
b. The size and type of the gear unit.
c. The gear ratio.
d. The serial number.
e. The gear rated power.
f. The rated input speed, in revolutions per minute.
g. The rated output speed, in revolutions per minute.
h. The gear service factor, as defined in this standard.
i. The purchaser's item number.

SECTION 3—ACCESSORIES

• 3.1 General

The purchaser will specify on the data sheets the accessories to be supplied by the vendor.

3.2 Couplings and Guards

3.2.1 Unless otherwise specified, flexible couplings and guards between drivers and driven equipment shall be supplied by the manufacturer of the driven equipment.

3.2.2 The make, type, and mounting arrangement of the coupling shall be agreed upon by the purchaser and the vendors of the driver and driven equipment unless otherwise specified. Unless otherwise specified, the torque rating of the coupling-to shaft juncture shall be at least equal to that required by API 671. A spacer coupling with a minimum 125-millimeter (5-inch) spacer shall be used, unless otherwise specified. Couplings shall be forged steel and designed to allow the necessary end float caused by expansion and other end movements of the shaft.

3.2.3 Information on shafts, keyway dimensions (if any), and shaft end movements due to end play and thermal effects shall be furnished to the vendor supplying the coupling.

3.2.4 Couplings shall be keyed in place. Keys, keyways, and fits shall conform to commercial class AGMA 9002. Cylindrical shafts shall have the interference fit specified in AGMA 9002. Other mounting methods shall be agreed upon by the purchaser and the vendor. Coupling hubs shall be furnished with tapped puller holes [at least 10 millimeters ($^3/_8$ inch) in size] to aid removal.

3.2.5 To assure accurate alignment of connected machinery, the total indicator reading of coupling registration and alignment surfaces shall be controlled as specified in 3.2.5.1 and 3.2.5.2.

3.2.5.1 The coupling surfaces normally used for checking alignment shall be concentric with the axis of coupling hub rotation within the following limits: 13 micrometers (0.0005 inch) total indicator reading per 25 millimeters (1 inch) of shaft diameter, with a minimum applicable tolerance of 25 micrometers (0.001 inch) total indicator reading and a maximum of 75 micrometers (0.003 inch) total indicator reading. All other diameters not used for location, registration, or alignment shall be to the coupling manufacturer's standard, provided that balance requirements are met.

3.2.5.2 The shaft end shall be machined such that when the coupling is mounted the runout of the coupling shall not exceed the limits of 3.2.5.1.

3.2.6 An easily removable coupling guard shall be placed over all exposed couplings furnished by the vendor. The coupling guard shall be of sufficiently rigid design to withstand deflection and consequent rubbing as a result of bodily contact and shall extend to within 12.7 millimeters ($^1/_2$ inch) of the stationary housing.

3.3 Mounting Plates

3.3.1 GENERAL

3.3.1.1 Unless otherwise specified the gear unit shall be furnished for mounting on a baseplate.

3.3.1.2 In 3.3.1.2.1 through 3.3.1.2.10, the term *mounting plate* refers to both baseplates and soleplates.

3.3.1.2.1 The surfaces on which the gear mounts (mounting pads) and the multiple mounted pads on baseplate shall be machined coplaner within 0.05 millimeter (0.002 inch).

Note: Field installation should duplicate these machined surface tolerances in order to assist in obtaining proper gear tooth contact.

3.3.1.2.2 When the equipment support weighs more than 250 kilograms (500 pounds), the mounting plates shall be furnished with axial and lateral jackscrews the same size as or larger than the vertical jackscrews. Vertical jackscrews in the equipment feet shall be arranged to prevent marring of shimming surfaces. The lugs holding these jackscrews shall be attached to the mounting plates so that the lugs do not interfere with the installation or removal of the equipment, jackscrews or shims. If the equipment is too heavy to use jackscrews, other means shall be provided.

3.3.1.2.3 Machinery supports shall be designed to limit a change of alignment caused by the worst combination of

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pressure, torque, and allowable piping stress to 50 micrometers (0.002 inch) at the coupling flange.

3.3.1.2.4 Mounting plates shall not be drilled for equipment to be mounted by others. Mounting plates shall be supplied with leveling screws. Mounting plates that are to be grouted shall have 50-millimeter-radiused (2-inch-radiused) outside corners (in the plan view). Mounting surfaces that are not to be grouted shall be coated with a rust preventative immediately after machining.

3.3.1.2.5 Unless otherwise specified, epoxy grout shall be used. The vendor shall commercially blast, in accordance with SSPC SP6, all grouting surfaces of the mounting plates and shall coat these surfaces with inorganic zinc silicate.

Note: The grout manufacturer should be consulted to ensure proper field preparation of the mounting plate for satisfactory bonding of the grout.

3.3.1.2.6 Anchor bolts shall not be used to fasten machinery to the mounting plates.

3.3.1.2.7 Anchor bolts will be furnished by the purchaser.

3.3.1.2.8 Fasteners for attaching the components to the mounting plates shall be supplied by the vendor.

3.3.1.2.9 Mounting plate shall extend at least 25 millimeters (1 inch) beyond the outer three sides of equipment feet.

3.3.1.2.10 The vendor of the mounting plates shall furnish stainless steel (AISI STD Type 300) shim packs 3 to 15 millimeters ($^1/_8$ to $^1/_2$ inch) thick between the equipment feet and the mounting plates. All shim packs shall straddle the hold-down bolts and vertical jackscrews and be at least $^1/_4$ inch larger on all sides than the footprints of the equipment.

• **3.3.1.3** When specified, the baseplate shall be suitable for column mounting (that is, of sufficient rigidity to be supported at specified points) without continuous grouting under structural members. The baseplate design shall be mutually agreed upon by the purchaser and the vendor.

3.3.1.4 The baseplate shall be provided with lifting lugs for a four-point lift. Lifting the baseplate complete with all equipment mounted shall not permanently distort or otherwise damage the baseplate or the machinery mounted on it.

3.3.1.5 The bottom of the baseplate between structural members shall be open. When the baseplate is installed on a concrete foundation, it shall be provided with at least one grout hole having a clear area of at least 0.01 square meter (19 square inches) and no dimension less than 75 millimeters (3 inches) in each bulkhead section. These holes shall be located to permit grouting under all load-carrying structural members. Where practical, the holes shall be accessible for grouting with the equipment installed. The holes shall have 15-millimeter ($^1/_2$-inch) raised-lip edges, and if located in an area where liquids could impinge on the exposed grout, metallic covers with a minimum thickness of 16 gauge shall be provided. Vent holes at least 15-millimeters ($^1/_2$-inch) in size shall be provided at the highest point in each bulkhead section of the baseplate.

3.3.1.6 Unless otherwise specified, nonskid metal decking covering all walk and work areas shall be provided on the top of the baseplate.

3.4 Controls and Instrumentation

3.4.1 GENERAL

3.4.1.1 Unless otherwise specified, controls and instrumentation shall be suitable for outdoor installation.

3.4.1.2 Unless otherwise specified, controls and instrumentation shall conform to the requirements specified in API 614 and API 670.

• **3.4.1.3** Thermometers, thermocouples, or resistance temperature detectors, as specified by the purchaser, shall be provided.

3.4.1.4 All conduit shall be designed and installed so that it can be easily removed without damage and located so that it does not hamper removal of bearings, seals, or equipment internals.

3.4.2 TEMPERATURE GAUGES

3.4.2.1 Dial-type temperature gauges shall be heavy duty and corrosion resistant. They shall be at least 125 millimeters (5 inches) in diameter and bimetallic-type or liquid-filled. Black printing on a white background is standard for gauges.

3.4.2.2 The sensing elements of temperature gauges shall be in the flowing fluid.

Note: This is particularly important for lines that may run partially full.

3.4.2.3 Temperature gauges that are located in pressurized or flooded lines shall be furnished with $^3/_4$-inch NPT AISI Standard Type 300 stainless steel separable solid-bar thermowells.

3.4.3 THERMOCOUPLES AND RESISTANCE TEMPERATURE DETECTORS

Where practical, the design and location of thermocouples and resistance temperature detectors shall permit replacement while the unit is operating. The lead wires of thermocouples and resistance temperature detectors shall be installed as continuous leads between the thermowell or detector and the terminal box. Conduit runs from thermocouple heads to a pull box or boxes located on the baseplate shall be provided.

• 3.4.4 PRESSURE GAUGES

Pressure gauges (not including built-in instrument air gauges) shall be furnished with AISI Standard Type 316 stainless steel bourdon tubes and stainless steel movements, 110-millimeter ($4^1/_2$-inch) dials [150-millimeter (6-inch)

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dials for the range over 55 bar (800 pounds per square inch)], and $^1/_2$-inch NPT male alloy steel connections. Black printing on a white background is standard for gauges. When specified, liquid-filled gauges shall be furnished in locations subject to vibration. Gauge ranges shall preferably be selected so that the normal operating pressure is at the middle of the gauge's range. In no case, however, shall the maximum reading on the dial be less than the applicable relief valve setting plus 10 percent. Each pressure gauge shall be provided with a device such as a disk insert or blow-out back designed to relieve excess case pressure.

• 3.4.5 VIBRATION AND POSITION DETECTORS

When specified, vibration and axial-position transducers shall be supplied, installed and calibrated in accordance with API 670.

3.4.6 RELIEF VALVES

3.4.6.1 The vendor shall furnish the relief valves that are to be installed on equipment or in piping that the vendor is supplying. Other relief valves will be furnished by the purchaser. Relief valves for all operating equipment shall meet the limiting relief valve requirements defined in API 520, Parts I and II, and in API 526. The vendor shall determine the size and set pressure of all relief valves related to the equipment. The vendor's quotation shall list all relief valves and shall clearly indicate those to be furnished by the vendor. Relief valve settings, including accumulation, shall take into consideration all possible types of equipment failure and the protection of piping systems.

• **3.4.6.2** When specified, thermal relief valves shall be provided for components that may be blocked in by isolation valves.

3.4.6.3 Unless otherwise specified, relief valves shall have steel bodies.

3.4.7 FLOW INDICATORS

3.4.7.1 Unless otherwise specified, the flow indicator shall be flanged, shall be of the bull's eye type, and shall have a steel body. The diameter of the bull's eye shall be at least one-half the inside diameter of the oil pipe and shall clearly show the minimum oil flow.

3.4.7.2 To facilitate viewing of the flow of oil through return lines, each flow indicator should be installed with its bull's eye glass in a vertical plane. Flow indicators in supply lines shall be the propeller type.

3.4.8 ELECTRICAL SYSTEMS

• **3.4.8.1** The characteristics of electrical power supplies for heaters and instrumentation will be specified.

3.4.8.2 Electrical equipment located on the unit shall be suitable for the hazard classification specified.

3.4.8.3 Wiring shall be resistant to oil, heat, moisture, and abrasion. Stranded conductors shall be used in areas subject to vibration. Measurement wiring may be solid conductor. Thermoplastic insulation shall be used and shall be covered by a neoprene or equal sheath for abrasion protection. Wiring shall be suitable for the environmental temperatures specified.

3.4.8.4 Unless otherwise specified, all leads shall be permanently tagged for identification.

3.4.8.5 Wiring (including temperature element leads) shall be installed in rigid metallic conduits and boxes, properly bracketed to minimize vibration and isolated or shielded to prevent interference between voltage levels. Conduits may terminate (and in the case of temperature element heads, shall terminate) with a flexible metallic conduit long enough to permit access to the unit for maintenance without removal of the conduit.

3.5 PIPING AND APPURTENANCES

3.5.1 Piping design and joint fabrication, examination, and inspection shall be in accordance with ASME B31.3. Welding of piping shall be performed by operators and procedures qualified in accordance with Section IX of the ASME Code.

3.5.2 Oil drains shall be sized to run no more than half-full when flowing at a velocity of 0.3 meter per second (1 foot per second) and shall be arranged to ensure good drainage (recognizing the possibility of foaming conditions). Horizontal runs shall slope continuously, at least 40 millimeters per meter ($^1/_2$ inch per foot) toward the reservoir. If possible, laterals (not more than one in any transverse plane) should enter drain headers at 45 degree angles in the direction of the flow.

3.5.3 Nonconsumable backup rings and sleeve type joints shall not be used. Pressure piping downstream of oil filters shall be free from internal obstructions or pockets that could accumulate dirt. Pipe joints downstream of the oil filter (filter to bearing housing) shall be butt-welded. Piping joints in return lines and upstream of the filter (reservoir to filter) may be socket-welded. Threaded connections shall be used for instrument connections and where tubing is used.

3.5.4 Unless otherwise specified, oil-supply piping and tubing, including fittings (excluding slip-on flanges), shall be stainless steel.

3.5.5 The vendor shall include in his quotation all integral piping considered necessary for the successful operation of the gear unit or units, as well as all integral piping in accordance with Appendix C and items indicated on the data sheets.

3.5.6 Piping shall preferably be fabricated by bending and welding to minimize the use of flanges and fittings. Welded

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flanges are permitted only at equipment connections, at the edge of any base, and for ease of maintenance. The use of flanges at other points is permitted only with the purchaser's specific approval. Other than tees and reducers, welded fittings are permitted only to facilitate pipe layout in congested areas. Threaded connections shall be held to a minimum. Pipe bushings shall not be used.

3.5.7 Pipe threads shall be taper threads in accordance with ASME B1.20.1. Flanges shall be in accordance with ASME B16.5. Slip-on flanges are permitted only with the purchaser's specific approval. For socket-welded construction, a 1.5-millimeter ($^1/_{16}$-inch) gap shall be left between the pipe end and the bottom of the socket.

3.5.8 Design of piping systems shall achieve the following:

a. Proper support and protection to prevent damage from vibration or from shipment, operation, and maintenance.
b. Proper flexibility and normal accessibility for operation, maintenance, and thorough cleaning.
c. Installation in a neat and orderly arrangement adapted to the contour of the machine without obstruction of access openings.
d. Elimination of air pockets.
e. Complete drainage through low points without disassembly of piping.

3.5.9 When external drain lines from the gear casing and drain lines from continuously lubricated couplings are provided by the vendor, they shall be provided with sight flow indicators per 3.4.7.

3.5.10 Carbon steel piping in lube-oil systems shall be seamless in accordance with ASTM A 106, ASTM A 192, or a purchaser-approved equivalent. Stainless steel piping shall be seamless in accordance with ASTM A 312. The schedules in API 614 apply. Valves shall be steel with bolted bonnets and glands (excluding gauges and instruments).

3.5.11 Where space does not permit the use of $^1/_2$-, $^3/_4$-, or 1-inch pipe, seamless steel tubing conforming to ASTM A 192 or stainless steel tubing conforming to ASTM A 269 may be furnished with steel fittings. Minimum tube wall thicknesses are listed in API 614. The make and model of fittings shall be subject to approval by the purchaser.

3.6 Special Tools

3.6.1 When special tools and fixtures are required to disassemble, assemble, or maintain the unit, they shall be included in the quotation and furnished as part of the initial supply of the machine. For multiple-unit installations, the requirements for quantities of special tools and fixtures shall be mutually agreed upon by the purchase and the vendor. These or similar special tools shall be used during shop assembly and post-test disassembly of the equipment.

3.6.2 When special tools are provided, they shall be packaged in a separate, rugged metal box or boxes and shall be marked *special tools for* (tag/item number). Each tool shall be stamped or tagged to indicate its intended use.

SECTION 4—INSPECTION, TESTING, AND PREPARATION FOR SHIPMENT

4.1 General

4.1.1 The vendor shall provide sufficient advance notice to the purchaser before conducting any inspection and test outlined in the purchase order or other agreements. After advance notification of the vendor by the purchaser, the purchaser's representative shall have entry to all vendor plants and sub vendor plants where manufacturing, testing, or inspection of the equipment is in progress. It shall be the responsibility of the vendor to notify subvendors of the purchaser's inspection requirements.

4.1.2 In each instance, the actual number of calendar days for notification prior to inspection shall be established by mutual consent of the purchaser and the vendor.

• **4.1.3** The purchaser will specify the extent of his participation in the inspection and testing.

a. *Witnessed* means that a hold shall be applied to the production schedule and that the inspection or test shall be carried out with the purchaser or his representative in attendance. For mechanical running or performance tests, this requires written notification of a successful preliminary test.
b. *Observed* means that the purchaser will be notified of the timing of the inspection or test; however, the inspection or test is performed as scheduled, and if the purchaser or his representative is not present, the vendor shall proceed to the next step. (The purchaser should expect to be in the factory longer than for a witnessed test.)

4.1.4 Equipment for the specified inspection and tests shall be provided by the vendor.

4.1.5 During assembly of the gear and oil system and before testing, each component (including cast-in passages of these components) and all piping and appurtenances shall be cleaned by pickling or by another appropriate method to remove foreign materials, corrosion products, and mill scale.

• **4.1.6** When specified, the purchaser may inspect for cleanliness of the equipment and all piping and appurte-

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nances furnished by or through the vendor before piping is finally assembled.

• **4.1.7** When specified, the hardness of parts, welds, and heat-affected zones shall be verified as being within the allowable values by testing of the parts, welds, or heat-affected zones. The method, extent, documentation, and witnessing of the testing shall be mutually agreed upon by the purchaser and the vendor.

4.1.8 The purchaser's representative shall have access to the vendor's quality control program for review.

4.2 Inspection

4.2.1 GENERAL

4.2.1.1 The vendor shall keep the following data available for at least 20 years:

a. Necessary or specified certification of materials, such as mill test reports.
b. Test data and results to verify that the requirements of the specification have been met.
c. Fully identified records of all heat treatment whether performed in the normal course of manufacture or as part of a repair procedure.
d. Results of all quality control tests and inspections.
• e. When specified, final assembly, maintenance, and running clearances.
f. Details of all repairs.
g. Gear and bearing rating calculations.
h. Data required to comply with applicable codes and standards.

4.2.1.2 Pressure-containing parts shall not be painted until specified inspection of the parts is complete.

4.2.1.3 In addition to the requirements of 3.5.1, the purchaser may specify the following:

a. Parts that shall be subjected to surface and subsurface examination.
b. The type of examination required, such as magnetic particle, liquid penetrant, radiographic, and ultrasonic.

• 4.2.2 MATERIAL INSPECTION

4.2.2.1 General

When radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds or materials is required or specified, the criteria in 4.2.2.2 through 4.2.2.5 shall apply unless other corresponding procedures and acceptance criteria have been specified by the purchaser. Cast iron may be inspected only in accordance with 4.2.2.4 and/or 4.2.2.5. Welds, cast steel, and wrought material shall be inspected in accordance with 4.2.2.2 through 4.2.2.5.

Note: Radiographic and ultrasonic inspection are not appropriate for cast iron due to differences in accurate interpretation.

4.2.2.2 Radiography

4.2.2.2.1 Radiography shall be in accordance with ASTM E 94 and ASTM E 142.

4.2.2.2.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, UW-51 (100-percent) and UW-52 (Spot) of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME Code.

4.2.2.3 Ultrasonic Inspection

4.2.2.3.1 Ultrasonic inspection shall be in accordance with Section V, Articles 5 and 23 of the ASME code.

4.2.2.3.2 The acceptance standard for welded fabrications shall be Section VIII, Division 1, Appendix 12, of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME code.

4.2.2.4 Magnetic Particle

4.2.2.4.1 Both wet and dry methods for magnetic particle inspection shall be in accordance with ASTM E 709.

4.2.2.4.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 6 and Section V, Article 25 of the ASME Code. The acceptability of defects in castings shall be based on comparison with the photographs in ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits specified below:

Maximum Severity of Defects in Castings

Type	Defect	Maximum Severity Level
I	Linear Discontinuities	1
II	Shrinkage	2
III	Inclusions	2
IV	Chills and Chaplets	1
V	Porosity	1
VI	Welds	1

4.2.2.5 Liquid Penetrant

4.2.2.5.1 Liquid penetrant inspection shall be in accordance with Section V, Article 6 of the ASME Code.

4.2.2.5.2 The acceptance standard used for welded fabrications shall be Section VIII, Division 1, Appendix 8 and Section V, Article 24 of the ASME Code. The acceptance standard used for castings shall be Section VIII, Division 1, Appendix 7 of the ASME Code.

4.2.2.6 Regardless of the generalized limits in 4.2.2, the vendor shall be responsible for the review of the design limits

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of the equipment in the event that more stringent requirements are necessary.

4.2.2.7 Defects that exceed the limits in 4.2.2 shall be removed to meet the quality standards cited, as determined by the inspection method specified.

4.2.3 WELDS IN ROTATING ELEMENTS

• **4.2.3.1** When specified, all welds in rotating elements, including those attaching gears to shafts, shall receive 100-percent inspection. Accessible surfaces of welds shall be inspected after back chipping or gouging and again after stress relieving. Magnetic particle or ultrasonic inspection is preferred. Other methods, such as dye penetrant and radiography, are acceptable only as mutually agreed upon by the purchaser and the vendor.

4.2.4 MATERIAL INSPECTION FOR ROTATING ELEMENTS

• **4.2.4.1** When specified, all gear and pinion teeth shall be 100-percent-inspected by the magnetic particle method in accordance with ASTM A 275. Cracks are not acceptable. Linear indications that result from nonmetallic inclusions in the tooth flanks or roots that are larger than 1.5 millimeters (0.06 inch) shall be reported to the purchaser for disposition. Linear indicators are defined as those having a length equal to or greater than three times the width. Acceptance or rejection shall be decided on a case by case basis and shall be mutually agreed upon by the purchaser and the vendor.

4.2.4.2 Forgings and hot-rolled steel shafts shall meet the criteria specified in 4.2.4.2.1 and 4.2.4.2.2.

4.2.4.2.1 Forgings and hot-rolled steel shafts shall be free from cracks, seams, laps, shrinkage, and other similar injurious defects.

• **4.2.4.2.2** When specified, all forgings and bar stock for major rotating elements shall be 100-percent-inspected by the ultrasonic method after rough machining in accordance with ASTM A 388. Acceptable criteria shall be mutually agreed upon by the purchaser and the vendor.

4.3 Testing

4.3.1 GENERAL

4.3.1.1 Gears shall be tested in accordance with 4.3.2 and 4.3.3. Other tests that may be specified by the purchaser are described in 4.3.4.

• **4.3.1.2** The purchaser reserves the right to observe the testing, dismantling, inspection, and reassembly of equipment as specified. The purchaser will specify tests that will be witnessed.

4.3.1.3 The vendor shall notify the purchaser not less than 5 working days before the date the equipment will be ready for testing. If the testing is rescheduled, the vendor shall notify the purchaser not less than 5 working days before the new test date.

4.3.1.4 The vendor's test stand lubricating oil systems shall provide for filtration, in accordance with 2.10.10, Item e. Before any test is initiated, the facilities downstream of the filter shall be cleaned and flushed to prevent scoring of the journals, bearings, and gear teeth.

4.3.1.5 At least 6 weeks before the first scheduled running test, the vendor shall submit to the purchaser, for his review and comment, detailed procedures for the mechanical running test and all specified running optional tests (4.3.4), including acceptance criteria for all monitored parameters.

4.3.2 PRESSURE TEST

4.3.2.1 Pressurized oil systems and associated piping shall be tested either hydrostatically or dynamically with liquid at a minimum of 1.5 times the maximum allowable working pressure but not less than 1.4 bar gauge (20 pounds per square inch gauge). The test liquid should be at a minimum temperature of 15.6°C (60°F) when testing carbon steels.

• **4.3.2.2** Tests shall be maintained for a sufficient period of time to permit complete examination of parts under pressure. The test shall be considered satisfactory when no leaks are observed for a minimum of 30 minutes.

4.3.3 MECHANICAL RUNNING TEST

4.3.3.1 The mechanical running test of the gear shall be conducted by operating at maximum continuous speed for not less than 1 hour after bearing temperature and lube-oil temperatures have stabilized.

• **4.3.3.2** When specified, an extended mechanical running test of the gear shall be conducted in the following sequence**:**

a. The gear shall be operated at maximum continuous speed for 4 hours after bearings and lube-oil temperatures have stabilized.
b. The speed shall be increased to 110 percent of maximum continuous speed and run for a minimum of 15 minutes.
c. Testing at any additional speeds, the duration of testing at each speed, and the data to be recorded will be specified by the purchaser at the time of the purchase.

4.3.3.3 The requirements of 4.3.3.3.1 through 4.3.3.3.6 shall be met before the mechanical running test is performed.

4.3.3.3.1 The contract bearings shall be used in the machine for the mechanical running test.

4.3.3.3.2 All oil pressures, viscosities, and temperatures shall be within the range of operating values recommended in

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the vendor's operating instructions for the specific unit being tested. For pressure lubrication systems, oil flow rates for each bearing housing shall be measured.

4.3.3.3.3 The joints and connections in the casing and the oil system shall be checked for tightness and any leaks shall be corrected.

4.3.3.3.4 All warning, protective, and control devices used during the test shall be checked and adjustments shall be made as required.

• **4.3.3.3.5** Testing with the contract coupling is preferred. If this is not practical, the mechanical running test shall be performed with coupling-hub and idling adapters in place, resulting in moments equal (±10 percent) of the moment of the contract coupling hub plus one-half that of the coupling spacer. When all testing is completed, the idling adapters shall be furnished to the purchaser as part of the special tools.

4.3.3.3.6 The mechanical running tests shall be made with a job lube system, if such a system has been purchased with the gear unit or units.

4.3.3.4 During the running tests, the mechanical operation of all equipment being tested and the operation of the test instrumentation shall be satisfactory. The measured unfiltered vibration shall not exceed the limits of 2.7.2.4 and 2.8.3 and shall be recorded throughout the testing speed range.

4.3.3.5 Vibration measurements shall be made with a recently calibrated measuring device operating within its frequency range, which shall include the frequency range covered in 2.8.3.1.

4.3.3.6 All purchased vibration probes and oscillation-demodulators shall be in use during the tests. If vibration probes are not furnished by the vendor or the purchased probes are not compatible with shop readout facilities, then shop probes and readouts that meet the accuracy requirements of API 670 shall be used.

4.3.3.7 After the mechanical running tests are completed, the tooth mesh shall be inspected for surface damage and proper contact pattern. (See 2.7.2.1 for contact-pattern acceptance criteria.)

• **4.3.3.8** When specified, all hydrodynamic bearings shall be removed, inspected, and reassembled after the mechanical running tests are completed.

4.3.3.9 If replacement or modification of bearings or seals or dismantling of the case to replace or modify other parts is required to correct mechanical or performance deficiencies, the initial test will not be acceptable, and the final shop tests shall be run after these replacements or corrections are made.

4.3.3.10 When spare gear elements are ordered to permit concurrent manufacture, each spare element shall also be given mechanical running tests in accordance with the requirements of this standard.

4.3.3.11 The vendor shall keep a detailed log of the final tests, making entries at 15-minute intervals for the duration of the tests. Each entry shall include the following information:

a. Oil temperatures and inlet pressures.
b. Outlet oil (drain) temperature, when available.
c. Vibration amplitude, unfiltered and filtered 1 time for operating speed of each rotor.
d. Bearing temperatures (when measurements are available).

• 4.3.4 OPTIONAL TESTS

The purchaser will specify in the inquiry or in the order whether any of the shop tests specified in 4.3.4.1 through 4.3.4.4 shall be performed.

4.3.4.1 Full-Speed/Full- or Part-Load Test

The gear unit shall be tested to transmit partial or full-rated power, as agreed upon by the purchaser and the vendor, at its rated input speed. The load shall be applied by a mechanical or hydraulic method (such as dynamometers and pony brakes) until the bearing temperatures and lube-oil temperatures have stabilized. Details of the test, including vibration limits, shall be negotiated before the order.

4.3.4.2 Full-Torque/Slow-Roll Test

The unit's full torque shall be calculated at the gear-rated power and the rated input or output speed. The full torque shall then be applied to its respective shaft by mechanical or hydraulic means at a speed convenient for the vendor's test stand equipment.

For example, 2,000 revolutions per minute, 1.49 megawatts (2,000 horsepower) is equivalent to 7,130 newton-meters (63,025 inch-pounds) of torque. At 50 revolutions per minute, 7,130 newton-meters (63,025 inch-pounds) torque is equivalent to 37.3 kilowatts (50 horsepower). The duration of the test shall be negotiated before the time of order.

Note: The full-torque/slow-roll test is designed to demonstrate only tooth contact and load-carrying capability. Vibration and temperature limitations, as outlined elsewhere in this standard, should not be applied.

4.3.4.3 Full-Torque/Static Test

One shaft of the gear shall be locked. The full torque, as calculated in 4.3.4.2, shall then be applied to the other shaft by mechanical or hydraulic means. This procedure shall be repeated at several mesh points of gear set. The number of load applications shall be negotiated before the time of order.

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4.3.4.4 Sound-Level Test

Unless otherwise specified, the sound-power test shall be performed in accordance with ISO 3746. When specified, a more rigorous test shall be performed per ISO 3744.

Note: Refer to ISO 3740 for guidance on which of the two standards is more appropriate for the given application.

4.4 Preparation for Shipment

• **4.4.1** Equipment shall be suitably prepared for the type of shipment specified, including blocking of the rotor when necessary. Blocked rotors shall be identified by means of corrosion-resistant tags attached with stainless steel wire. The preparation shall make the equipment suitable for 6 months of outdoor storage from the time of shipment, with no disassembly required before operation, except for inspection of bearings and seals. If storage for a longer period is contemplated, the purchaser will consult with the vendor regarding the recommended procedures to be followed.

4.4.2 The vendor shall provide the purchaser with the instructions necessary to preserve the integrity of the storage preparation after the equipment arrives at the job site and before start-up.

4.4.3 The equipment shall be prepared for shipment after all testing and inspection has been completed and the equipment has been approved by the purchaser. The preparation shall include as a minimum that specified in 4.4.3.1 through 4.4.3.10.

4.4.3.1 Exterior surfaces, except for machined surfaces, shall be given at least one coat of the manufacturer's standard paint. The paint shall not contain lead or chromates.

4.4.3.2 Exterior machined surfaces shall be coated with a suitable rust preventive.

4.4.3.3 The interior of the gear unit shall be clean; free from scale, welding spatter, and foreign objects; and sprayed or flushed with a suitable rust preventive that can be removed with solvent. The rust preventive shall be applied through all openings while the machine is slow-rolled.

4.4.3.4 Internal steel areas of bearing housings and carbon steel oil systems' auxiliary equipment such as reservoirs, vessels, and piping shall be coated with a suitable oil-soluble rust preventive.

4.4.3.5 Exposed shafts and shaft couplings shall be wrapped with waterproof, moldable waxed cloth or volatile-corrosion inhibitor paper. The seams shall be sealed with oil-proof adhesive tape.

4.4.3.6 Flanged openings shall be provided with metal closures at least 4.8 millimeters ($^3/_{16}$ inch) thick, with rubber gaskets and at least four full-diameter bolts. For studded openings, all nuts needed for the intended service shall be used to secure closures.

4.4.3.7 Threaded openings shall be provided with steel caps or solid-shank steel plugs whose metallurgy is equal to or better than that of the pressure casing. In no case shall non-metallic (such as plastic) plugs or caps be used.

4.4.3.8 Lifting points and lifting lugs shall be clearly identified on the equipment package. The recommended lifting arrangement shall be identified on boxed equipment.

4.4.3.9 The equipment shall be identified with item and serial numbers. Material shipped separately shall be identified with securely affixed corrosion-resistant metal tags indicating the item and serial number of the equipment for which it is intended. In addition, crated equipment shall be shipped with duplicate packing lists, one inside and one on the outside of the shipping container.

• **4.4.3.10** If spare gear elements are purchased, the gear or gears shall be prepared for unheated indoor storage for a period of at least 3 years. The gear shall be treated with a rust preventive and shall be housed in a vapor-barrier envelope with a slow-release volatile-corrosion inhibitor. The gear shall be suitably crated for domestic or export shipment as specified. A purchaser-approved resilient material 3.0 millimeters ($^1/_8$ inch) [not tetrafluorothylene (TFE) or polytetrafluorothylene (PTFE)] thick shall be used between the cradle and the rotor at the support areas. The rotor shall not be supported at the journals.

4.4.4 Auxiliary piping connections furnished on the purchased equipment shall be impression stamped or permanently tagged to agree with the vendor's connection table or general arrangement drawing. Service and connection drawings shall be indicated.

4.4.5 Bearing assemblies and the exposed ends of shafts shall be fully protected against the entry of moisture and dirt. If vapor-phase-inhibitor crystals in bags are installed in large cavities to absorb moisture, the bags must be attached in an accessible area for ease of removal. Where applicable, bags shall be installed in wire cages attached to flanged covers and bag locations shall be indicated by corrosion-resistant tags attached with stainless steel wire.

4.4.6 One copy of the manufacturer's standard installation instructions shall be packed and shipped with the equipment.

4.4.7 Connections on piping removed for shipment shall be match-marked for ease of reassembly.

• **4.4.8** When specified, the fit-up and assembly of machine-mounted piping, intercoolers, and so forth shall be completed in the vendor's shop prior to shipment.

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SECTION 5—VENDOR'S DATA

5.1 General

5.1.1 The information to be furnished by the vendor is specified in 5.2 and 5.3. The vendor shall complete and forward the Vendor Drawing and Data Requirements form (see Appendix E) to the address or addresses noted on the inquiry or order. This form shall detail the schedule for transmission of drawings, curves, and data as agreed to at the time of the order, as well as the number and type of copies required by the purchaser.

5.1.2 The data shall be identified on transmittal (cover) letters and in title blocks or title pages with the following information:

a. The purchaser/user's corporate name.
b. The job/project number.
c. The equipment item number and service name.
d. The inquiry or purchase order number.
e. Any other identification specified in the inquiry or purchase order.
f. The vendor's identifying proposal number, shop order number, serial number, or other reference required to identify return correspondence completely.

5.2 Proposals

• 5.2.1 GENERAL

The vendor shall forward the original proposal and the specified number of copies to the addressee specified in the inquiry documents. As a minimum, the proposal shall include the data specified in 5.2.2 through 5.2.4, as well as a specific statement that the system and all its components are in strict accordance with this standard. If the system and components are not in strict accordance, the vendor shall provide details to enable the purchaser to evaluate any proposed alternative designs. All correspondence shall be clearly identified in accordance with 5.1.2.

5.2.2 DRAWINGS

5.2.2.1 The drawings indicated on the vendor drawings and data requirement form (see Appendix E) shall be included in the proposal. If typical drawings, schematics, and bills of material are used, they shall be marked to indicate the applicable weight and dimension data and to reflect the actual equipment and scope proposed. As a minimum, the following data shall be furnished:

a. An outline drawing for the system showing overall dimensions, maintenance clearance dimensions, overall weights, and the direction of rotation.
b. Cross-sectional drawings showing the details of the proposed equipment.
c. Schematics of all auxiliary systems including lube-oil.

5.2.3 TECHNICAL DATA

The following data shall be included in the proposal:

a. The purchaser's data sheets, with complete vendor's information entered thereon.
b. Noise data sheet in the form requested by the purchaser.
c. The Vendor Drawing and Data Requirements form (see Appendix E) indicating the schedule according to which the vendor agrees to transmit all the data specified as part of the contract.
d. A schedule for shipment of the equipment, in weeks after receipt of the order.
e. A list of major wearing components, showing interchangeability with the purchaser's other units.
f. A list of spare parts recommended for start-up and normal maintenance purposes.
g. A list of the special tools furnished for maintenance. The vendor shall identify any metric items included in the offering.
h. A statement of any special weather protection and winterization required for start-up, operation, and periods of idleness under the site conditions specified on the data sheets. The list shall show the protection to be furnished by the purchaser, as well as that included in the vendor's scope of supply.
i. A complete tabulation of utility requirements, such as those for steam, water electricity, air, gas and lube-oil, including the quantity of lube-oil required and the supply pressure, the heat load to be removed by the oil, and the nameplate power rating and operating power requirements of auxiliary drivers. (Approximate data shall be defined and clearly indicated as such.)
j. A description of the tests and inspection procedures for materials, as required by 2.11.1.3.
k. A description of any special requirements specified in the purchaser's inquiry and as outlined in 2.4.6, 2.10.10, 2.11.1.1, 2.11.1.2, and 3.4.6.1.
l. Any start-up, shutdown, or operating restrictions required to protect the integrity of the equipment.
m. If L/d exceeds the ratio allowed as specified in Table 3, a typical analytic justification shall be included in accordance with 2.4.3.7.

5.2.4 OPTIONS

The vendor shall furnish a list of the procedures for any special or optional tests that have been specified by the purchaser or proposed by the vendor.

5.3 Contract Data

5.3.1 GENERAL

5.3.1.1 The contract data to be furnished is specified in Appendix E. Each drawing, bill of material, and data sheet shall have a title block in its lower right-hand corner that

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shows the date of certification, a reference to all identification data specified in 5.1.2, the revision number and date, and the title.

5.3.1.2 The purchaser will promptly review the vendor's data when he receives them; however, this review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the data have been reviewed, the vendor shall furnish certified copies in the quantity specified.

5.3.1.3 A complete list of vendor data shall be included with the first issue of the major drawings. This list shall contain titles, drawing numbers, and a schedule for transmission of all the data the vendor will furnish (see Appendix E).

5.3.2 DRAWINGS

The drawings furnished shall contain sufficient information so that, with the drawings and the manuals specified in 5.3.6, the purchaser can properly install, operate, and maintain the ordered equipment. Drawings shall be clearly legible, shall be identified in accordance with 5.3.1.1, and shall be in accordance with ASME Y14.2M. As a minimum, each drawing shall include the details listed in Appendix E for that drawing.

5.3.3 TECHNICAL DATA

The data shall be submitted in accordance with Appendix E and identified in accordance with 5.3.1.1. Any comments on the drawings or revisions of the specifications that necessitate a change in the data shall be noted by the vendor. These notations will result in the purchaser's issue of completed, corrected data sheets as part of the order specifications.

5.3.4 PROGRESS REPORTS

The vendor shall submit progress reports to the purchaser at the intervals specified on the Vendor Drawing and Data Requirements form (see Appendix E). The reports shall include engineering, purchasing, manufacturing, and testing schedules for all major components. Planned and actual dates and the percentage completed shall be indicated for each milestone in the schedule.

5.3.5 PARTS LISTS AND RECOMMENDED SPARES

5.3.5.1 The vendor shall submit complete parts lists for all equipment and accessories supplied. The lists shall include manufacturer's unique part numbers, materials of construction, and delivery times. Materials shall be identified as specified in 2.11.1.2. Each part shall be completely identified and shown on cross-sectional or assembly-type drawings so that the purchaser may determine the interchangeability of the part with other equipment. Parts that have been modified from standard dimensions and/or finish to satisfy specific performance requirements shall be uniquely identified by part number for interchangeability and future duplication purposes. Standard purchased items shall be identified by the original manufacturer's name and part number.

5.3.5.2 The vendor shall indicate on the above parts lists which parts are recommended spares for startup and which parts are recommended for normal maintenance (see Item f of 5.2.3). The vendor shall forward the lists to the purchaser promptly after receipt of the reviewed drawings and in time to permit order and delivery of the parts before field start-up. The transmittal letter shall be identified with data specified in 5.1.2.

5.3.6 INSTALLATION, OPERATION, MAINTENANCE, AND TECHNICAL DATA MANUALS

5.3.6.1 General

The vendor shall provide sufficient written instructions and a list of all drawings to enable the purchaser to correctly install, operate, and maintain all of the equipment ordered. This information shall be compiled in a manual or manuals with a cover sheet that contains all reference-identifying data specified in 5.1.2, an index sheet that contains section titles, and a complete list of referenced and enclosed drawings by title and drawing number. The manual shall be prepared for the specified installation; a typical manual is not acceptable.

5.3.6.2 Installation Manual

Any special information required for proper installation design that is not on the drawings shall be compiled in a manual that is separate from the operating and maintenance instructions. This manual shall be forwarded at a time that is mutually agreed upon in the order but not later than the final issue of prints. The manual shall contain information such as special alignment and grouting procedures, utility specifications (including quantities), and all other installation design data, including the drawings and data specified in 5.2.2 and 5.2.3. The manual shall also include sketches that show the location of the center of gravity and rigging provisions to permit the removal of the top half of the casings, rotors, and any subassemblies that weigh more than 135 kilograms (300 pounds).

5.3.6.3 Operating and Maintenance and Technical Data Manual

The manual containing operating and maintenance and technical data shall be provided at the time of shipment. This manual shall include a section that provides special instructions for operation at specified extreme environmental conditions, such as temperatures. As a minimum, the manual shall also include all of the data listed in Appendix E.

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APPENDIX A—GENERAL-PURPOSE GEAR UNIT DATA SHEETS

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
PARALLEL SHAFT
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
P.O No. _____ Date _____
Requisition No. _____
Inquiry No. _____
Revision _____ Date _____ By _____

Applicable To: ○ Proposal ○ Purchase ○ As Built

For _____ Manufacturer _____
Site _____ Model No. _____
Unit _____ Serial No. _____
Service _____ Driver Type _____
No. Required _____ Driven Equipment _____

NOTE: Numbers Within ( ) Refer To Applicable API Standard 677 Paragraphs:

○ Information To Be Completed By Purchaser ☐ Information To Be Completed By Manufacturer

Driven Equip. Power (2.1.4): Normal _____ Max _____
Driver Power: Rated _____ Max _____
Gear Rated Power (2.4.1) _____
Torque @ Max. Cont. Speed _____ kg·m
Max.Torque (2.4.1) _____ kg·m _____ RPM
Rated Speed, RPM (1.4):
Input _____ ○ Specified ○ Nominal
Output _____ ○ Specified ○ Nominal
Allow. Var. In Gear Ratio (1.4)(+)(-) _____ **%**
Max Continuous Speed (1.4) _____ RPM
Trip Speed (1.4) _____ RPM
Gear Service Factor (2.4.2) _____ (Min)
Pin/Gear Hardness (Table 3) _____ **/** _____
Shaft Assembly Designation (2.2) _____
HS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
LS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
HS Shaft End: ○ Cylindrical ○ Taper ○ Hyd.Taper
○ 1-Key ○ 2-Keys ○ Integral Flange
LS Shaft End: ○ Cylindrical ○ Taper ○ Hyd.Taper
○ 1-Key ○ 2-Keys ○ Integral Flange
External Loads (2.1.16) _____
Other Operating Conditions (2.8.1.5)(2.10.3) _____

○ INSTALLATION DATA

○ Indoor ○ Heated ○ Under Roof
○ Outdoor ○ Unheated ○ Partial Sides
○ Grade ○ Mezzanine ○ _____
○ Winterization Req'd ○ Tropicalization Req'd
Electrical Area (2.1.9) Class _____ Grp _____ Div _____
Max Allow SPL (2.1.5) _____ dBA @ _____ m
Elevation _____ m Barometer _____ (kPa)
Range Of Ambient Temperatures: (BAR)

	Dry Bulb	Wet Bulb
Normal	_____ C	_____ C
Maximum	_____ C	_____ C
Minimum	_____ C	_____ C

Unusual Conditions (2.1.14) (2.11.1.5)**:**
○ Dust ○ Fumes ○ _____
Notes: _____

☐ BASIC GEAR DATA

Mechanical Rating _____ kW _____ RPM
Thermal Rating _____ kW _____ RPM
Full Load Horsepower Loss _____
Mechanical Efficiency _____ %
Pitch Line Velocity _____ m/sec
Tooth Pitting Index, "K" (2.4.3) **:**
Actual _____ Allowable _____
Tangential Load, "Wt" (2.4.3.2) _____ N
Bending Stress Number, "St" (2.4.4.2)

	Pinion	Gear
Actual	_____	_____
Allowable	_____	_____

Material Index Number (Fig 3, Table 3) _____
Anticipated Sound Press Level _____ dBA @ _____ m
Journal Static Weight Loads**:**
Pinion _____ kg Gear _____ kg
WR^2 Referred To LS Shaft _____ kg · m^2
Breakaway Torque _____ kg · m @Low Speed Shaft
Overhung Load Factor (Table 4) _____

☐ CONSTRUCTION FEATURES

TYPE OF GEAR ☐ Reducer ☐ Increaser
☐ Single Stage ☐ Double Stage
☐ Single Helical ☐ Double Helical
☐ Other _____

TEETH

Number Of Teeth: Pinion _____ Gear _____
Gear Ratio _____ Center Distance _____ mm
Pitch Dia. mm**:** Pinion _____ Gear _____
Finish _____ (RA) AGMA Geometry Factor "J" **:**
Pinion _____ Gear _____
Helix Angle _____ Deg Normal Press Angle _____ Deg
Net Face Width, "F" _____ mm Pinion L / D _____
Normal Diametral Pitch, "PND" _____ Backlash _____ mm

MANUFACTURING METHODS	Pinion	Gear
Teeth Generating Process		
Teeth Finishing Process		
Teeth Hardening Method		

Pinion To Shaft (2.7.3) ☐ Integral ☐ Shrunk-On
Gear To Shaft ☐ Integral ☐ Shrunk-On

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
PARALLEL SHAFT
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ ADDITIONAL REQUIREMENTS

MOUNTING PLATES (3.3)
○ Gear Furnished With (3.3.1):
○ Baseplate ○ Soleplate
○ Baseplate Suitable for Column Mounting (3.3.1.3)
○ Grout Type (3.3.1.2.5): ○ Epoxy ○ Other _____

● **PAINTING** (4.4.3.1) ○ _____

MISCELLANEOUS
○ Torsional Analysis By (2.8.1.8)**:** ○ Gear Vendor ○ Other
○ Lateral Analysis By (2.8.1.3) (2.8.1.8) :
○ Gear Vendor ○ Other
○ Spare Set Of Gear Rotors
○ Orientation Of Oil Inlet And Drain Connections (2.6.2.1) :
_____

○ VIBRATION DETECTORS (3.4.5)
Per API 670 Except Where Indicated Otherwise Below

RADIAL (2.7.4.2) (2.9.4.3)
Manufacturer _____
○ No. At Each Shaft Bearing _____ Total No. _____
○ Oscillator-Demodulators Supplied By _____
○ Manufacturer _____
○ Monitor Supplied By _____
○ Location _____ Enclosure _____
○ Manufacturer _____
○ Alarm _____ Shutdown _____

AXIAL (2.7.4.2) (2.9.4.3)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Oscillator-Demodulators Supplied By _____
○ Manufacturer _____
○ Shutdown**:** Time Delay _____ Seconds
○ Monitor Supplied By _____
○ Location _____ Enclosure _____
○ Manufacturer _____
○ Alarm _____ Shutdown _____
○ Shutdown**:** Time Delay _____ Seconds

ACCELEROMETER (2.9.4.4)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Monitor Supplied By _____

Notes: _____

☐ RADIAL BEARINGS

	Pinion	Gear
Type		
Diameter, mm		
Length, mm		
Journal Velocity, m/sec		
Loading, kPa		
Clearance (min-max), mm		
Span, mm		
L10 , Hrs (Roller Elm't)		

☐ THRUST BEARINGS

Location		
Manufacturer		
Type		
Size		
Area, mm^2		
Loading, kPa		
Rating, kPa		
L10 , Hrs (Roller Elm't)		
Int.Thrust Load, N (+)(-)		
Ext.Thrust Load, N (+)(-)		

☐ COUPLINGS

Manufacturer		
Model		
Cplg.Rating, kW/100RPM		
Cylindrical / 1-Key	☐	☐
Cylindrical / 2-Keys	☐	☐
Tapered / 1-Key	☐	☐
Tapered / 2-Keys	☐	☐
Tapered / Keyless	☐	☐

☐ MATERIALS

Gear Casing _____ Oil Seals _____
Radial Bearings _____
Thrust Bearing(s) _____
HS Shaft _____ LS Shaft _____
Pinion(s) _____ Hardness _____
Gear Rim(s) _____ Hardness _____
Low Temp. Operation (2.11.5)

☐ PIPING CONNECTIONS

	No.	Size	Type
Service			
Lube Oil Inlet			
Lube Oil Outlet			
Casing Drain			
Vent			
Casing Purge			

Notes: _____

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PARALLEL SHAFT
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ INSTRUMENTS

- ○ Low Oil Pressure Alarm Switch (2.10.10.j) _____
- ○ High Oil Temp. Alarm Switch (2.10.10.l) _____
- ○ Temperature Measuring Devices (3.4.1.3): _____
  - ○ Thermometers _____
  - ○ Thermocouples _____
  - ○ Resistance Temp. Detectors _____
  - ○ Liquid Filled Pressure Gages (3.4.4) _____
- ○ Thermal Relief Valves (3.4.6.2) _____

○ CONTRACT DATA

- ○ Test Data Prior To Shipment _____
- ○ Progress Reports (5.3.4) _____

_____

○ SHIPMENT (4.4.1)

	Contract Unit	Spares
Export Boxing (4.4.3.10)	○	○
Domestic Boxing (4.4.3.10)	○	○
Outdoor Storage Over 6 Months	○	○
Fit-Up & Assembly Of Mounted Accessories (4.4.8)	○	○

○ COUPLINGS AND GUARDS

	High Speed	Low Speed
Coupling Furnished By		
Coupling Type		
Cplg Rating kW/100 RPM		
Coupling manufacturer		
Coupling Lubrication		
Mount Cplg Halves (3.2.1)		
Taper, mm/m		
Taper gauge furnished by		
Limited End Float		
Cplg Guard Furnished By		

Notes: _____

_____

○ LUBRICATION REQUIREMENTS (2.10)

- ○ Oil System Furnished By (2.10.8) :
  - ○ Gear Vendor ○ Other _____
- ○ Oil Visc.: _____ cP @ 40 C _____ cP @ 100 C (2.10.8)
- ○ Lubrication Requirements (2.10.10) (Fig C-1 Or C-2)
- ○ Standby Oil Pump (2.10.10.b) _____
- ○ Oil Pump Casing (2.10.10.c): ○ Cast Iron ○ Steel
- ○ Oil Cooler (2.10.10.d) ○ Water Cooled ○ Air Cooled
- ○ Heaters Required (2.10.10.f):
  - ○ Electric With Thermostats ○ Steam
- ○ Duplex Filters (2.10.10.e)

□ LUBRICATION REQUIREMENTS

System Type (2.10.9) : □ Self-Contained Splash
□ Circulating □ Pressurized

- □ Filters, Micron Rating (2.10.10.e) _____
- □ Minimum Startup Oil Temperature _____ C
- □ Oil Flow (2.10.10.a) _____ m^3/ hr
- □ Oil Pressure (2.10.10.a) _____ kPa
- □ Unit Power Loss _____ kW

Reservoir (2.10.10.g) : □ Gear Casing □ Separate

- □ Lube Oil Inlet Size _____ mm
- □ Lube Oil Outlet Size mm

□ WEIGHTS AND DIMENSIONS

Net Weight: Gear _____ kg □ Auxiliaries _____ kg
Max. Maintenance Weight (Identify) _____ kg
Total Shipping Weight(s) _____ kg
Total Shipping Dimensions _____ X _____ X _____

Notes: _____

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PARALLEL SHAFT
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

	○ INSPECTIONS AND TESTS (4.1.3)	Req'd	Wit-nessed	Ob-served	Test Log
4	Shop Inspection (4.1.3)	○	○	○	
5	Cleanliness Inspection (4.1.6)	○	○	○	
6	Hardness Verification Inspection (4.1.7)	○	○	○	○
7	Mag. Particle Inspection (4.2.4.1)	○	○	○	
8	Ultrasonic Inspection (4.2.4.2.2)	○	○	○	
9	Weld Inspection (4.2.3.1)	○	○	○	
10	Dismantle-Reassembly Inspection	○	○	○	
11	Contact Check (2.7.2.1)	●	○	○	○
12	Contact Check Tape Lift (2.7.2.1)	○	○	○	○
13	Bearing Visual Check (4.3.3.8)	○	○	○	○
14	Axial Stability Check (2.7.2.4)	●	○	○	○
15	Residual Unbalance Check (2.8.2.3)	○	○	○	○
16	Mechanical Run Test (4.3.3)	●	○	○	○
17	Extended Mechanical Run Test (4.3.3.2)	○	○	○	○
18	Mechanical Run Test (Spare				
19	Rotors) (4.3.3.10)	●	○	○	○
20	Part Or Full Load And Full Speed				
21	Test (4.3.4.1)	○	○	○	○
22	Full Torque, Slow Roll Test (4.3.4.2)	○	○	○	○
23	Full Torque Static Test (4.3.4.3)	○	○	○	○
24	Sound Level Test (4.3.4.4)	○	○	○	○
25	Mechanical Run Test Coupling (4.3.3.3.5) :				
26	Couplings Installed	○	○	○	○
27	Couplings Hubs With Idlers	○	○	○	○
28	Use Shop Lube System				
29	(4.3.3.3.2) (4.3.3.11)	○	○	○	○
30	Use Job Lube System (4.3.3.3.2)				
31	(4.3.3.11) (4.3.3.3.6)	○	○	○	○
32	Use Shop Vibration Probes,Etc.(4.3.3.6)	○	○	○	
33	Use Job Vibration Probes, Etc.(4.3.3.6)	○	○	○	
34	Final Assembly, Maintenance &				
35	Running Clearance (4.2.1.e)	○	○	○	○
36	Oil System Cleanliness	●	○	○	
37	Oil System-Casing Joint				
38	Tightness (4.3.3.3.3)	●	○	○	
39	Warning And Protection				
40	Devices (4.3.3.3.4)	●	○	○	
41	Oil System Leak Test (4.3.2.2)	●	○	○	

42 Notes: _____

43 _____
44 _____
45 _____
46 _____
47 _____
48 _____
49

Notes: _____

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PARALLEL SHAFT
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
P.O No. _____ Date _____
Requisition No. _____
Inquiry No. _____
Revision _____ Date _____ By _____

Applicable To: ○ Proposal ○ Purchase ○ As Built

For _____ Manufacturer _____
Site _____ Model No. _____
Unit _____ Serial No. _____
Service _____ Driver Type _____
No. Required _____ Driven Equipment _____

NOTE: Numbers Within ( ) Refer To Applicable API Standard 677 Paragraphs:

○ Information To Be Completed By Purchaser ☐ Information To Be Completed By Manufacturer

Driven Equip. Power (2.1.4): Normal _____ Max _____
Driver Power: Rated _____ Max _____
Gear Rated Power (2.4.1) _____
Torque @ Max. Cont. Speed _____ Lb Ft
Max.Torque (2.4.1) _____ Lb Ft @ _____ RPM
Rated Speed, RPM (1.4):
Input _____ ○ Specified ○ Nominal
Output _____ ○ Specified ○ Nominal
Allow. Var. In Gear Ratio (1.4)(+)(-) _____ %
Max Continuous Speed (1.4) _____ RPM
Trip Speed (1.4) _____ RPM
Gear Service Factor (2.4.2) _____ (Min)
Pin/Gear Hardness (Table 3) _____ / _____
Shaft Assembly Designation (2.2) _____
HS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
LS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
HS Shaft End: ○ Cylindrical ○ Taper ○ Hyd.Taper
○ 1-Key ○ 2-Keys ○ Integral Flange
LS Shaft End: ○ Cylindrical ○ Taper ○ Hyd.Taper
○ 1-Key ○ 2-Keys ○ Integral Flange
External Loads (2.1.16) _____
Other Operating Conditions (2.8.1.5)(2.10.3) _____

○ INSTALLATION DATA

○ Indoor ○ Heated ○ Under Roof
○ Outdoor ○ Unheated ○ Partial Sides
○ Grade ○ Mezzanine ○ _____
○ Winterization Req'd ○ Tropicalization Req'd
Electrical Area (2.1.9) Class _____ Grp _____ Div _____
Max Allow SPL (2.1.5) _____ dBA @ _____ Ft
Elevation _____ Ft Barometer _____ PSIA
Range Of Ambient Temperatures:

	Dry Bulb	Wet Bulb
Normal	_____ F	_____ F
Maximum	_____ F	_____ F
Minimum	_____ F	_____ F

Unusual Conditions (2.1.14) (2.11.1.5):
○ Dust ○ Fumes ○ _____

Notes: _____

☐ BASIC GEAR DATA

Mechanical Rating _____ HP @ _____ RPM
Thermal Rating _____ HP @ _____ RPM
Full Load Horsepower Loss _____
Mechanical Efficiency _____ %
Pitch Line Velocity _____ FPM
Tooth Pitting Index, "K" (2.4.3) :
Actual _____ Allowable _____
Tangential Load, "Wt" (2.4.3.2) _____ Lb
Bending Stress Number, "St" (2.4.4.2)

	Pinion	Gear
Actual	_____	_____
Allowable	_____	_____

Material Index Number (Fig 3, Table 3) _____
Anticipated Sound Press Level _____ dBA @ _____ Ft
Journal Static Weight Loads:
Pinion _____ Lbs Gear _____ Lbs
WR² Referred To LS Shaft _____ Lb Ft²
Breakaway Torque _____ Ft Lb @ LS Shaft
Overhung Load Factor (Table 4) _____

☐ CONSTRUCTION FEATURES

TYPE OF GEAR
☐ Reducer ☐ Increaser
☐ Single Stage ☐ Double Stage
☐ Single Helical ☐ Double Helical
☐ Other _____

TEETH
Number Of Teeth: Pinion _____ Gear _____
Gear Ratio _____ Center Distance _____ In
Pitch Dia. In.: Pinion _____ Gear _____
Finish _____ (RA) AGMA Geometry Factor "J" :
Pinion _____ Gear _____
Helix Angle _____ Deg Normal Press Angle _____ Deg
Net Face Width, "F" _____ In Pinion L / D _____
Normal Diametral Pitch, "PND" _____ Backlash _____ In

MANUFACTURING METHODS	Pinion	Gear
Teeth Generating Process		
Teeth Finishing Process		
Teeth Hardening Method		

Pinion To Shaft (2.7.3) ☐ Integral ☐ Shrunk-On
Gear To Shaft ☐ Integral ☐ Shrunk-On

--`,,``,,,,``,,,,

GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
PARALLEL SHAFT
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ ADDITIONAL REQUIREMENTS

MOUNTING PLATES (3.3.)
○ Gear Furnished With (3.3.1):
○ Baseplate ○ Soleplate
○ Baseplate Suitable for Column Mounting (3.3.1.3)
○ Grout Type (3.3.1.2.5): ○ Epoxy ○ Other _____

● **PAINTING** (4.4.3.1) ○ _____

MISCELLANEOUS
○ Torsional Analysis By (2.8.1.8): ○ Gear Vendor ○ Other
○ Lateral Analysis By (2.8.1.3) (2.8.1.8):
○ Gear Vendor ○ Other
○ Spare Set Of Gear Rotors
○ Orientation Of Oil Inlet And Drain Connections (2.6.2.1):
_____

○ VIBRATION DETECTORS (3.4.5)
Per API 670 Except Where Indicated Otherwise Below

RADIAL (2.7.4.2) (2.9.4.3)
Manufacturer _____
○ No. At Each Shaft Bearing _____ Total No. _____
○ Oscillator-Demodulators Supplied By _____
○ Manufacturer _____
○ Monitor Supplied By _____
○ Location _____ Enclosure _____
○ Manufacturer _____
○ Alarm _____ Shutdown _____

AXIAL (2.7.4.2) (2.9.4.3)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Oscillator-Demodulators Supplied By _____
○ Manufacturer _____
○ Shutdown**:** Time Delay _____ Seconds
○ Monitor Supplied By _____
○ Location _____ Enclosure _____
○ Manufacturer _____
○ Alarm _____ Shutdown _____
○ Shutdown**:** Time Delay _____ Seconds

ACCELEROMETER (2.9.4.4)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Monitor Supplied By _____

Notes: _____

☐ RADIAL BEARINGS

	Pinion	Gear
Type		
Diameter, In		
Length, In		
Journal Velocity, FPS		
Loading, PSI		
Clearance (min-max), In		
Span, In		
L10 , Hrs (Roller Elm't)		

☐ THRUST BEARINGS

Location		
Manufacturer		
Type		
Size		
Area, In^2		
Loading, PSI		
Rating, PSI		
L10 , Hrs (Roller Elm't)		
Int.Thrust Load, Lb (+)(-)		
Ext.Thrust Load, Lb(+)(-)		

☐ COUPLINGS

Manufacturer		
Model		
Cplg.Rating,HP/100RPM		
Cylindrical / 1-Key	☐	☐
Cylindrical / 2-Keys	☐	☐
Tapered / 1-Key	☐	☐
Tapered / 2-Keys	☐	☐
Tapered / Keyless	☐	☐

☐ MATERIALS

Gear Casing _____ Oil Seals _____
Radial Bearings _____
Thrust Bearing(s) _____
HS Shaft _____ LS Shaft _____
Pinion(s) _____ Hardness _____
Gear Rim(s) _____ Hardness _____
Low Temp. Operation (2.11.5)

☐ PIPING CONNECTIONS

	No.	Size	Type
Service			
Lube Oil Inlet			
Lube Oil Outlet			
Casing Drain			
Vent			
Casing Purge			

Notes: _____

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
PARALLEL SHAFT
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

INSTRUMENTS

- ○ Low Oil Pressure Alarm Switch (2.10.10.j): _____
- ○ High Oil Temp. Alarm Switch (2.10.10.l): _____
- ○ Temperature Measuring Devices (3.4.1.3):
  - ○ Thermometers _____
  - ○ Thermocouples _____
  - ○ Resistance Temp. Detectors _____
  - ○ Liquid Filled Pressure Gages (3.4.4) _____
- ○ Thermal Relief Valves (3.4.6.2) _____

CONTRACT DATA

- ○ Test Data Prior To Shipment _____
- ○ Progress Reports (5.3.4) _____

SHIPMENT (4.4.1)

	Contract Unit	Spares
Export Boxing (4.4.3.10)	○	○
Domestic Boxing (4.4.3.10)	○	○
Outdoor Storage Over 6 Months	○	○
Fit-Up & Assembly Of Mounted Accessories (4.4.8)	○	○

COUPLINGS AND GUARDS

	High Speed	Low Speed
Coupling Furnished By		
Coupling Type		
Cplg Rating HP/100 RPM		
Coupling Manufacturer		
Coupling Lubrication		
Mount Cplg Halves (3.2.1)		
Taper		
Taper Gauge Furnished By		
Limited End Float		
Cplg Guard Furnished By		

Notes: _____

LUBRICATION REQUIREMENTS (2.10)

- ○ Oil System Furnished By (2.10.8) :
  - ○ Gear Vendor ○ Other _____
- ○ Oil Visc.: _____ SSU@100 F _____ SSU@210 F (2.10.8)
- ○ Lubrication Requirements (2.10.10) (Fig C-1 Or C-2)
- ○ Standby Oil Pump (2.10.10.b) _____
- ○ Oil Pump Casing (2.10.10.c): ○ Cast Iron ○ Steel
- ○ Oil Cooler (2.10.10.d) ○ Water Cooled ○ Air Cooled
- ○ Heaters Required (2.10.10.f):
  - ○ Electric With Thermostats ○ Steam
- ○ Duplex Filters (2.10.10.e)

LUBRICATION REQUIREMENTS

System Type (2.10.9) : ☐ Self-Contained Splash ☐ Circulating ☐ Pressurized

- ☐ Filters, Micron Rating (2.10.10.e) _____
- ☐ Minimum Startup Oil Temperature _____ F
- ☐ Oil Flow (2.10.10.a) _____ GPM
- ☐ Oil Pressure (2.10.10.a) _____ PSIG
- ☐ Unit Power Loss _____ HP

Reservoir (2.10.10.g) : ☐ Gear Casing ☐ Separate

- ☐ Lube Oil Inlet Size _____ In
- ☐ Lube Oil Outlet Size _____ In

WEIGHTS AND DIMENSIONS

Net Weight: Gear _____ Lbs ☐ Auxiliaries _____ Lbs
Max. Maintenance Weight (Identify) _____ Lbs
Total Shipping Weight(s) _____ Lbs
Total Shipping Dimensions _____ X _____ X _____

Notes: _____

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
PARALLEL SHAFT
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ INSPECTIONS AND TESTS (4.1.3)

	Req'd	Wit-nessed	Ob-served	Test Log
Shop Inspection (4.1.3)	○	○	○	
Cleanliness Inspection (4.1.6)	○	○	○	
Hardness Verification Inspection (4.1.7)	○	○	○	○
Mag. Particle Inspection (4.2.4.1)	○	○	○	
Ultrasonic Inspection (4.2.4.2.2)	○	○	○	
Weld Inspection (4.2.3.1)	○	○	○	
Dismantle-Reassembly Inspection	○	○	○	
Contact Check (2.7.2.1)	●	○	○	○
Contact Check Tape Lift (2.7.2.1)	○	○	○	○
Bearing Visual Check (4.3.3.8)	○	○	○	○
Axial Stability Check (2.7.2.4)	●	○	○	○
Residual Unbalance Check (2.8.2.3)	○	○	○	○
Mechanical Run Test (4.3.3)	●	○	○	○
Extended Mechanical Run Test (4.3.3.2)	○	○	○	○
Mechanical Run Test (Spare Rotors) (4.3.3.10)	●	○	○	○
Part Or Full Load And Full Speed Test (4.3.4.1)	○	○	○	○
Full Torque, Slow Roll Test (4.3.4.2)	○	○	○	○
Full Torque Static Test (4.3.4.3)	○	○	○	○
Sound Level Test (4.3.4.4)	○	○	○	○
Mechanical Run Test Coupling (4.3.3.3.5) :				
Couplings Installed	○	○	○	○
Couplings Hubs With Idlers	○	○	○	○
Use Shop Lube System (4.3.3.3.2) (4.3.3.11)	○	○	○	○
Use Job Lube System (4.3.3.3.2) (4.3.3.11) (4.3.3.3.6)	○	○	○	○
Use Shop Vibration Probes, Etc.(4.3.3.6)	○	○	○	
Use Job Vibration Probes, Etc.(4.3.3.6)	○	○	○	
Final Assembly, Maintenance & Running Clearance (4.2.1.e)	○	○	○	○
Oil System Cleanliness	●	○	○	
Oil System-Casing Joint Tightness (4.3.3.3.3)	●	○	○	
Warning And Protection Devices (4.3.3.3.4)	●	○	○	
Oil System Leak Test (4.3.2.2)	●	○	○	

Notes: _____

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
P.O No. _____ Date _____
Requisition No. _____
Inquiry No. _____
Revision _____ Date _____ By _____

Applicable To: ○ Proposal ○ Purchase ○ As Built

For _____ Manufacturer _____
Site _____ Model No. _____
Unit _____ Serial No. _____
Service _____ Driver Type _____
No. Required _____ Driven Equipment _____

NOTE: Numbers Within ( ) Refer To Applicable API Standard 677 Paragraphs:
○ Information To Be Completed By Purchaser ☐ Information To Be Completed By Manufacturer

Driven Equip. Power (2.1.4): Normal _____ Max _____
Driver Power: Rated _____ Max _____
Gear Rated Power (2.4.1) _____ kW
Torque @ Max. Cont. Speed _____ kg· m
Max.Torque (2.4.1) _____ kg· m _____ RPM
Rated Speed, RPM (1.4):
Input _____ ○ Specified ○ Nominal
Output _____ ○ Specified ○ Nominal
Allow. Var. In Gear Ratio (1.4)(+)(-) _____ **%**
Max Continuous Speed (1.4) _____ RPM
Trip Speed (1.4) _____ RPM
Gear Service Factor (2.4.2) _____ (Min)
Pin/Gear Hardness (Table 3) _____ **/** _____
Shaft Assembly Designation (2.2) _____
HS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
LS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
HS Shaft End: ○ Cylindrical ○ Taper/Keyed
○ 1-Key ○ 2-Keys
LS Shaft End: ○ Cylindrical ○ Taper/Keyed
○ 1-Key ○ 2-Keys
External Loads (2.1.16) _____
Other Operating Conditions (2.8.1.5)(2.10.3) _____

☐ BASIC GEAR DATA

Mechanical Rating _____ kW _____ RPM
Thermal Rating _____ kW _____ RPM
Full Load Horsepower Loss _____ kW
Mechanical Efficiency _____ %
Pitch Line Velocity _____ m/sec
Tooth Pitting Index, "K" (2.4.3.2 and 2.4.3.4) **:**
Actual _____ Allowable _____
Tangential Load, "Wt" (2.4.3.2 and 2.4.3.4) _____ kg
Bending Stress Number, "St" (2.4.4.3)

	Pinion	Gear
Actual		
Allowable		

Material Index Number (Fig 3, Table 3) _____
Anticipated Sound Press Level _____ dBA @ _____ m
Mounting of Bevel Gears (2.7.1.6): ☐ Straddle ☐ Overhung
Journal Static Weight Loads**:**
Pinion _____ kg Gear _____ kg
WR**2** Referred To LS Shaft _____ kg· m²
Breakaway Torque _____ N· m
Overhung Load Factor (Table 4) _____

○ INSTALLATION DATA

○ Indoor (2.1.14) ○ Heated ○ Under Roof
○ Outdoor ○ Unheated ○ Partial Sides
○ Grade ○ Mezzanine ○ _____
○ Winterization Req'd ○ Tropicalization Req'd
Electrical Area (2.1.9) Class _____ Grp _____ Div _____
Max Allow SPL (2.1.5) _____ dBA @ _____ m
Elevation _____ m Barometer _____ kPa abs
Range Of Ambient Temperatures:

	Dry Bulb	Wet Bulb
Normal	_____ C	_____ C
Maximum	_____ C	_____ C
Minimum	_____ C	_____ C

Unusual Conditions (2.1.14) (2.11.1.5)**:**
○ Dust ○ Fumes ○ _____
Notes: _____

☐ CONSTRUCTION FEATURES

TYPE OF GEAR ☐ Reducer ☐ Increaser
☐ Single Stage ☐ Double Stage
Other _____

TEETH
Number Of Teeth: Pinion _____ Gear _____
Gear Ratio _____ Mounting Distance _____ mm
Pitch Dia. mm**:** Pinion _____ Gear _____
Finish _____ (RA) AGMA Geometry Factor "J" **:**
Pinion _____ Gear _____
Helix Angle _____ Deg Transverse Press Angle _____ Deg
Face Width, "F" _____ mm Pinion L/D (2.4.3.6) _____
Normal Diametral Pitch, "PND" _____ Backlash _____ mm

MANUFACTURING METHODS	Pinion	Gear
Teeth Generating Process		
Teeth Finishing Process		
Teeth Hardening Method		

Pinion To Shaft (2.7.3) ☐ Integral ☐ Shrunk-On
Gear To Shaft ☐ Integral ☐ Shrunk-On

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ ADDITIONAL REQUIREMENTS

MOUNTING PLATES (3.3)
○ Gear Furnished With (3.3.1):
○ Baseplate ○ Soleplate
○ Baseplate Suitable For Column Mounting (3.3.1.3)
○ Grout Type (3.3.1.2.5): ○ Epoxy ○ Other _____
MOUNTING FLANGES
○ Clearance Fit With Jackbolts
○ Register Fit
● **PAINTING** (4.4.3.1) ○ _____
MISCELLANEOUS
○ Torsional Analysis By (2.8.1.8): ○ Gear Vendor ○ Other
○ Lateral Analysis By (2.8.1.3) (2.8.1.8):
○ Gear Vendor ○ Other
○ Spare Set Of Gear Rotors
○ Orientation Of Oil Inlet And Drain Connections (2.6.2.1):

○ VIBRATION DETECTORS (3.4.5)
Per API 670 Except Where Indicated Otherwise Below

ACCELEROMETER (2.9.4.4)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Monitor Supplied By

○ INSTRUMENTS
○ Low Oil Pressure Alarm Switch (2.10.10.j): _____
○ High Oil Temp. Alarm Switch (2.10.10.l): _____
○ Temperature Measuring Devices (3.4.1.3): _____
○ Thermometers _____
○ Thermocouples _____
○ Resistance Temp. Detectors _____
○ Liquid Filled Pressure Gauges (3.4.4) _____
○ Thermal Relief Valves (3.4.6.2) _____

○ CONTRACT DATA
○ Test Data Prior To Shipment _____
○ Progress Reports (5.3.4) _____

○ SHIPMENT (4.4.1)

	Contract Unit	Spares
Export Boxing (4.4.3.10)	○	○
Domestic Boxing (4.4.3.10)	○	○
Outdoor Storage Over 6 Months	○	○
Fit-Up & Assembly Of Mounted Accessories (4.4.8)	○	○

Notes: _____

☐ RADIAL BEARINGS

	Pinion	Gear
Manufacturer		
Type		
Class		
Cage Speed, m/sec		
☐ B10 ☐ L10 Hours		

NOTES _____

☐ THRUST BEARINGS

	Pinion	Gear
Manufacturer		
Type		
Class		
Cage Speed, m/sec		
☐ B10 ☐ L10 Hours		
Down Thrust Capacity, N		
Up Thrust Capacity, N		

NOTES _____

☐ COUPLINGS

Manufacturer		
Model		
Cplg.Rating,kW/100RPM		
Cylindrical / 1-Key	☐	☐
Cylindrical / 2-Keys	☐	☐
Tapered / 1-Key	☐	☐
Tapered / 2-Keys	☐	☐
Tapered / Keyless	☐	☐

☐ MATERIALS

Gear Casing _____ Oil Seals _____
Radial Bearings _____
Thrust Bearing(s) _____
HS Shaft _____ LS Shaft _____
Pinion(s) _____ Hardness _____
Gear Rim(s) _____ Hardness _____
Low Temp. Operation (2.11.5)

☐ PIPING CONNECTIONS

	No.	Size	Type
Service			
Lube Oil Inlet			
Lube Oil Outlet			
Casing Drain			
Vent			
Casing Purge			

Notes: _____

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
SI UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ COUPLINGS AND GUARDS

	High Speed	Low Speed
Coupling Furnished By		
Coupling Type		
Cplg Rating kW/100 RPM		
Coupling Manufacturer		
Coupling Lubrication		
Mount Cplg Halves (3.2.1)		
Taper		
Taper Gauge Furnished By		
Limited End Float		
Cplg Guard Furnished By		

Notes: _____

○ LUBRICATION REQUIREMENTS (2.10)

○ Oil System Furnished By (2.10.8) :
○ Gear Vendor ○ Other _____
○ Oil Visc.: _____ cP @40 C _____ cP@100 C (2.10.8)
○ Lubrication Requirements (2.10.10) (Fig C-1 Or C-2)
● Main Oil Pump ○ Shaft Driven ○ Motor Driven
○ Standby Oil Pump (2.10.10.b) _____
Oil Pump Casings (2.10.10.c): ○ Cast Iron ○ Steel
● Oil Cooler (2.10.10.d) ○ Water Cooled ○ Air Cooled
○ Heaters Required (2.10.10.f):
○ Electric With Thermostats ○ Steam
○ Duplex Filters (2.10.10.e)

○ INSPECTIONS AND TESTS (4.1.3)

	Req'd	Wit-nessed	Ob-served	Test Log
Shop Inspection (4.1.3)	○	○	○	
Cleanliness Inspection (4.1.6)	○	○	○	
Hardness Verification Inspection (4.1.7)	○	○	○	○
Mag. Particle Inspection (4.2.4.1)	○	○	○	
Ultrasonic Inspection (4.2.4.2.2)	○	○	○	
Weld Inspection (4.2.3.1)	○	○	○	
Dismantle-Reassembly Inspection	○	○	○	
Contact Check (2.7.2.2)	●	○	○	○
Contact Check Tape Lift (2.7.2.2)	○	○	○	○
Bearing Visual Check (4.3.3.8)	○	○	○	○
Bevel Backlash Check (Appendix G)	○	○	○	○
Residual Unbalance Check (2.8.2.3)	○	○	○	○
Mechanical Run Test (4.3.3)	●	○	○	○
Extended Mechanical Run Test (4.3.3.2)	○	○	○	○
Mechanical Run Test (Spare Rotors) (4.3.3.10)	●	○	○	○
Sound Level Test (4.3.4.4)	○	○	○	○
Mechanical Run Test Coupling (4.3.3.3.5):				
Couplings Installed	○	○	○	○
Coupling Hubs With Idlers	○	○	○	○

□ LUBRICATION REQUIREMENTS

System Type (2.10.9) : □ Self-Contained Splash
□ Circulating □ Pressurized
□ Filters, Micron Rating (2.10.10.e) _____
□ Cooler Not Required
□ Minimum Startup Oil Temperature _____ C
□ Oil Flow (2.10.10.a) _____ GP m3/hr
□ Oil Pressure (2.10.10.a) _____ kPa
□ Unit Power Loss _____ kW
Reservoir (2.10.10.g) : □ Gear Casing □ Separate
□ Lube Oil Inlet Size _____ mm
□ Lube Oil Outlet Size _____ mm

□ WEIGHTS AND DIMENSIONS

Net Weight: Gear _____ kg □ Auxiliaries _____ kg
Max. Maintenance Weight (Identify) _____ kg
Total Shipping Weight(s) _____ kg
Total Shipping Dimensions _____ X _____ X _____

○ INSPECTIONS AND TESTS (4.1.3)

Use Shop Lube System (4.3.3.3.2) (4.3.3.11)	○	○	○	○
Use Job Lube System (4.3.3.3.2) (4.3.3.11) (4.3.3.3.6)	○	○	○	○
Final Assembly, Maintenance & Running Clearance (4.2.1.e)	○	○	○	○
Oil System Cleanliness	●	○	○	
Oil System-Casing Joint Tightness (4.3.3.3.3)	●	○	○	
Warning And Protection Devices (4.3.3.3.4)	●	○	○	
Oil System Leak Test (4.3.2.2)	●	○	○	

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
P.O No. _____ Date _____
Requisition No. _____
Inquiry No. _____
Revision _____ Date _____ By _____

Applicable To: ○ Proposal ○ Purchase ○ As Built

For _____ Manufacturer _____
Site _____ Model No. _____
Unit _____ Serial No. _____
Service _____ Driver Type _____
No. Required _____ Driven Equipment _____

NOTE: Numbers Within ( ) Refer To Applicable API Standard 677 Paragraphs:
○ Information To Be Completed By Purchaser ☐ Information To Be Completed By Manufacturer

Driven Equip. Power (2.1.4): Normal _____ Max _____
Driver Power: Rated _____ Max _____
Gear Rated Power (2.4.1) _____
Torque @ Max. Cont. Speed _____ Lb Ft
Max.Torque (2.4.1) _____ Lb Ft @ _____ RPM
Rated Speed, RPM (1.4):
Input _____ ○ Specified ○ Nominal
Output _____ ○ Specified ○ Nominal
Allow. Var. In Gear Ratio (1.4)(+)(-) _____ **%**
Max Continuous Speed (1.4) _____ RPM
Trip Speed (1.4) _____ RPM
Gear Service Factor (2.4.2) _____ (Min)
Pin/Gear Hardness (Table 3) _____ **/** _____
Shaft Assembly Designation (2.2) _____
HS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
LS Shaft Rotation Facing Cpl'g. End (2.3.3): ○ CW ○ CCW
HS Shaft End: ○ Cylindrical ○ Taper/Keyed
○ 1-Key ○ 2-Keys
LS Shaft End: ○ Cylindrical ○ Taper/Keyed
○ 1-Key ○ 2-Keys
External Loads (2.1.16) _____
Other Operating Conditions (2.8.1.5)(2.10.3) _____

☐ BASIC GEAR DATA

Mechanical Rating _____ HP @ _____ RPM
Thermal Rating _____ HP @ _____ RPM
Full Load Horsepower Loss _____
Mechanical Efficiency _____ %
Pitch Line Velocity _____ FPM
Tooth Pitting Index, "K" (2.4.3.2 and 2.4.3.4) **:**
Actual _____ Allowable _____
Tangential Load, "Wt" (2.4.3.2 and 2.4.3.4) _____ Lbs
Bending Stress Number, "St" (2.4.4.3)

	Pinion	Gear
Actual		
Allowable		

Material Index Number (Fig 3, Table 3) _____
Anticipated Sound Press Level _____ dBA @ _____ Ft
Mounting of Bevel Gears (2.7.1.6): ☐ Straddle ☐ Overhung
Journal Static Weight Loads**:**
Pinion _____ Lbs Gear _____ Lbs
WR² Referred To LS Shaft _____ Lb Ft²
Breakaway Torque _____ Ft Lb @ LS Shaft
Overhung Load Factor (Table 4) _____

○ INSTALLATION DATA

○ Indoor (2.1.14) ○ Heated ○ Under Roof
○ Outdoor ○ Unheated ○ Partial Sides
○ Grade ○ Mezzanine ○ _____
○ Winterization Req'd ○ Tropicalization Req'd
Electrical Area (2.1.9) Class _____ Grp _____ Div _____
Max Allow SPL (2.1.5) _____ dBA @ _____ Ft
Elevation _____ Ft Barometer _____ PSIA
Range Of Ambient Temperatures:

	Dry Bulb	Wet Bulb
Normal	_____ F	_____ F
Maximum	_____ F	_____ F
Minimum	_____ F	_____ F

Unusual Conditions (2.1.14) (2.11.1.5):
○ Dust ○ Fumes ○ _____
Notes: _____

☐ CONSTRUCTION FEATURES

TYPE OF GEAR ☐ Reducer ☐ Increaser
☐ Single Stage ☐ Double Stage
☐ Other _____

TEETH
Number Of Teeth: Pinion _____ Gear _____
Gear Ratio _____ Mounting Distance _____ In
Pitch Dia. In**:** Pinion _____ Gear _____
Finish _____ (RA) AGMA Geometry Factor "J" **:**
Pinion _____ Gear _____
Helix Angle _____ Deg Transverse Press Angle _____ Deg
Face Width, "F" _____ In Pinion L/D (2.4.3.6) _____
Normal Diametral Pitch, "PND" _____ Backlash _____ In

MANUFACTURING METHODS	Pinion	Gear
Teeth Generating Process		
Teeth Finishing Process		
Teeth Hardening Method		

Pinion To Shaft (2.7.3) ☐ Integral ☐ Shrunk-On
Gear To Shaft ☐ Integral ☐ Shrunk-On

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

○ ADDITIONAL REQUIREMENTS

MOUNTING PLATES (3.3)
○ Gear Furnished With (3.3.1):
○ Baseplate ○ Soleplate
○ Baseplate Suitable For Column Mounting (3.3.1.3)
○ Grout Type (3.3.1.2.5): ○ Epoxy ○ Other _____
MOUNTING FLANGES
○ Clearance Fit With Jackbolts
○ Register Fit
● **PAINTING** (4.4.3.1) ○ _____
MISCELLANEOUS
○ Torsional Analysis By (2.8.1.8): ○ Gear Vendor ○ Other
○ Lateral Analysis By (2.8.1.3) (2.8.1.8):
○ Gear Vendor ○ Other
○ Spare Set Of Gear Rotors
○ Orientation Of Oil Inlet And Drain Connections (2.6.2.1):

○ VIBRATION DETECTORS (3.4.5)
Per API 670 Except Where Indicated Otherwise Below
ACCELEROMETER (2.9.4.4)
○ Manufacturer _____ No. Required _____
○ Location _____
○ Monitor Supplied By

○ INSTRUMENTS
○ Low Oil Pressure Alarm Switch (2.10.10.j): _____
○ High Oil Temp. Alarm Switch (2.10.10.l): _____
○ Temperature Measuring Devices (3.4.1.3):
○ Thermometers _____
○ Thermocouples _____
○ Resistance Temp. Detectors _____
○ Liquid Filled Pressure Gages (3.4.4) _____
○ Thermal Relief Valves (3.4.6.2) _____

○ CONTRACT DATA
○ Test Data Prior To Shipment _____
○ Progress Reports (5.3.4) _____

○ SHIPMENT (4.4.1)

	Contract Unit	Spares
Export Boxing (4.4.3.10)	○	○
Domestic Boxing (4.4.3.10)	○	○
Outdoor Storage Over 6 Months	○	○
Fit-Up & Assembly Of Mounted Accessories (4.4.8)	○	○

Notes: _____

☐ RADIAL BEARINGS

	Pinion	Gear
Manufacturer		
Type		
Class		
Cage Speed, FPM		
☐ B10 ☐ L10 Hours		

Notes: _____

☐ THRUST BEARINGS

	Pinion	Gear
Manufacturer		
Type		
Class		
Cage Speed, FPM		
☐ B10 ☐ L10 Hours		
DownThrust Capacity, Lb		
Up Thrust Capacity, Lb		

Notes: _____

☐ COUPLINGS

Manufacturer		
Model		
Cplg.Rating,HP/100RPM		
Cylindrical / 1-Key	☐	☐
Cylindrical / 2-Keys	☐	☐
Tapered / 1-Key	☐	☐
Tapered / 2-Keys	☐	☐
Tapered / Keyless	☐	☐

☐ MATERIALS
Gear Casing _____ Oil Seals _____
Radial Bearings _____
Thrust Bearing(s) _____
HS Shaft _____ LS Shaft _____
Pinion(s) _____ Hardness _____
Gear Rim(s) _____ Hardness _____
Low Temp. Operation (2.11.5)

☐ PIPING CONNECTIONS

	No.	Size	Type
Service			
Lube Oil Inlet			
Lube Oil Outlet			
Casing Drain			
Vent			
Casing Purge			

Notes: _____

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GENERAL PURPOSE GEAR UNITS
API 677 SECOND EDITION
BEVEL MOUNTED
DATA SHEET
U.S. CUSTOMARY UNITS

Job No. _____ Item No. _____
Date _____ By _____
Revision No. _____ By _____

COUPLINGS AND GUARDS

	High Speed	Low Speed
Coupling Furnished By		
Coupling Type		
Cplg Rating HP/100 RPM		
Coupling Manufacturer		
Coupling Lubrication		
Mount Cplg Halves (3.2.1)		
Taper		
Taper Gauge Furnished By		
Limited End Float		
Cplg Guard Furnished By		

Notes: _____

LUBRICATION REQUIREMENTS (2.10)

○ Oil System Furnished By (2.10.8) :
○ Gear Vendor ○ Other _____
○ Oil Visc.: _____ SSU@100 F _____ SSU@210 F (2.10.8)
○ Lubrication Requirements (2.10.10) (Fig C-1 Or C-2)
● Main Oil Pump ○ Shaft Driven ○ Motor Driven
○ Standby Oil Pump (2.10.10.b) _____
Oil Pump Casings (2.10.10.c): ○ Cast Iron ○ Steel
● Oil Cooler (2.10.10.d) ○ Water Cooled ○ Air Cooled
○ Heaters Required (2.10.10.f):
○ Electric With Thermostats ○ Steam
○ Duplex Filters (2.10.10.e)

INSPECTIONS AND TESTS (4.1.3)

	Req'd	Wit-nessed	Ob-served	Test Log
Shop Inspection (4.1.3)	○	○	○	
Cleanliness Inspection (4.1.6)	○	○	○	
Hardness Verification Inspection (4.1.7)	○	○	○	○
Mag. Particle Inspection (4.2.4.1)	○	○	○	
Ultrasonic Inspection (4.2.4.2.2)	○	○	○	
Weld Inspection (4.2.3.1)	○	○	○	
Dismantle-Reassembly Inspection	○	○	○	
Contact Check (2.7.2.2)	●	○	○	○
Contact Check Tape Lift (2.7.2.2)	○	○	○	○
Bearing Visual Check (4.3.3.8)	○	○	○	○
Bevel Backlash Check (Appendix G)	○	○	○	○
Residual Unbalance Check (2.8.2.3)	○	○	○	○
Mechanical Run Test (4.3.3)	●	○	○	○
Extended Mechanical Run Test (4.3.3.2)	○	○	○	○
Mechanical Run Test (Spare Rotors) (4.3.3.10)	●	○	○	○
Sound Level Test (4.3.4.4)	○	○	○	○
Mechanical Run Test Coupling (4.3.3.3.5):				
Couplings Installed	○	○	○	○
Coupling Hubs With Idlers	○	○	○	○

LUBRICATION REQUIREMENTS

System Type (2.10.9) : ☐ Self-Contained Splash
☐ Circulating ☐ Pressurized
☐ Filters, Micron Rating (2.10.10.e) _____
☐ Cooler Not Required
☐ Minimum Startup Oil Temperature _____ F
☐ Oil Flow (2.10.10.a) _____ GPM
☐ Oil Pressure (2.10.10.a) _____ PSIG
☐ Unit Power Loss _____ HP
Reservoir (2.10.10.g) : ☐ Gear Casing ☐ Separate
☐ Lube Oil Inlet Size _____ In
☐ Lube Oil Outlet Size _____ In

WEIGHTS AND DIMENSIONS

Net Weight: Gear _____ Lbs ☐ Auxiliaries _____ Lbs
Max. Maintenance Weight (Identify) _____ Lbs
Total Shipping Weight(s) _____ Lbs
Total Shipping Dimensions _____ X _____ X _____

INSPECTIONS AND TESTS (4.1.3)

Use Shop Lube System (4.3.3.3.2) (4.3.3.11)	○	○	○	○
Use Job Lube System (4.3.3.3.2) (4.3.3.11) (4.3.3.3.6)	○	○	○	○
Final Assembly, Maintenance & Running Clearance (4.2.1.e)	○	○	○	○
Oil System Cleanliness	●	○	○	
Oil System-Casing Joint Tightness (4.3.3.3.3)	●	○	○	
Warning And Protection Devices (4.3.3.3.4)	●	○	○	
Oil System Leak Test (4.3.2.2)	●	○	○	

Notes: _____

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APPENDIX B—LATERAL CRITICAL SPEED MAP AND MODE SHAPES FOR A TYPICAL ROTOR

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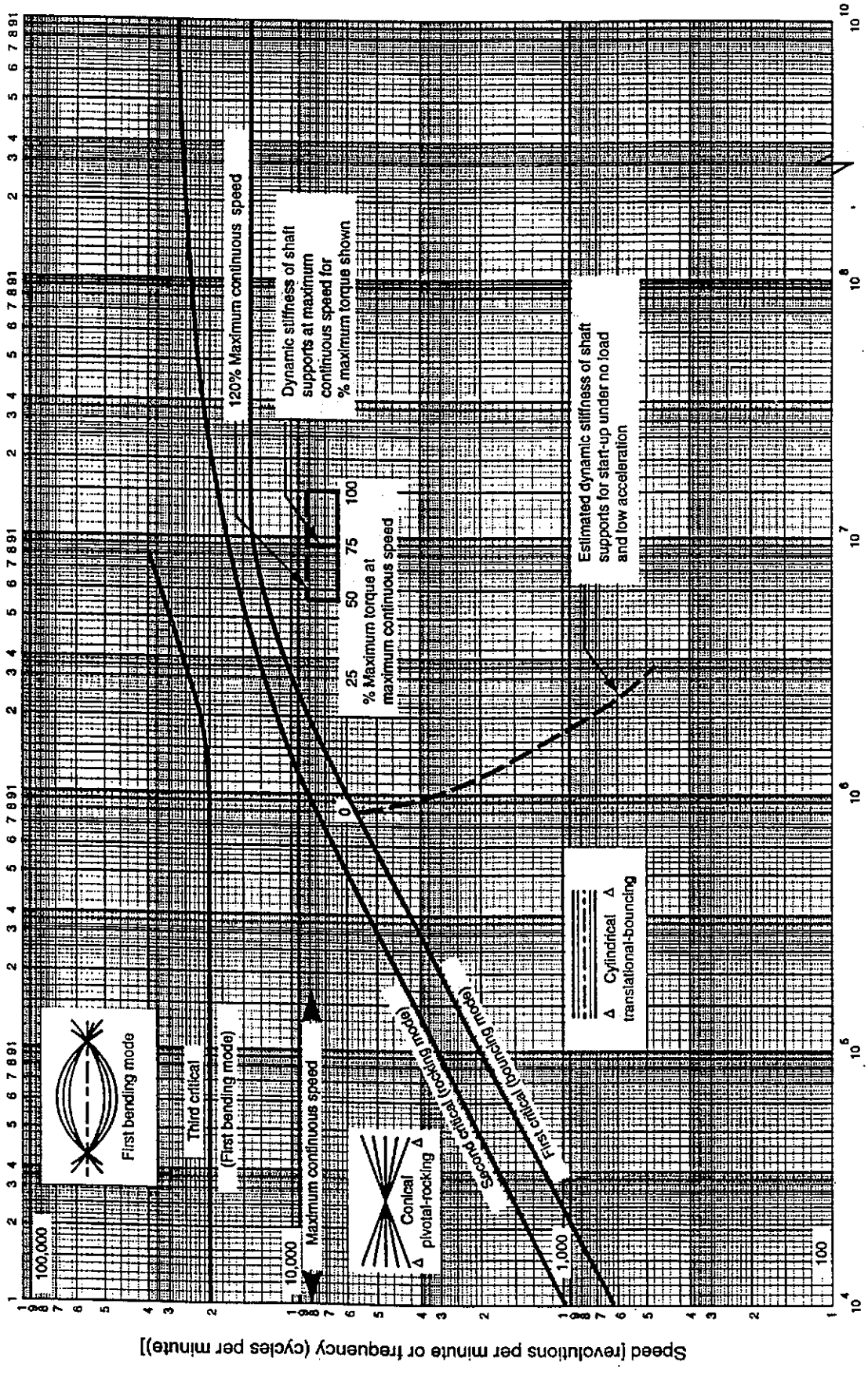
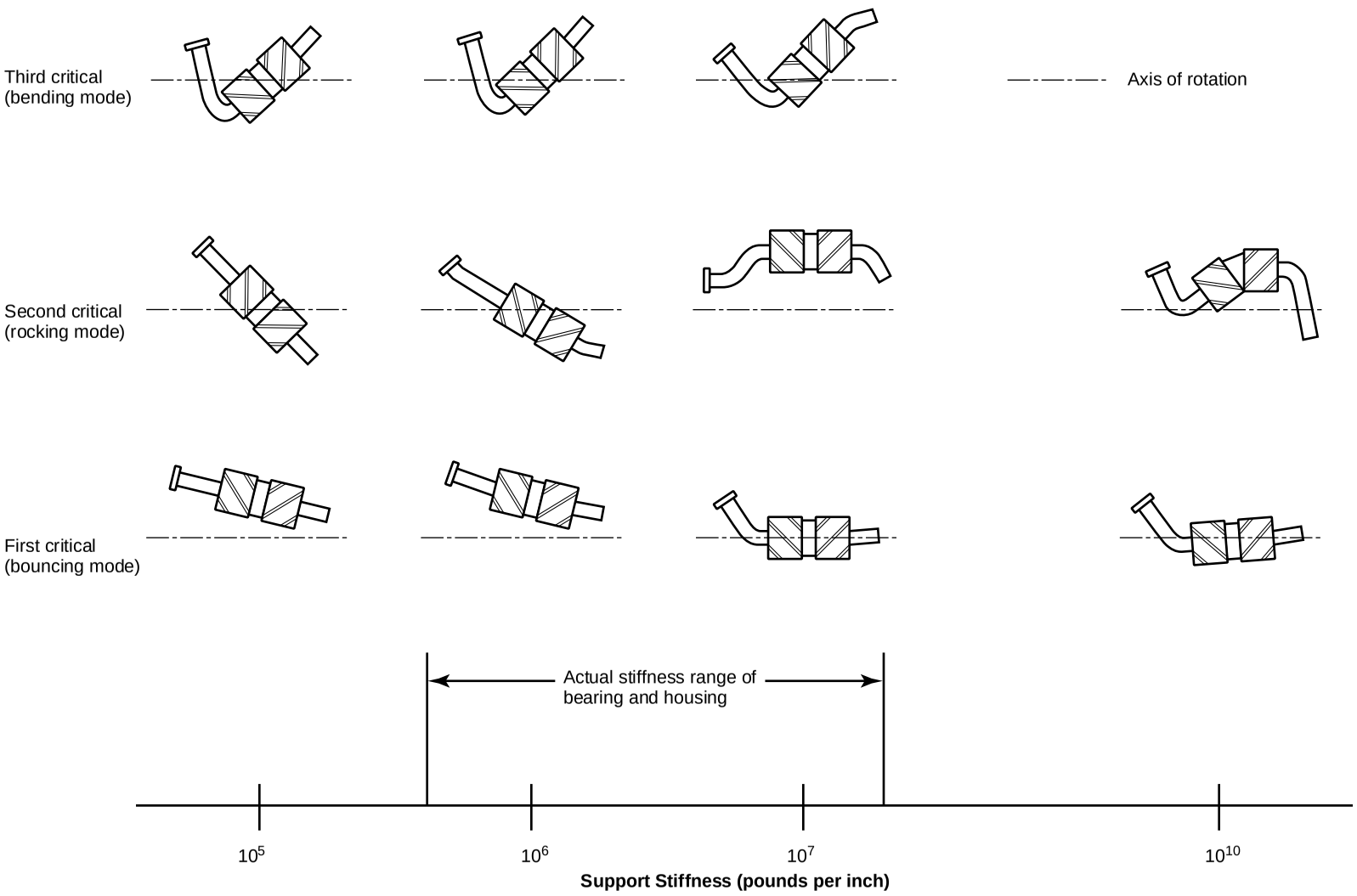


Figure B-1—Lateral Critical Speed Map for a Typical Rotor

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Note: The mode shapes in this figure are normalized, which exaggerates the deflections of the rotor. The actual maximum deflection may be so small that it is insignificant. These mode shapes apply only to operation directly on a critical speed. Mode shapes will vary with the geometry of the rotor.

Figure B-2—Mode Shapes Versus Support Stiffness for a Typical Rotor

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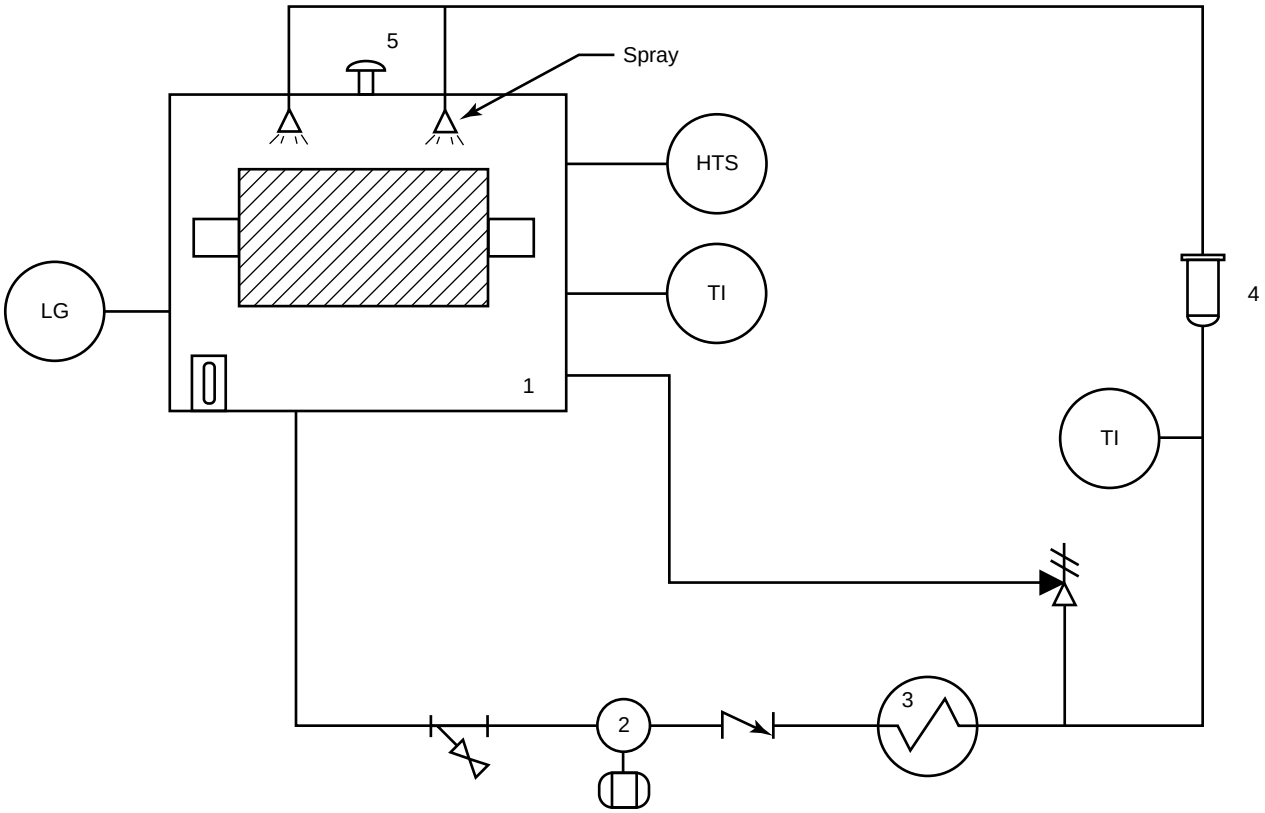
APPENDIX C—TYPICAL PRESSURIZED LUBE-OIL SYSTEM

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Legend

1. Gear and sump
2. Circulating oil pump
3. Heat exchanger
4. Cartridge filter
5. Breather element

LG—Level indicator

HTS—High temperature switch

TI—Temperature indicator

Note: Used with an antifriction bearing design gearbox to increase gear unit's thermal capacity.

Check valve

Strainer

XX — Instrument

Relief Valve

Motor

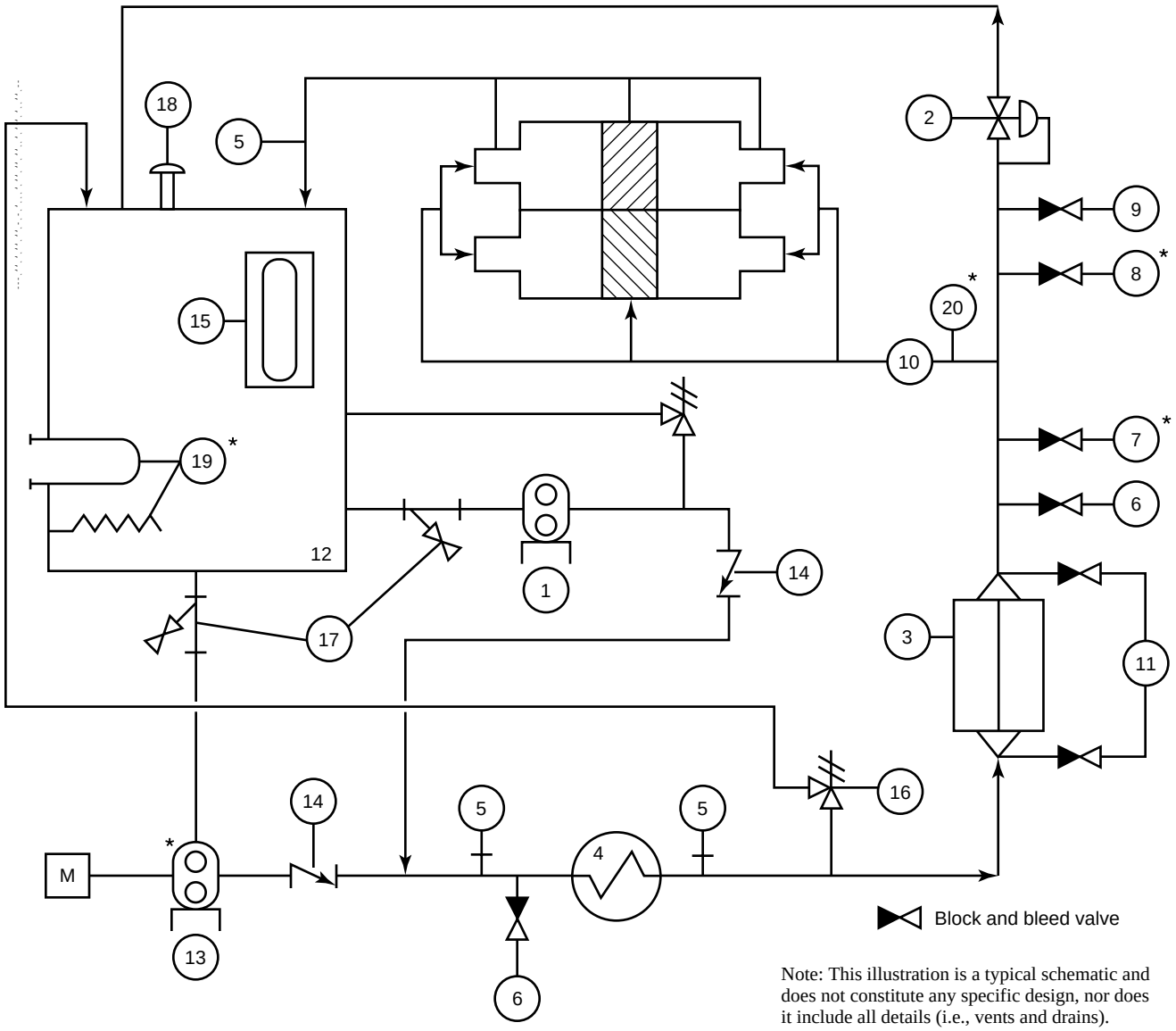
Filter

Figure C-1—Typical Lube-Oil Circulation System for Gear Units

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Note: This illustration is a typical schematic and does not constitute any specific design, nor does it include all details (i.e., vents and drains).

1. Shaft driven main oil pump
2. Pressure regulating valve
* 3. Duplex full flow filter
4. Oil cooler
5. Temperature gauge
6. Pressure gauge
* 7. Low pressure alarm switch
* 8. Low pressure auxiliary oil pump start-up switch
9. Low pressure trip switch
10. Sight flow indicator
11. Pressure differential indicator
12. Oil reservoir (separate)
13. Auxiliary oil pump
* 14. Check valve
15. Oil-level indicator
16. Relief valve
17. Suction strainer
18. Filler/breather
* 19. Sump heater
* 20. High oil temperature alarm switch

* Optional items

Figure C-2—Typical Pressurized Lube-Oil System for Hydrodynamic Bearing in Gear Units

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APPENDIX D—MATERIALS AND MATERIAL SPECIFICATIONS FOR GENERAL PURPOSE GEAR UNITS

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Table D-1—Material Specifications for General Purpose Gear Units

Part	Material[a]	Form
Casing		
Cast	ASTM A 27, Grade 65-35	Cast
	ASTM A 48, Class 30[b]	Cast
Fabricated	ASTM A 575, A576	Hot-rolled bars
	ASTM A 131	Plate
	ASTM A 283	Plate
	ASTM A 284, Grade B	Plate
	ASTM A 285	Plate
	ASTM A 516	Plate
	ASTM A 6	Plate or shapes
	ASTM A 36	Plate or shapes
	AISI 1010	Plate or shapes
	AISI 1020	Plate or shapes
Gear		
Forged	AISI 4140	Nitrided or through hardened
	AISI 4340	Nitrided or through hardened
	AISI E4140H	Through hardened
	AISI E4145	Through hardened
	AISI E4340[c]	Through hardened
	AISI 2317	Carburized
	AISI 3310	Carburized
	AISI 4320	Carburized
	AISI 4620	Carburized
	AISI 8620	Carburized
	AISI 9310	Carburized
Fabricated		
Web	ASTM A 283, Grade B	Plate
	ASTM A 285, Grades B and C	Plate
	ASTM A 36	Plate or shapes
Hub	AISI 1020	Forged or hot rolled
	AISI 4140	Forged or hot rolled
	AISI 4340	Forged or hot rolled
Rim	AISI 1045	Through hardened
	AISI 4130	Through hardened
	AISI 4135	Through hardened
	AISI E4140H	Through hardened
	AISI E4145	Through hardened
	AISI 4140	Nitrided or through hardened
	AISI 4340	Nitrided or through hardened
	AISI 2317	Carburized
	AISI 3310	Carburized
	AISI 4320	Carburized
	AISI 4620	Carburized
	AISI 8620	Carburized
	AISI 9310	Carburized
Forged Pinion	AISI 4140	Nitrided or through hardened
	AISI 4340	Nitrided or through hardened
	AISI E4145H	Through hardened
	AISI E4340H[d]	Through hardened
	AISI 2317	Carburized
	AISI 3310	Carburized
	AISI 4320	Carburized
	AISI 4620	Carburized
	AISI 8620	Carburized
	AISI 9310	Carburized
	AISI 1030	Forged or hot rolled
	AISI 1045	Forged or hot rolled
	AISI 4140	Forged or hot rolled
	AISI 4145H	Forged or hot rolled
	AISI 4340	Forged or hot rolled

[a]Descriptions of AISI designations can be found in ASTM DS 56B.
[b]Minimum.
[c]Vacuum degas on critical applications.
[d]Generally vacuum degas.

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APPENDIX E—VENDOR DRAWING AND DATA REQUIREMENTS

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TYPICAL
VENDOR DRAWING AND
DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PURCHASE ORDER NO. _____ DATE _____
REQUISITION NO. _____ DATE _____
INQUIRY NO. _____ DATE _____
PAGE 1 OF 2 BY _____

FOR _____ REVISION _____
SITE _____ UNIT _____
SERVICE _____ NO. REQUIRED _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

DISTRIBUTION RECORD

Final—Received from vendor	Final—Due from vendor[c]	Review—Returned to vendor	Review—Received from vendor	Review—Due from vendor[c]	DESCRIPTION
					1. Certified dimensional outline drawing and list of connections
					2. Cross-sectional drawings and part numbers[d]
					3. Rotor assembly drawings and part numbers[d]
					4. Thrust-bearing assembly drawing and part numbers[d]
					5. Journal-bearing assembly drawings and bill of materials
					6. Coupling assembly drawing and bill of materials[d,e]
					7. Lube-oil schematic and bills of materials
					8. Lube-oil arrangement drawings and list of connections[e]
					9. Lube-oil component drawings and data[d]
					10. Electrical and instrumentation schematics and bill of materials
					11. Electrical and instrumentation arrangement drawing and list of connections
					12. Tooth-contact drawing and specifications[d]
					13. Tooth-contact check records
					14. Record of deviations from manufacturing process control system[f]
					15. Vibration analysis data
					16. Lateral critical speed analysis report[f]
					17. Torsional analysis report[f]
					18. Input and output shaft position diagram
					19. Weld procedures[f]
					20. Hydrostatic test logs (oil system)
					21. Mechanical running test logs[f]
					22. Rotor balancing logs
					23. Rotor mechanical and electrical runout[f]
					24. As-built data sheets
					25. As-built dimensions or data[d]
					26. Installation manual
					27. Operating, maintenance, and technical manual
					28. Spare parts recommendation and price list

[a]Proposal drawings and data do not have to be certified or as-built.
[b]Purchaser will indicate in this column the desired time frame for submission of materials, using the nomenclature given at the end of the form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and shall include this form with his proposal.
[d]These items are normally provided only in the instruction manual.
[e]If furnished by the vendor.
[f]If specified.

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GENERAL-PURPOSE STEAM TURBINE VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PAGE 2 OF 2 BY _____
DATE _____ REV NO. _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

DISTRIBUTION RECORD

Final—Received from vendor	Final—Due from vendor[c]	Review—Returned to vendor	Review—Received from vendor	Review—Due from vendor[c]	DESCRIPTION
					29. Progress reports and delivery schedule
					30. Preservation, packaging, and shipping procedures
					31. List of special tools furnished for maintenance
					32. Material safety data sheets (OSHA Form 20)
					33. Nondestructive test procedures and acceptance criteria
					34. Tabulation of utility requirements
					35. Optional test data and reports

[a]Proposal drawings and data do not have to be certified or as-built.
[b]Purchaser will indicate in this column the desired time frame for submission of materials, using the nomenclature given at the end of the form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and shall include this form with his proposal.
[d]These items are normally provided only in the instruction manual.
[e]If furnished by the vendor.
[f]If specified.

Notes:

1. Where necessary to meet the scheduled shipping date, the vendor shall proceed with manufacture upon receipt of the order and without awaiting the purchaser's approval of drawings.
2. The vendor shall send all drawings and data to _____
   _____
3. All drawings and data shall show project, purchase order, and item numbers as well as plant location and unit. One set of the drawings and instructions necessary for field installation, in addition to the copies specified above, shall be forwarded with shipment.

Nomenclature:

S—number of weeks before shipment.
F—number of weeks after firm order.
D—number of weeks after receipt of approved drawings.

Vendor _____
Date _____ Vendor Reference_____
Signature _____

(Signature acknowledges receipt of all instructions)

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Description

1. Certified dimensional outline drawing and list of connections including the following:
   a. The size, rating, and location of all customer connections.
   b. Approximate overall and handling weights.
   c. Overall dimensions, and maintenance and dismantling clearances.
   d. Centerline height of each shaft.
   e. Dimensions of baseplates (if furnished) complete with diameters, number, and locations of bolt holes and the thicknesses of sections through which the bolts must pass, and recommended clearances.
   f. Grouting details.
   g. Center of gravity and lifting points.
   h. Shaft end separation and alignment data.
   i. Direction of rotation for each shaft.
   j. Winterization, tropicalization, and/or noise attenuation details, when required.
2. Cross-sectional drawings and part numbers.
3. Rotor assembly drawings and part numbers.
4. Thrust-bearing assembly drawings and bills of materials.
5. Journal-bearing assembly drawings and bills of materials.
6. Coupling assembly drawings and bills of materials.
7. Lube-oil schematic and bill of materials including the following:
   a. Oil flows, temperatures, and pressures at each use point.
   b. Control, alarm, and trip settings (pressure and recommended temperatures).
   c. Total heat loads.
   d. Utility requirements, including electrical, water, and air.
   e. Pipe, valve, and orifice sizes.
   f. Instrumentation, safety devices, control schemes, and wiring diagrams.
8. Lube-oil arrangement drawing and list of connections.
9. Lube-oil component drawings and data including the following:
   a. Pumps and drivers.
   b. Coolers, filters, and reservoir.
   c. Instrumentation.
   d. Spare parts lists and recommendations.
10. Electrical and instrumentation schematics and bills of materials including the following:
    a. Vibration alarm and shutdown limits.
    b. Bearing temperature alarm and shutdown limits.
    c. Lube-oil temperature alarm and shutdown limits.
    d. Driver.
11. Electrical and instrumentation arrangement drawing and list of connections.
12. Tooth-contact drawing and specifications.
13. Tooth-contact check records.
14. Record of deviations from manufacturing process control system.
15. Vibration analysis data, including the following:
    a. The number of teeth on each gear.
    b. The number of teeth on each gear-type coupling (when furnished).
16. Lateral critical speed analysis report.
17. Torsional analysis report.
18. Input and output shaft position diagram.
19. Weld procedures.
20. Hydrostatic test logs (oil system).
21. Mechanical running test logs.
22. Rotor balancing logs.
23. Rotor mechanical and electrical runout.

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24. As-built data sheets.

25. As-built dimensions or data.

26. Installation manual—including the following as a minimum:
    a. Storage procedures.
    b. Foundation plan.
    c. Grouting.
    d. Setting equipment, rigging procedures, component weights, and lifting diagram.
    e. Shaft alignment diagram.
    f. Piping recommendations.
    g. Composite outline drawing for driven equipment/gear/driver train, including anchor bolt locations.
    h. Dismantling clearances.

27. Operating, maintenance, and technical manual—including the following as a minimum:

    Section 1—Operation:
    a. Start-up including tests and checks before start-up.
    b. Routine operational procedures.
    c. Lube-oil recommendations.

    Section 2—Disassembly and assembly:
    a. Gears in casing.
    b. Journal bearings.
    c. Thrust bearings (including clearance and preload on antifriction bearings).
    d. Seals.
    e. Thrust collars, if applicable.
    f. Allowable wear of running clearances.
    g. Fits and clearances for rebuilding.
    h. Bolt torque values.
    i. Illustrated maintenance procedures and intervals.

    Section 3—Vibration data:
    a. Vibration analysis data.
    b. Lateral critical speed analysis.
    c. Torsional critical speed analysis.

    Section 4—As-built data:
    a. As-built data sheets.
    b. As-built dimensions or data (see Item 25).
    c. Noise data sheets.
    d. Performance data.

    Section 5—Drawing and data requirements:
    a. A drawing list, including latest revision numbers and dates.
    b. Certified dimensional outline drawing and list of connections.
    c. Cross-sectional drawing and bill of materials.
    d. Shaft seal drawing and bill of materials.
    e. Rotor assembly drawings and bills of materials.
    f. Thrust-bearing assembly drawings and bills of materials.
    g. Journal-bearing assembly drawings and bills of materials.
    h. Lube-oil schematic and bills of materials.
    i. Lube-oil arrangement drawing and list of connections.
    j. Lube-oil component drawings and data, and bill of materials.
    k. Electrical and instrumentation schematics, wiring diagrams, and bills of materials.
    l. Electrical and instrumentation arrangement drawing and list of connections.
    m. Coupling assembly drawing and bill of materials.
    n. Complete parts list referenced to cross-section drawings.

28. Spare part recommendations and price list.

29. Progress reports and delivery schedule.

30. Preservation, packaging, and shipping procedures.

31. List of special tools furnished for maintenance.

32. Material safety data sheets (OSHA Form 20).

33. Nondestructive test procedures and acceptance criteria.

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APPENDIX F—COUPLINGS FOR GEAR UNITS

F.1 General

F.1.1 This appendix is intended to provide a guide to the selection of coupling types and the location of thrust bearings in equipment trains that employ gears. This appendix is not intended to supersede this standard or the information contained in the data sheets.

F.1.2 Gear units must be connected to driving and driven machines by means of couplings that will not impose harmful forces on the rotating elements of the gear unit. This is necessary to maintain uniform distribution of the tooth loading across the face of the gears throughout varying thermal and load conditions. Excessive moment forces exerted across a coupling will cause the pinion or gear to cock in its bearings, resulting in a shift in the tooth loading toward one end of the gear teeth. Excessive axial force transmitted across a coupling will cause one helix of a double-helical gear to be more heavily loaded than the other if the arrangement of machinery is such that the axial force is transmitted across the gear-tooth mesh to reach an opposing thrust bearing.

F.2 Coupling Types

F.2.1 Many different types of couplings have been developed over the years, and many variations within a particular generic type of coupling have evolved. The coupling types listed in F.2.2 through F.2.5 are not intended to be all inclusive but are intended to include the most popular types used in conjunction with gear units.

F.2.2 Generic types (popular designations) of couplings include gear-tooth or gear-type, grid-type, flexible-disk, and diaphragm couplings.

F.2.3 Rubber-bushed couplings (which are not recommended for high-speed applications) include the rubber-in-shear, flexible-shaft (quill), and rigid-flange (solid) types.

F.2.4 Any of the couplings listed in F.2.2 and F.2.3 may be arranged to have a limited end float, that is, to limit the axial freedom of one connected shaft with respect to the other.

F.2.5 Of the couplings listed in F.2.2 and F.2.3, those considered to be torsionally flexible, or soft, are the grid-type, rubber-in-shear, and flexible-shaft couplings.

F.2.6 Several of the coupling types listed in F.2.2 and F.2.3 have the potential for transmitting high axial forces as thermal or load changes cause connected shafts to grow toward each other or to try to separate. Only in the gear-tooth and grid types is this axial force indeterminate, since in these types the slip force is directly related to the power being transmitted and the coefficient of friction that the coupling is exhibiting at the moment. This coefficient or friction depends on many variables, including the following:

a. The driving force on the coupling teeth.
b. The speed of rotation.
c. The condition of the coupling tooth surfaces.
d. The hardness of the coupling tooth surfaces.
e. The degree of misalignment between the connected shafts.
f. The smoothness of operation of the connected machines.
g. The lubrication of the coupling teeth.

Experiments have been performed to try to determine the friction force exhibited by these couplings, but these attempts have been successful only in proving how widely variable the force is. Machinery designers currently assume a coefficient of friction of 0.25, although they know that under adverse maintenance conditions this value may reach 0.3 or more. With a tendency toward over-design in the name of reliability, the sizing of thrust bearings becomes a problem that may result in an inefficient gear unit. A definite hazard of this approach lies in the possibility of overloading the teeth of a double-helical gear in the event of coupling hang-up caused by sludging or excessive wear.

F.3 Diaphragm Couplings

Flexible-disk and diaphragm-type couplings have a distinct advantage with regard to axial force problems, since the force required to displace one half of a coupling with respect to the other half is quite predictable. Once the machine is properly installed and correct axial settings are obtained, the maximum axial force that may be transmitted across a gear mesh or carried by opposing thrust bearings is known. Thrust bearings can be selected for optimum conditions. The axial forces to be transmitted across the gear mesh can readily be included with other factors in the sizing of the gear unit.

F.4 Limited-End-Float Couplings

F.4.1 The use of limited-end-float couplings makes it possible to retain the known advantages of gear-type couplings while eliminating or reducing the potential problem of excessive thrust. In machinery trains in which only one unit (such as a compressor) requires thrust bearings to maintain the internal axial clearances between the stator and the rotor, one limited-end-float coupling between the compressor and the gear, and a second limited-end-float coupling between the gear and the motor eliminate the need for thrust bearings on either the gear or the motor. This arrangement minimizes the load on the compressor thrust bearing since the most it will feel from the connected machinery will be equal to the motor centering force. (See Figure F-1, Panel A).

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F.4.2 In a machinery train that involves a steam- or gas-turbine prime mover, a double-helical gear, and a compressor, both the turbine and the compressor require thrust bearings. In this case, the use of a single limited-end-float coupling can eliminate the thrust bearing from the gear. Selecting the coupling that has the higher calculated thrust-transmitting potential (usually the low-speed coupling) as a limited-end-float coupling and eliminating the thrust bearing from the gear can drastically reduce the thrust on the machine connected to that gear shaft. Selecting smaller thrust bearings with improved machine efficiency is possible. (See Figure F-1, Panel B).

F.5 Flexible-Shaft Couplings

As machinery trains increase in size and power, the use of flexible-shaft couplings between the high-speed gear and one or both of the connected machines is gaining in popularity. The flexible shafts are usually arranged to pass through hollow pinions or gear shafts, thereby greatly shortening the overall length of the machine, compared with one using more conventional couplings. For turbine-driven generators (Figure F-1, Panel C) or motor-driven compressors (Figure F-1, Panel D), both connections may use a quill arrangement (a flexible shaft through a hollow shaft). For turbine-driven geared compressors (Figure F-1, Panel E), one connection must incorporate axial freedom to accommodate thermal expansion. In this case, an arrangement in which the quill is fixed at one end and flexibly coupled with axial float at the other end should be considered.

F.6 Rigid-Flange Couplings

The only place where a rigid-flange connection between a gear unit and coupled machine is permitted is where the coupled machine has only one radial bearing, as is the case with a single-bearing motor or generator (see Figure F-1, Panel F). Such a coupling arrangement is practical only for relatively low-speed units and where the connected shafts can be installed and maintained in excellent alignment with proper elevation and offset of the single bearing pedestal.

F.7 Torsionally Flexible Couplings

Analysis of the rotor dynamics of a complex multimass system sometimes indicates the need to use torsionally flexible couplings between connected rotors to obtain the designed torsional tuning of the system. Several choices are available to the designer, each with characteristics that must be considered with the entire system in mind; for example, the linearity of the spring rate is different for each of the three types (grid-type, rubber-in-shear, and flexible-shaft) of torsionally flexible couplings in popular use. The ability to accommodate angular and offset misalignment, as well as end-float characteristics, also varies. Selecting the right coupling for the job must be guided by experience in the design and application of machinery systems.

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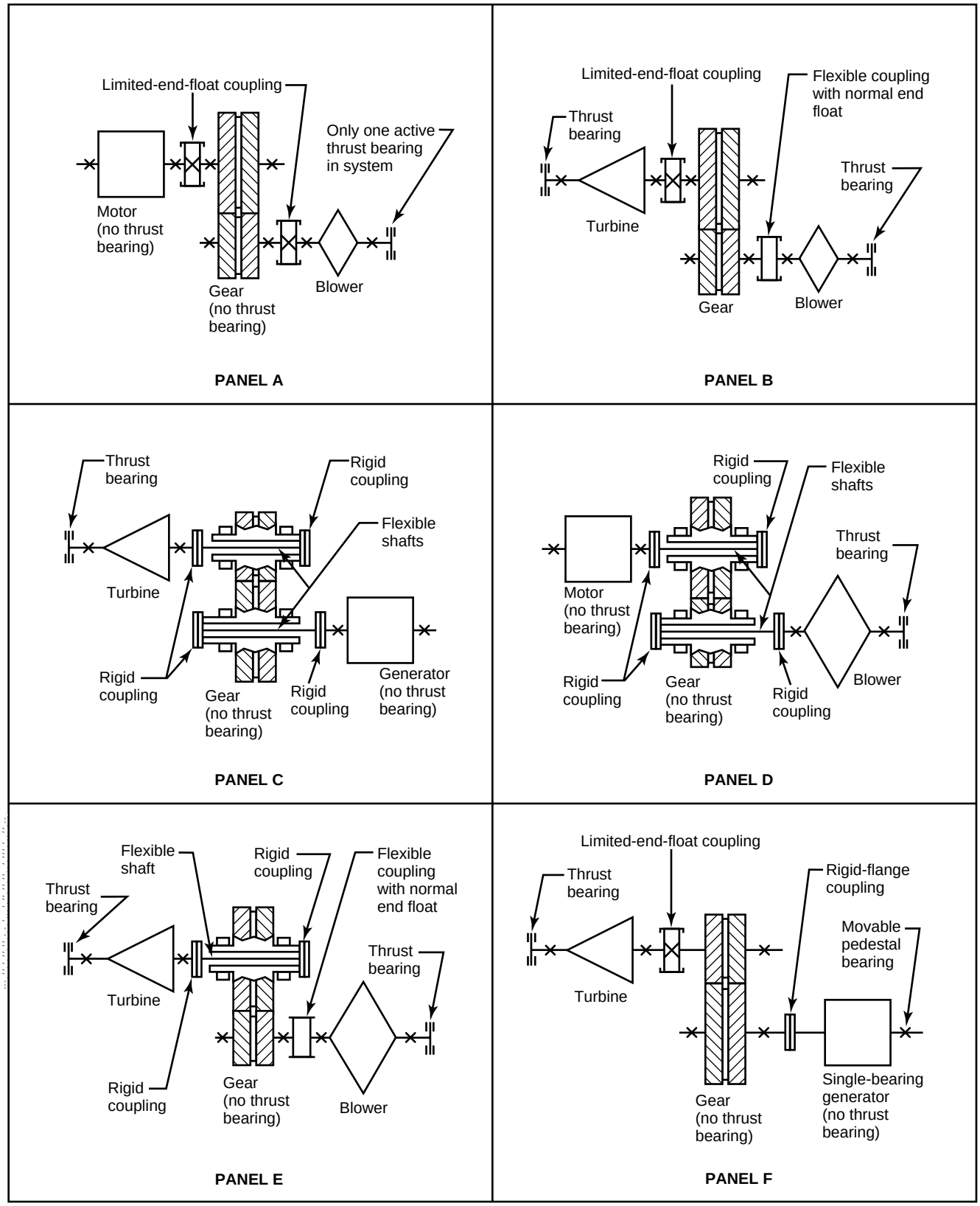


Figure F-1—Couplings for Gear Units

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APPENDIX G—SPIRAL BEVEL GEAR TOOTH CONTACT ARRANGEMENT REQUIREMENTS FOR INSPECTION

G.1 General Information

Spiral bevel gearing requires attention to the orientation of the bevel tooth contact patterns of the gear set at the manufacturer. An inspector at the final testing and assembly will need to recognize the proper gear tooth contact-bearing patterns to ensure that the gear set is properly aligned and ready for installation with the driven equipment (i.e., fan, blower, or agitator).

Determining gear set contact-bearing surface patterns requires an understanding of gear tooth nomenclature. There are nine basic elements of the gear tooth, which are described in Figure G-1, that will aid in certifying that a gear set is properly manufactured and assembled. This nomenclature will be used for all the exhibits and narratives in this appendix.

The TOE of a bevel gear tooth is the portion of the tooth surface at the inner end.

The HEEL of a bevel gear tooth is the portion of the tooth surface at the outer end.

The TOP of a gear tooth is the upper or addendum portion of the tooth surface.

The FLANK of a gear tooth is the lower or dedendum portion of the tooth surface.

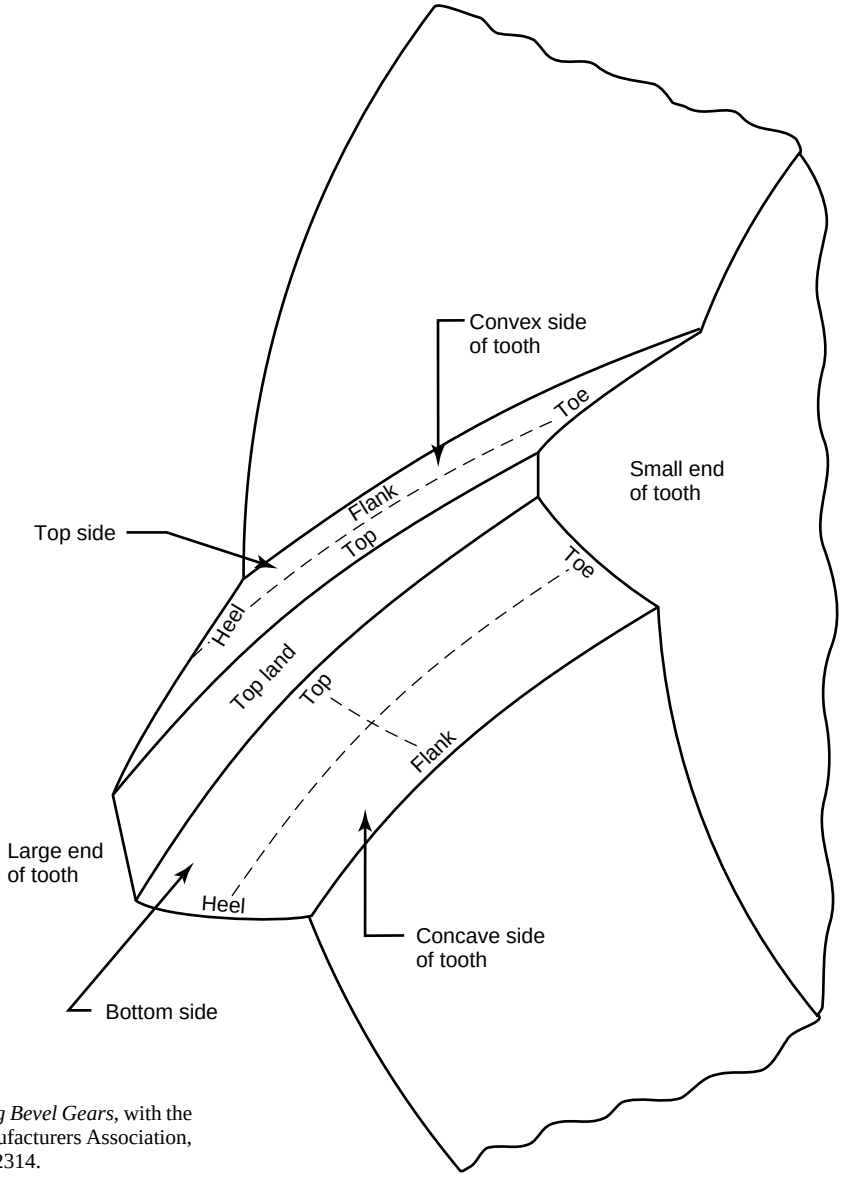
The TOP LAND of a gear tooth is the surface of the top of the tooth.

The TOP SIDE of a tooth.

The BOTTOM SIDE of a tooth.

The CLEARANCE is the distance from the bottom of a tooth space to the top of a mating tooth.

The WORKING AREA is all that portion of the tooth surface above the clearance.



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Figure G-1—Gear Tooth Nomenclature

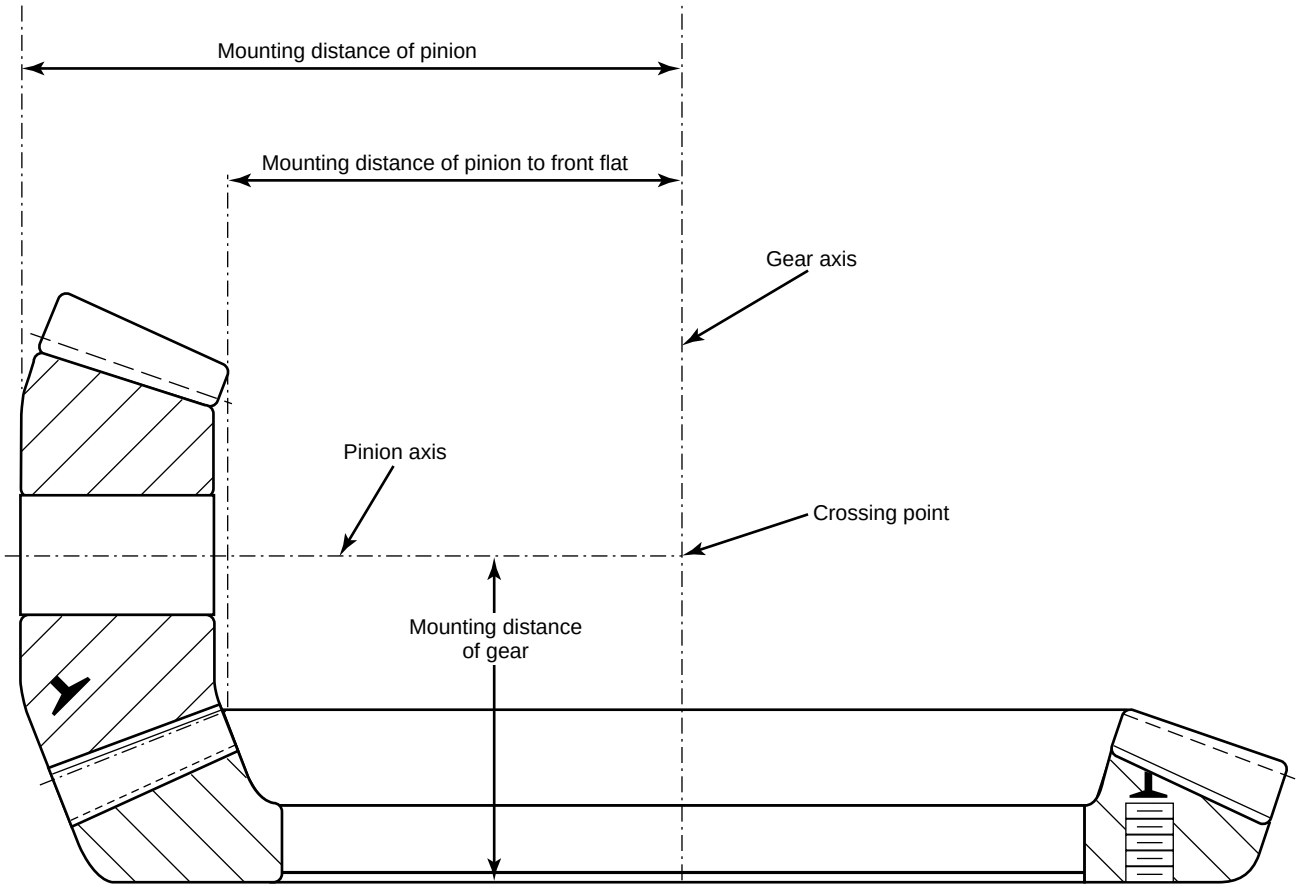
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G.2 Mounting Distance

The quality of performance that is designed and manufactured into a set of bevel gears can only be achieved by correct mounting. Each gear and pinion must be positioned relative to the other to provide proper tooth contact pattern and backlash. Desirable tooth contact patterns and improper contact patterns are illustrated. Typical corrective adjustments (not dimensioned) are indicated in Figure G-2.

The axial position of a bevel gear and pinion in assembly is given by a dimension called the mounting distance (MD) as shown in Figure G-2. This measurement is the linear dimension from the axial locating surface of a given member to the crossing point[a] which is the point of intersection of the pinion and gear axes. Normally, the back of the pinion or gear is used to establish mounting distance; however, for convenience in assembling some gears, a front surface may be used. In all cases, the distance will be given to a flat surface square with the axis of the gear or pinion. The mounting distance is then etched on the surface of the gear set.

[a]Crossing point is the apparent point of intersection of the gear and pinion axis on a drawing showing the two axes. For bevel gears, it is the point of intersection of the two axes.



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Figure G-2—Determining Mounting Distances

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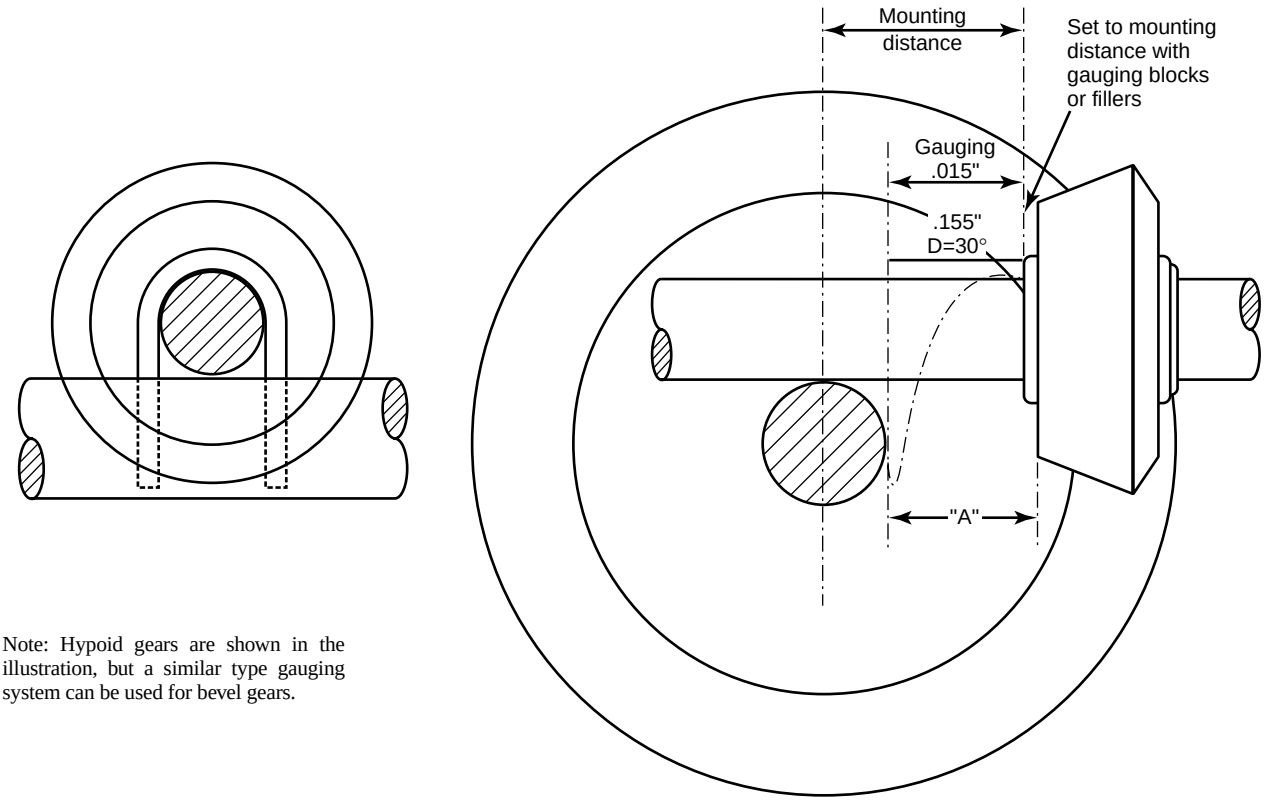
G.3 Positioning Gear and Pinion in Bevel Gear Drives

The pinion shaft assembly, including bearings, is usually contained in a cylindrical cartridge. Pinion mounting distance is set by adjusting thickness of shims between cartridge flange and gear drive housing. A gauge is used to set the pinion member to its proper mounting distance. This mounting distance is measured by means of a special gauge manufactured for this purpose. This gauge is provided by the manufacturer and is a special shop tool. Reference points for measuring distances are shown in Figure G-3.

In the case of large bevel gears where direct measurement of mounting distance is difficult, a flat is hand-ground on the back cone surfaces (back angles) of the gear and pinion when in proper position on the testing machine. When the gears are assembled, they must be positioned so that the hand-ground flats on the back cone surfaces are flush. These surfaces are marked "X" and must be assembled in a manner similar to that used for lapped gears.

The gear member will now be positioned by either of two methods. Once the pinion is positioned, proper location of the gear may now be determined by measuring the backlash. If the backlash does not conform to specifications, the gear member must be repositioned axially.

To adjust, the caps are removed from the housing at the shaft ends and shims are removed from one end as needed, and equivalent thickness shims are added to the opposite end. Caps are then replaced. This procedure allows repositioning of gear/shaft assembly while maintaining bearing clearances.



Note: Hypoid gears are shown in the illustration, but a similar type gauging system can be used for bevel gears.

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Figure G-3—Reference Points in Bevel Gear Drives

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G.4 Positioning Gear and Pinion in Bevel Gear Drives

G.4.1 Backlash is the difference in width of the gear tooth and the gear tooth space in the mating gear (see Figure G-4). It is measured at the tightest point of mesh.

Backlash is necessary to achieve correct operation of the gears and varies with the size of the tooth and the operating conditions. Bevel gears are cut to have a definite amount of backlash when correctly assembled together. Excessive or insufficient backlash can result in noise, excessive wear, and damage. Backlash can be changed by changing the position of one or both members.

Setting the correct backlash is an important part of the gear assembly procedure. Unless specified otherwise by purchaser, the normal backlash is etched or stamped on one or both of the members. Table G-1 shows recommended values but in many cases, manufacturing or operating conditions make it necessary to go outside these values. Deviations to the normal backlash settings will be discussed with purchaser during the engineering of the gear. Values are given for normal backlash at the tightest point of mesh. Normal backlash is measured in a direction normal to the surface of the tooth. It can be checked by locking the pinion against rotation, placing a dial indicator against the gear tooth perpendicular to the tooth surface at the extreme heel of the tooth, and rotating the gear. To establish backlash in the transverse plane rotation, the normal backlash must be divided by the cosine of the spiral angle and the cosine of the pressure angle of the gear teeth. Transverse rotation is approximately 30 percent higher than normal backlash (see Figure G-5).

The graph in Figure G-6 illustrates the amount of axial movement necessary for either the pinion or gear member to obtain a change in backlash.

G.4.2 ASSEMBLING BEVEL GEARS, AXIAL MOVEMENT VS. BACKLASH

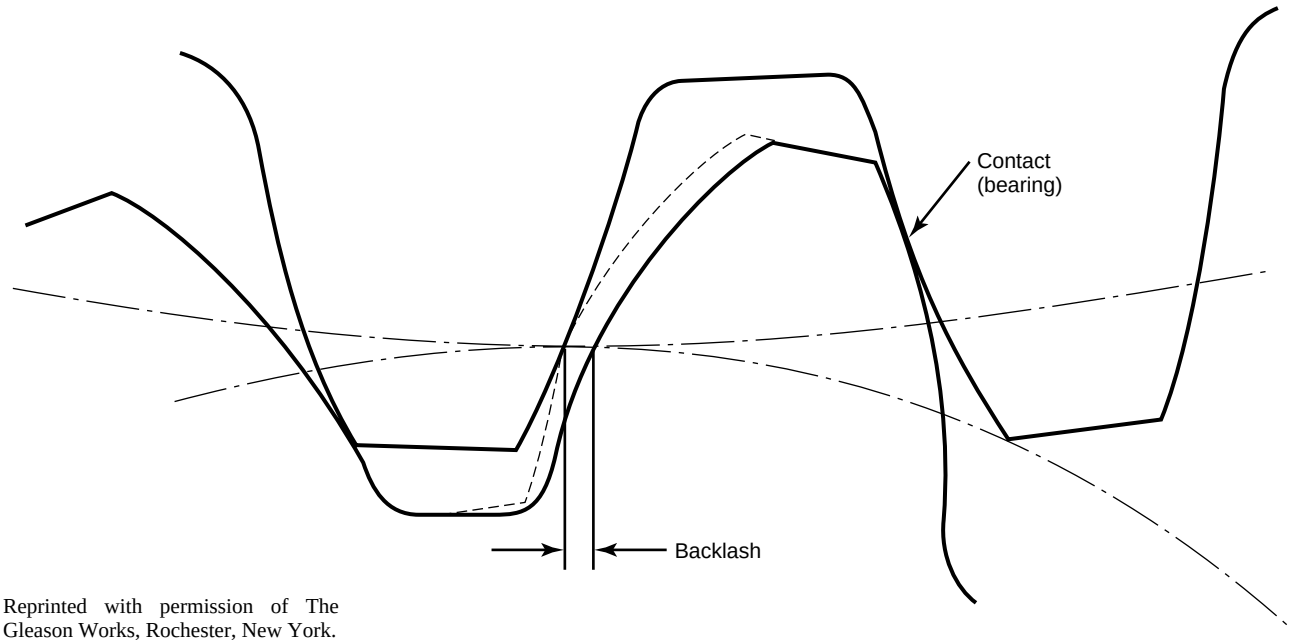
The amount of axial movement for either pinion or gear member necessary to obtain a change in backlash may be determined by the following formulas:

$$\Delta P = \frac{\Delta B_p}{2\tan\phi\sin\gamma}$$

$$\Delta G = \frac{\Delta B_g}{2\tan\phi\sin\Gamma}$$

Where:

ΔB_p = change of backlash for pinion (mm/mil).
ΔB_g = change of backlash for gear (mm/mil).
ΔP = axial movement of pinion (mm/mil).
ΔG = axial movement of gear (mm/mil).
ϕ = pressure angle.
γ = pitch angle pinion.
Γ = pitch angle gear.



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Figure G-4—Backlash in the Plane of Rotation

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Figure G-5—Measurement of Normal Backlash

Table G-1—Recommended Values of Normal Backlash at Tight Points of Mesh (All Values in Inches)

Diametral Pitch	Normal Backlash at Tight Points of Mesh
1.00 to 1.25	.020 – .030
1.25 to 1.50	.018 – .026
1.50 to 1.75	.016 – .022
1.75 to 2.00	.014 – .018
2.00 to 2.50	.012 – .016
2.50 to 3.00	.010 – .013
3.00 to 3.50	.008 – .011
3.50 to 4.00	.007 – .009
4.00 to 5.00	.006 – .008
5.00 to 6.00	.005 – .007
6.00 to 8.00	.004 – .006
8.00 to 10.00	.003 – .005
10.00 to 16.00	.002 – .004
16.00 to 20.00	.001 – .003

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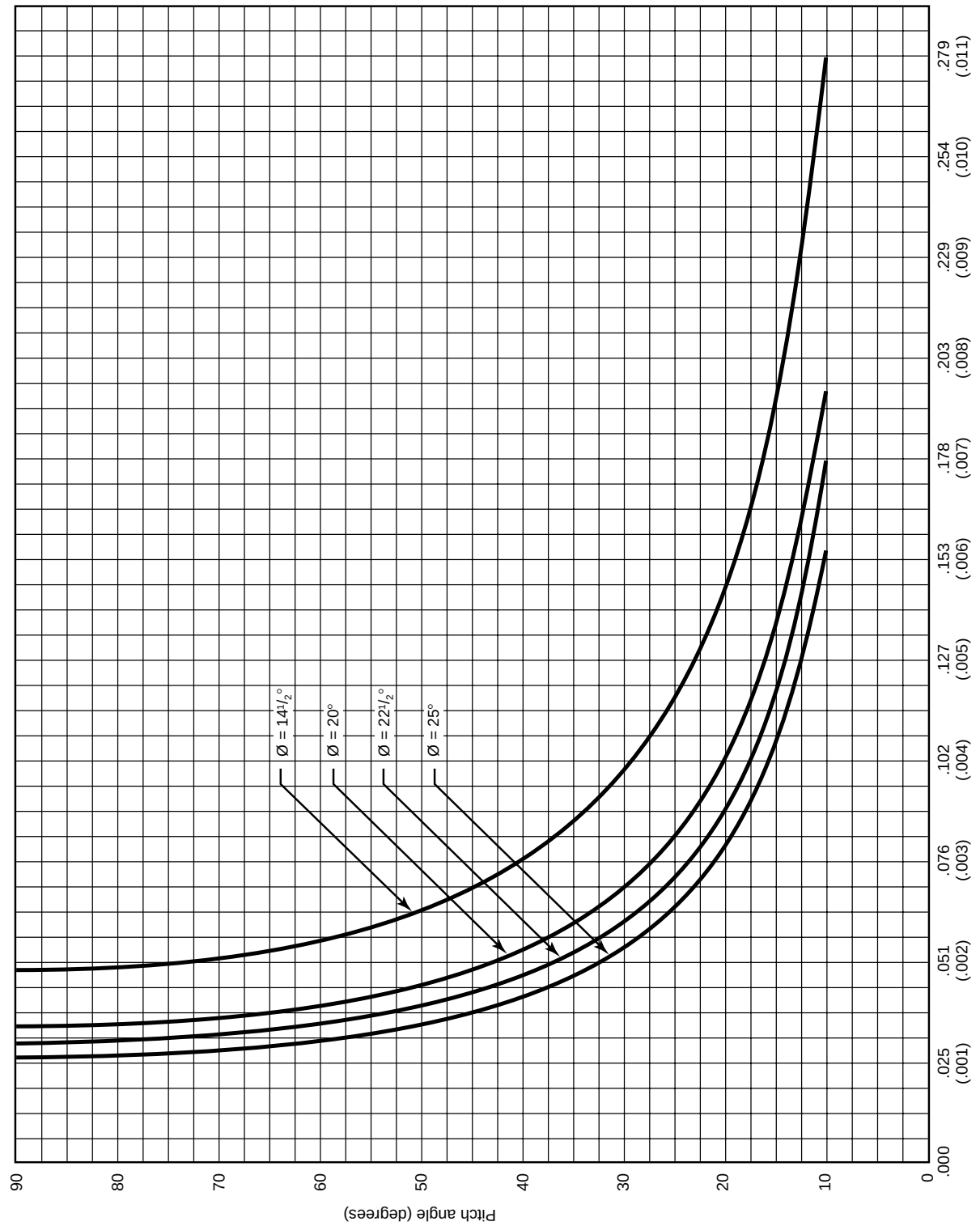


Figure G-6—Axial Movement Necessary to Obtain a Change in Backlash

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G.5 Tooth Contact for Bevel Gears

Inspection of the gear tooth contact surfaces is much more critical in bevel and spiral bevel gear units than in parallel shaft gear units due to the interaction of backlash and bearing patterns. Identifying on the test stand the contact pattern acceptability is an important part of ensuring that any gear unit will provide good dependable service. An inspector should be familiar with the desired tooth contact patterns that will be seen during a gear test procedure. Acceptable contact can be identified and improper alignments can be corrected prior to shipment of the gear unit. Acceptable tooth contact for both bevel and parallel shaft gear units are presented as guides to inspectors.

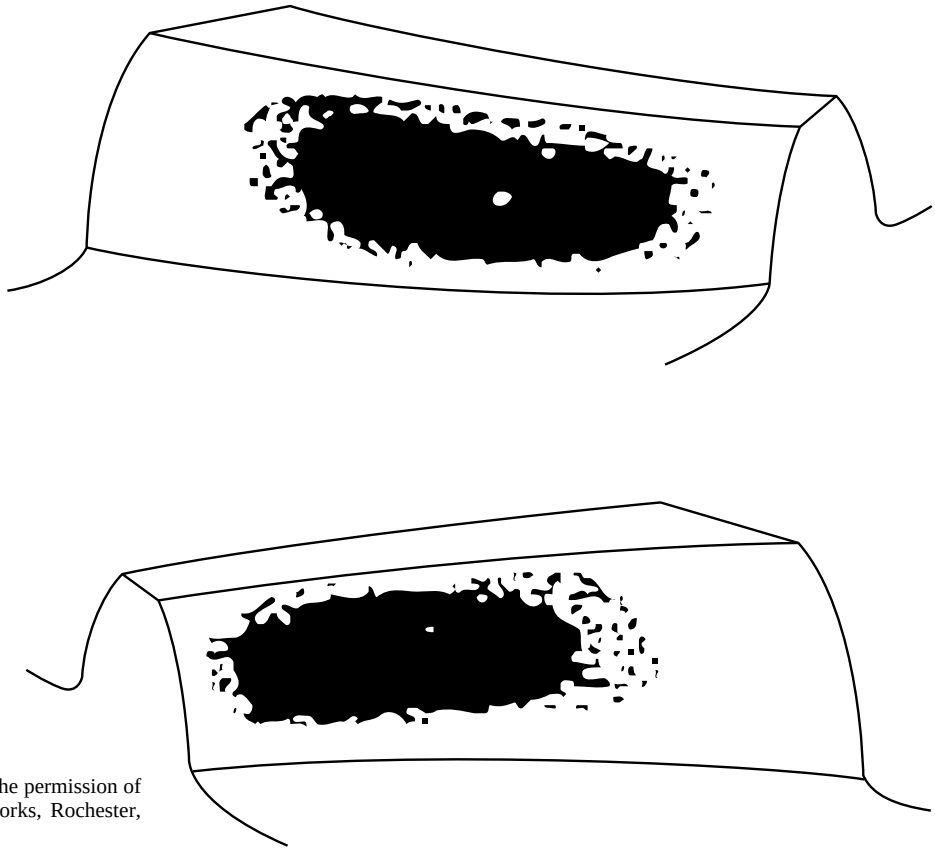
Determining tooth contact in bevel and spiral bevel gear units shall be done in the manufacturer's facility on a suitable bevel gear test machine. This machine should permit adjustment of the mounting distances of both gear members to duplicate mounting in the job housing. If a test machine is not available, tooth contact shall be checked in the job housing.

Using a suitable marking compound, check the tooth contact pattern. Suitable compounds include coating one member with prussian blue, copper sulfate, or finely ground red lead in oil. If the markings on the gear set have been followed, the bearing pattern will conform to accepted standards. A permanent record of tooth-bearing contact can be established by using clear cellophane tape. A tape transfer showing the bearing contact may be lifted directly from the tooth profile and attached to a notated sheet of white paper.

Gears are cut with a contact pattern that will cover about half the length of the tooth, the location will slightly favor the toe end of the tooth (see Figures G-7 and G-8). Under a load condition, the bearing pattern will shift somewhat toward the heel of the tooth, thus becoming more central on the gear teeth. Under no circumstances must the pattern be concentrated on the ends of the teeth.

Situations where tooth contact patterns are not centrally located on the gear teeth will lead to premature failure of the gear in the field. The types of patterns where the tooth contact is not central to the gear teeth implies that the gear members are not mounted correctly in relation to each other. The types of mounting errors will produce different types of tooth contact patterns on the gear teeth. Three common types of contact error (see Figure G-9) are:

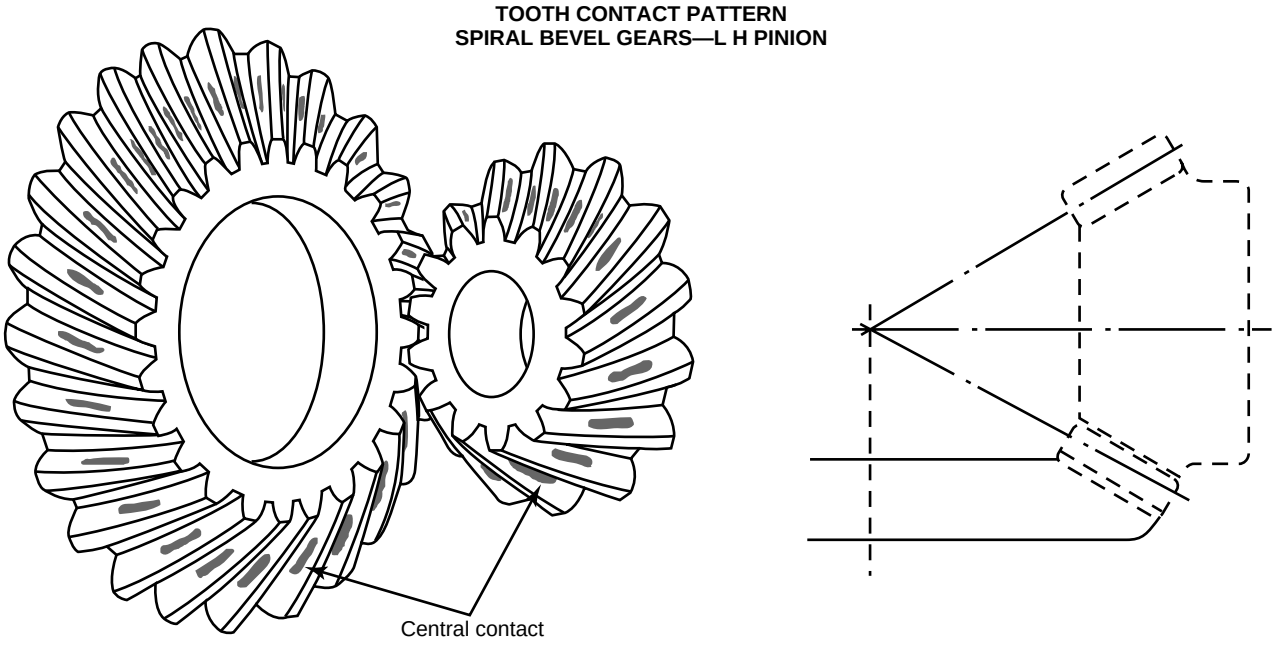
a. Profile error (pinion axial position error).
b. Cross contact error.
c. Shaft angle error.

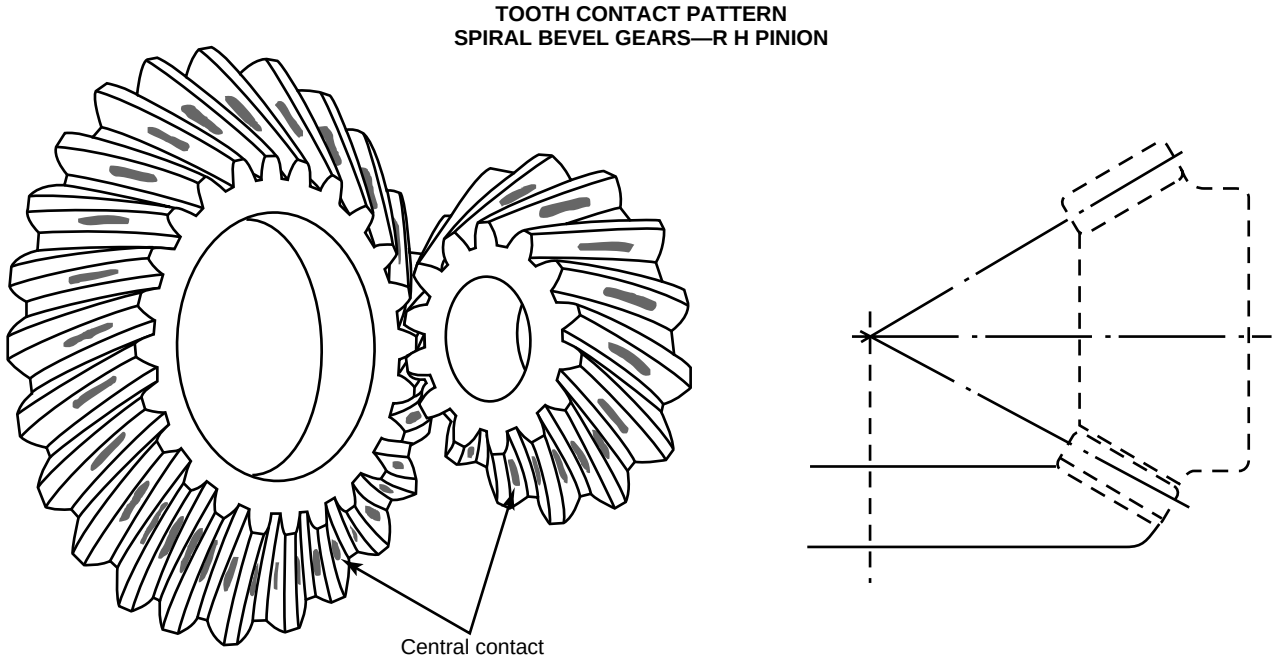


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Figure G-7—Desirable Bearing Pattern

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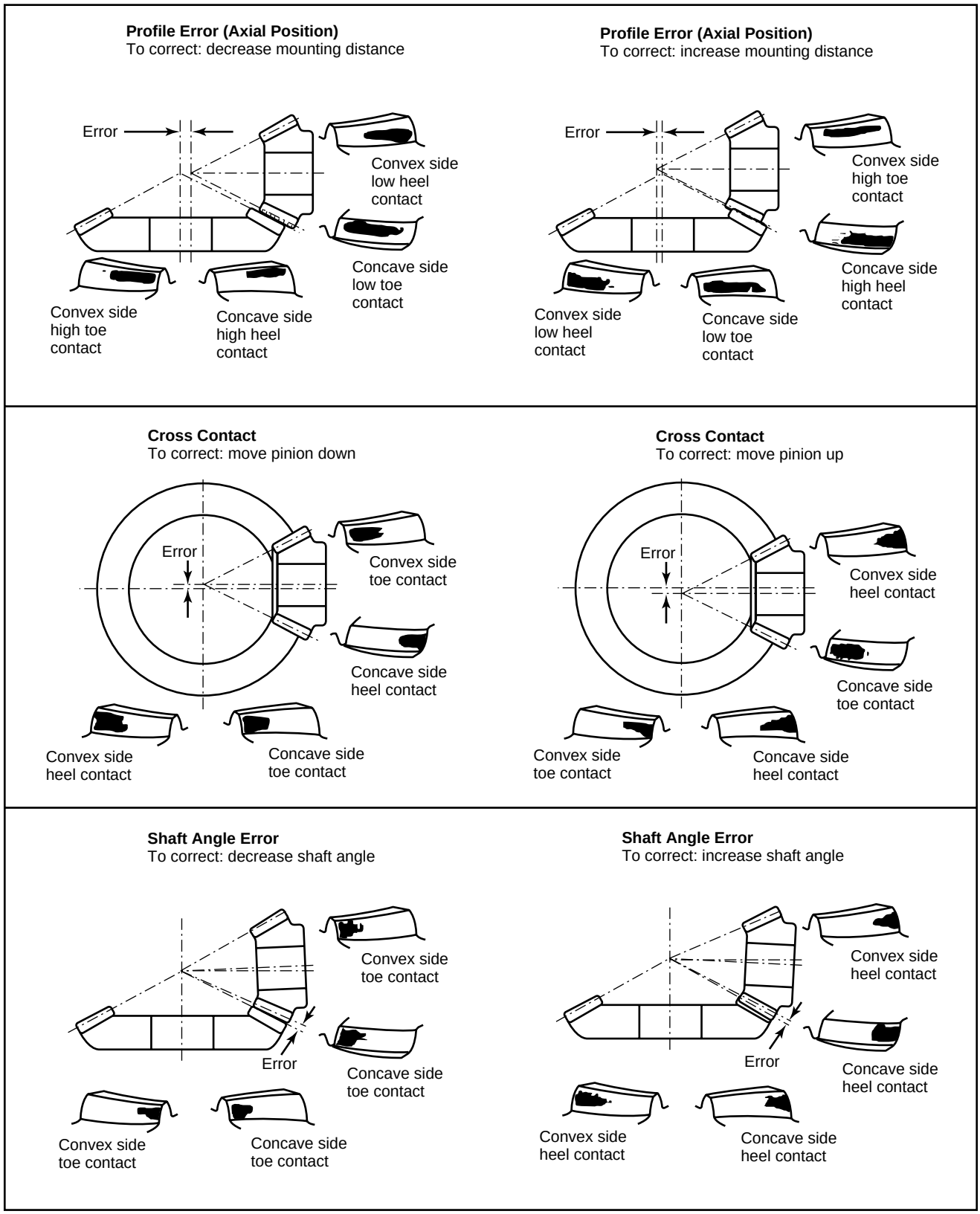




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Figure G-8—Preferred Contact Resulting from Correct Mounting Position

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Figure G-9—Common Types of Contact Error

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APPENDIX H—PROCEDURE FOR DETERMINATION OF RESIDUAL UNBALANCE

H.1 Scope

This appendix describes the procedure to be used to determine residual unbalance in machine rotors. Although some balancing machines may be set up to read out the exact amount of unbalance, the calibration can be in error. The only sure method of determining residual unbalance is to test the rotor with a known amount of unbalance.

H.2 Definition

Residual unbalance is the amount of unbalance remaining in a rotor after balancing. Unless otherwise specified, residual unbalance shall be expressed in gm-mm (gram-millimeters) or oz.-in. (ounce-inches).

H.3 Maximum Allowable Residual Unbalance

H.3.1 The maximum allowable residual unbalance per plane shall be calculated using the equation in 2.8.2.1.

H.3.2 If the actual static weight load on each journal is not known, assume that the total rotor weight is equally supported by the bearings. For example, a two bearing rotor weighing 2,700 kilograms (6,000 pounds) would be assumed to impose a static weight load of 1,360 kilograms (3,000 pounds) on each journal.

H.4 Residual Unbalance Check

H.4.1 GENERAL

H.4.1.1 When the balancing machine readings indicate that the rotor has been balanced to within the specified tolerance, a residual unbalance check shall be performed before the rotor is removed from the balancing machine.

H.4.1.2 To check the residual unbalance, a known trial weight is attached to the rotor sequentially in six (or twelve, if specified by the purchaser) equally spaced radial positions, each at the same radius. The check is run in each correction plane, and the readings in each plane are plotted on a graph using the procedure specified in H.4.2.

H.4.2 PROCEDURE

H.4.2.1 Select a trial weight and radius that will be equivalent to between one and two times the maximum allowable residual unbalance [that is, if U_{max} is 1,440 gm-mm (2 oz.-in.), the trial weight should cause 1,440 to 2,880 gm-mm (2 to 4 oz.-in.) of unbalance].

H.4.2.2 Starting at the last known heavy spot in each correction plane, mark off the specified number of radial positions (six or twelve) in equal (60 or 30 degree) increments around the rotor. Add the trial weight to the last known heavy spot in one plane. If the rotor has been balanced very precisely and the final heavy spot cannot be determined, add the trial weight to any one of the marked radial positions.

H.4.2.3 To verify that an appropriate trial weight has been selected, operate the balancing machine and note the units of unbalance indicated on the meter. If the meter pegs, a smaller trial weight should be used. If little or no meter reading results, a larger trial weight should be used. Little or no meter reading generally indicates that the rotor was not balanced correctly, the balancing machine is not sensitive enough, or a balancing machine fault exists (i.e., a faulty pickup). Whatever the error, it must be corrected before proceeding with the residual unbalance check.

H.4.2.4 Locate the weight at each of the equally spaced positions in turn and record the amount of unbalance indicated on the meter for each position. Repeat the initial position as a check. All verification shall be performed using only one sensitivity range on the balance machine.

H.4.2.5 Plot the readings on the residual unbalance work sheet and calculate the amount of residual unbalance (see Figures H-1 and H-2). The maximum meter reading occurs when the trial weight is added at the rotor's heavy spot; the minimum reading occurs when the trial weight is opposite the heavy spot. Thus, the plotted readings should form an approximate circle (see Figures H-3 and H-4). An average of the maximum and minimum meter readings represents the effect of the trial weight. The distance of the circle's center from the origin of the polar plot represents the residual unbalance in that plane.

H.4.2.6 Repeat the steps described in H.4.2.1 through H.4.2.5 for each balance plane. If the specified maximum allowable residual unbalance has been exceeded in any balance plane, the rotor shall be balanced more precisely and checked again. If a correction is made in any balance plane, the residual unbalance check shall be repeated in all planes.

H.4.2.7 For stack component balanced rotors, a residual unbalance check shall be performed after the addition and balancing of the first rotor component, and at the completion of balancing of the entire rotor, as a minimum.

Note: This ensures that time is not wasted and rotor components are not subjected to unnecessary material removal in attempting to balance a multiple component rotor with a faulty balancing machine.

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Equipment (Rotor) No.: _____

Purchase Order No.: _____

Correction Plane (inlet, drive-end, etc.—use sketch): _____

Balancing Speed: _____ rpm

N—Maximum Allowable Rotor Speed: _____ rpm

W—Weight of Journal (closest to this correction plane): _____ kg (lbs)

U_{max}—Maximum Allowable Residual Unbalance =
6350 *W/N* (4 *W/N*)
6350 × _____ kg/_____ rpm; 4 × _____lbs/_____rpm _____ gm-mm (oz.-in.)

Trial unbalance ($2 \times U_{max}$) _____ gm-mm (oz.-in.)

R—Radius (at which weight will be placed): _____ mm (in.)

Trial Unbalance Weight = Trial Unbalance/*R*
_____gm-mm/_____mm/_____oz.-in./_____inches _____ g (oz.)

Conversion Information: 1 ounce = 28.350 grams

Test Data

Position	Trial Weight Angular Location	Balancing Machine Amplitude Readout
1		
2		
3		
4		
5		
6		
7		

Rotor Sketch

Test Data—Graphic Analysis

Step 1: Plot data on the polar chart (Figure H-2). Scale the chart so the largest and smallest amplitude will fit conveniently.

Step 2: With a compass, draw the best fit circle through the six points and mark the center of this circle.

Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. _____ units

Step 4: Record the trial unbalance from above. _____ gm-mm (oz.-in.)

Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). _____ gm-mm (oz.-in.)

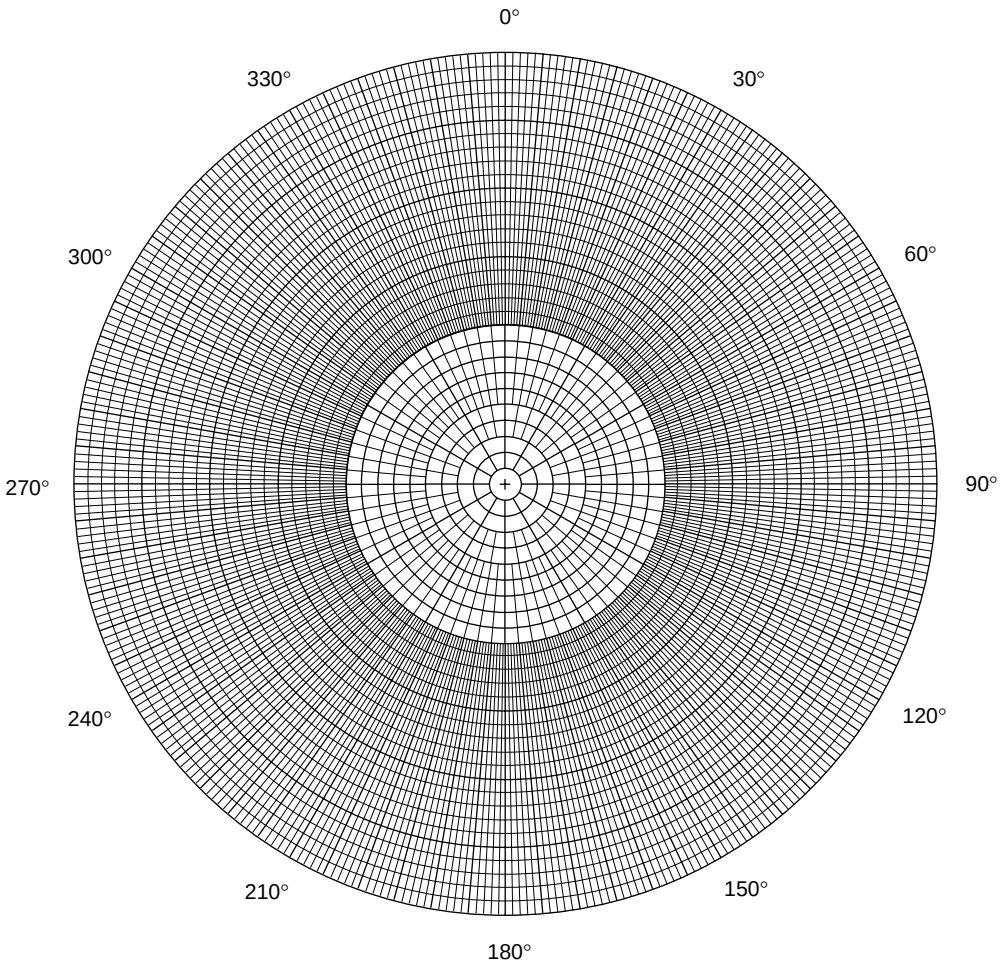
Step 6: Divide the answer in Step 5 by the answer in Step 3. _____ Scale Factor

You now have a correlation between the units on the polar chart and the gm-in. of actual balance.

Notes:
1. The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor.
2. The balancing machine amplitude readout for position "7" should be the same as position "1" indicating repeatability. Slight variations may result from imprecise positioning of the trial weight.

Figure H-1—Residual Unbalance Work Sheet

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance.

NOTE: Several possibilities for the drawn circle not including the origin of the polar chart include: operator error during balancing, a faulty balancing machine pickup or cable, or the balancing machine is not sensitive enough.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance.

Record actual residual unbalance _____ gm-mm (oz.-in.)

Record allowable residual unbalance (from Figure H-1) _____ gm-mm (oz.-in.)

Correction plane_____ for Rotor No._____ (has/has not) passed.

By _____ Date _____

Figure H-2—Residual Unbalance Work Sheet (continued)

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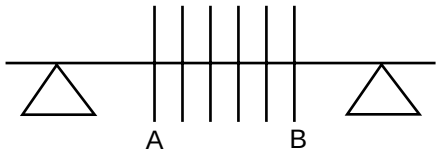
Equipment (Rotor) No.:	*C–101*	
Purchase Order No.:		
Correction Plane (inlet, drive-end, etc.—use sketch):	*A*	
Balancing Speed:	*800*	rpm
N—Maximum Allowable Rotor Speed:	*10,000*	rpm
W—Weight of Journal (closet to this correction plane):	*908*	lbs (~~kg~~)
U_{max}—Maximum Allowable Residual Unbalance = 6350 *W/N* (4 *W/N*) ~~6350 × ____ kg/ ____ rpm; 4~~ × *908* lbs/ *10,000* rpm	*0.36*	oz.-in. (~~gm-mm~~)
Trial unbalance (2 × U_{max})	*0.72*	oz.-in. (~~gm-mm~~)
R—Radius (at which weight will be placed):	*6.875*	in. (~~mm~~)
Trial Unbalance Weight = Trial Unbalance/*R* ~~____ gm-mm/ ____ mm/~~ *0.72* oz.-in./ *6.875* inches	*0.10*	oz. (~~g~~)

Conversion Information: 1 ounce = 28.350 grams

Test Data

Position	Trial Weight Angular Location	Balancing Machine Amplitude Readout
1	*0°*	*14.0*
2	*60°*	*12.0*
3	*120°*	*14.0*
4	*180°*	*23.5*
5	*240°*	*23.0*
6	*300°*	*15.5*
7	*0°*	*14.0*

Rotor Sketch



C-101

Test Data—Graphic Analysis

Step 1: Plot data on the polar chart (Figure H-4). Scale the chart so the largest and smallest amplitude will fit conveniently.

Step 2: With a compass, draw the best fit circle through the six points and mark the center of this circle.

Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record.	*35*	units
Step 4: Record the trial unbalance from above.	*0.72*	oz.-in. (~~gm-mm~~)
Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance).	*1.44*	oz.-in. (~~gm-mm~~)
Step 6: Divide the answer in Step 5 by the answer in Step 3.	*0.041*	Scale Factor

You now have a correlation between the units on the polar chart and the gm-in. of actual balance.

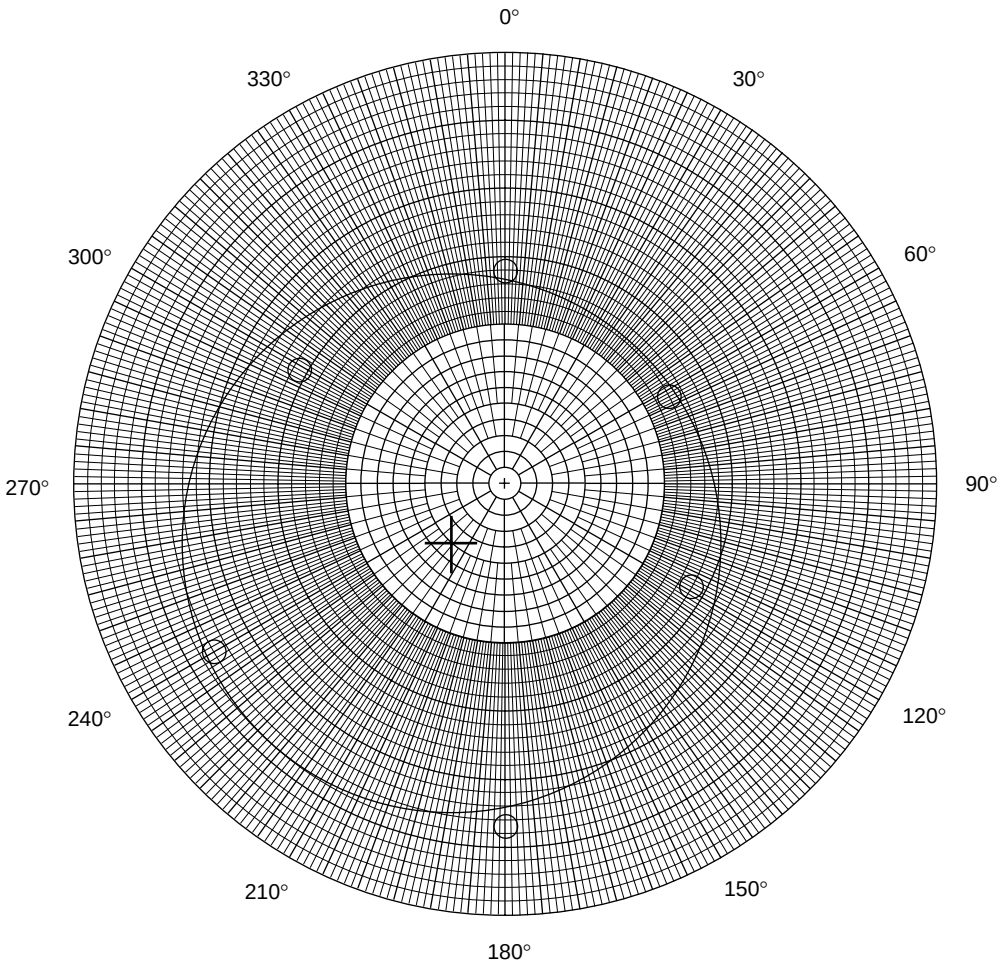
Notes:
1. The trial weight angular location should be referenced to a keyway or some other permanent marking on the rotor.
2. The balancing machine amplitude readout for position "7" should be the same as position "1" indicating repeatability. Slight variations may result from imprecise positioning of the trial weight.

Figure H-3—Sample Calculations for Residual Unbalance

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance.

NOTE: Several possibilities for the drawn circle not including the origin of the polar chart include: operator error during balancing, a faulty balancing machine pickup or cable, or the balancing machine is not sensitive enough.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual unbalance.

Record actual residual unbalance _____*5 (0.041) = 0.21*_____ (oz.-in.)(~~gm-mm~~)

Record allowable residual unbalance (from Figure H-3) _____*0.36*_____ (oz.-in.)(~~gm-mm~~)

Correction plane _____*A*_____ for Rotor No. _____*C-101*_____ (has/~~has not~~) passed.

By _____*John Inspector*_____ Date _____*11-31-94*_____

Figure H-4—Sample Calculations for Residual Unbalance (continued)

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